BORROWED CREDIBILITY

HOW SCIENCE DECIDES WHAT TO BELIEVE

REPUTATION CAN GUIDE ATTENTION.

EVIDENCE SHOULD GOVERN BELIEF.

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

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Preface - Against Cynicism

A scientific paper rarely arrives alone. It arrives wearing clothes: an institutional affiliation, a list of coauthors, a journal target, a funding history, perhaps a famous laboratory name. These details help a busy reader decide where to spend attention. They also create a subtle temptation to treat the history of the messenger as evidence about the truth of the message. This book is about that temptation and, more importantly, about the institutional conditions that make the temptation rational.

The argument is deliberately constructive. Modern research institutions have produced extraordinary knowledge, and most of the mechanisms criticized here were created to solve genuine coordination problems. Peer review exists because expertise matters. Selective journals exist because attention is scarce. Grant panels exist because resources are finite. Reputation exists because trust cannot be rebuilt from zero in every interaction. The problem is not that science uses shortcuts. The problem is that shortcuts can become self-reinforcing, hard to audit, and increasingly detached from the particular claim being judged.

The book therefore treats credibility as infrastructure. Like any infrastructure, it has throughput constraints, legacy components, failure modes, and incentives. Once framed this way, the question changes. Instead of asking whether prestigious institutions are good or bad, we can ask which decisions legitimately need pedigree information, which should be insulated from it, how quickly direct evidence should override it, and how legal, economic, and technical reforms can reduce the cost of verification.

What this book is, and is not

This is not a book about the stupidity of experts, the corruption of universities, or the secret conspiracy of a scientific priesthood.

Those stories are emotionally satisfying because they make a complex problem legible: there are villains, there are victims, and there is a truth waiting outside the gates. Reality is harder. Modern research institutions are full of intelligent people who usually have reasonable motives, often work under severe constraints, and participate in procedures that were invented for defensible reasons. The trouble is that systems can become unreliable without anyone intending them to be unreliable.

The central problem of modern science is not that it uses reputation. It could not function without it. Human beings cannot independently verify every claim on which their work depends. A physicist trusts instrument makers. A physician trusts clinical trial systems. An economist trusts datasets assembled by others. A software researcher trusts compilers, benchmarks, libraries, and thousands of claims embedded in prior work. The total evidential chain behind even an ordinary scientific sentence is far too large for one person to reconstruct. We therefore compress uncertainty. We use names, journals, universities, professional societies, citations, grants, awards, and networks as shortcuts for information we do not have time to inspect directly.

The shortcut becomes dangerous when it changes category. A prestigious institution can be evidence that a researcher passed difficult filters. It cannot be evidence that a particular claim is true. A highly cited paper can be evidence that a claim became influential. It cannot, by itself, be evidence that the claim survived serious attempts at falsification. A famous scientist may deserve a stronger prior than an unknown one in some contexts, but the prior must remain a prior. When the reputation of the speaker begins to substitute for the evidential state of the statement, credibility is no longer merely earned. It is borrowed.

This book studies that transfer. It asks how institutional prestige moves from university to person, from person to paper, from paper to theory, from theory to grant program, and from grant program back to institution. It asks why small early differences can become enormous later differences even among researchers of similar ability. It asks how peer review can remain essential while being noisy, expensive, and vulnerable to prestige effects. It asks why a system that formally rewards novelty often produces strong incentives for conformity. It asks why research that never receives funding may matter as much to the epistemic health of society as research that turns out to be wrong.

A final caution concerns tone. Criticism of academic institutions is easily absorbed into two incompatible narratives: one in which experts are presumed corrupt, and another in which institutional criticism itself is treated as anti-scientific. Both are intellectually lazy. Institutions deserve criticism precisely because expertise matters. The more society depends on specialized knowledge, the more important it becomes to understand how that knowledge is selected, financed, communicated, and corrected. A strong research system should be confident enough to expose its own mechanisms to scrutiny.

The practical ambition is modest but consequential. A reader should finish the book with a way to distinguish at least four questions that are commonly collapsed: Who is likely to be competent? Which work deserves scarce attention? Which claim is currently well supported? Which institution should receive resources for the next attempt? Pedigree can be relevant to the first two, sometimes to the fourth, and only indirectly to the third. Much institutional confusion comes from moving an answer from one question into another without noticing the transfer.

The recurring researcher in the chapters that follow, Elena Varga, is a composite case rather than a real person. Her career is constructed from ordinary features of contemporary research: journal triage, grant thresholds, short-term contracts, competitive review, collaboration networks, publication incentives, replication, and the growing use of automated evaluation. The purpose of the composite is explanatory. It lets the same institutional mechanism be seen at different moments without pretending that one biography could represent a field. Empirical claims are anchored separately in published studies and institutional practice.

Credibility as infrastructure

The argument is deliberately non-antagonistic because the remedies depend on understanding what the existing mechanisms accomplish. If one believes that peer review is merely censorship, the obvious solution is to abolish it. If one understands peer review as a scarce and imperfect allocation of expert attention, the problem becomes more interesting: how can its useful functions be preserved while reducing its power to reproduce hierarchy? If metrics are treated as inherently corrupt, one is tempted to remove them entirely. If metrics are understood as compressed signals with context-dependent error, one can ask which decisions they should inform, which decisions they should never determine, and how to audit their misuse.

A similar caution applies to artificial intelligence. It is easy to imagine an AI system that reads every paper, checks every citation, reproduces every analysis, detects every conflict, and assigns the correct credibility score. Such a system would be useful, but the fantasy hides the central difficulty. AI systems inherit data, incentives, benchmarks, institutional objectives, and failure modes. An AI reviewer trained on historical peer review may learn historical prestige bias more efficiently than humans express it. An AI that is rewarded for confident judgments may manufacture certainty where uncertainty should have been preserved. Automation does not remove institutional design. It makes institutional design executable.

The opportunity, however, is real. Many institutions of scientific credibility were built when evaluation was extraordinarily expensive. Journals filtered because printing and attention were scarce. Reviewers sampled because nobody could read everything. Hiring committees relied on pedigree because independently auditing a candidate's entire intellectual contribution was impossible. Funding panels ranked proposals as though differences in scores were more precise than the underlying judgments because budgets required discrete choices. Artificial intelligence can lower some of these evaluation costs. It can trace claims through literature, compare manuscripts against large bodies of evidence, inspect code, test whether citations support the sentences that invoke them, look for contradictions, and maintain dynamic maps of which findings have or have not been independently replicated.

The goal should therefore not be to replace trust with computation. Trust will remain unavoidable. The goal is to reduce the amount of trust that must be borrowed when direct inspection becomes affordable.

That is the positive thesis of this book. Science does not need a revolution against expertise. It needs a better architecture for credibility: one in which reputation is informative but not sovereign, one in which institutions can be trusted without being treated as infallible, one in which disagreement is not automatically interpreted as incompetence, and one in which the cost of checking a claim falls faster than the cost of producing one.

Thinking of credibility as infrastructure changes the reform question. Infrastructure is judged by what it enables, how it fails, and whether failure can be localized and repaired. A prestigious journal, a grant panel, or a university brand is valuable because it allows decisions to be made in a world of limited attention. The danger begins when the infrastructure becomes invisible and its outputs are mistaken for properties of nature. Once the distinction is explicit, reform becomes less moralistic. We can ask whether a signal is calibrated, whether two signals are redundant, whether access to a signal is unnecessarily concentrated, and whether a cheaper direct test could replace it. These are engineering questions about institutional reliability, not accusations about the motives of the people working inside the system.

This perspective also prevents a common reform error: confusing moral clarity with operational clarity. It is easy to say that science should reward rigor, originality, openness, independence, and social value. It is much harder to build procedures that can recognize those properties without creating a new bureaucracy or a new prestige game. Institutional design has to specify who observes what, at what cost, with which appeal route, and with what consequences for people whose work does not fit the dominant template. The argument of this book therefore proceeds slowly from the existing shortcuts to the mechanisms that might replace some of their functions. The aim is not to purify science of judgment, but to make judgment more local to the evidence and more corrigible when it fails.

Thinking infrastructurally also changes what counts as a successful reform. A rule that improves fairness but doubles review time may fail because reviewers are already scarce. A transparency mandate that ignores privacy, commercial confidentiality, or participant consent may reduce trust rather than increase it. A new metric can remove one bias and immediately create another target for optimization. The relevant unit of analysis is therefore not an isolated policy but a coupled system of incentives, information flows, costs, and correction pathways.

Prologue - The Paper From Nowhere

The recurring scientist in the chapters that follow is Elena Varga. She is not a real person and no episode is presented as a disguised biography. Elena is a composite used to keep the institutional sequence visible while the empirical examples remain tied to published evidence. Think of her as one manuscript, one grant application, and one career moving through a system whose local decisions are usually reasonable but whose cumulative effects are harder to see.

Keep the manuscript in mind as it travels through the book. At first it is merely a text. Then an affiliation appears and expectations change. It enters peer review and encounters experts who are also members of a competitive field. It seeks funding and meets a threshold that can alter an entire career. It becomes published, cited, taught, perhaps translated into policy. At each stage, credibility is added, transferred, or withheld.

Now imagine a second version of the same journey in which the infrastructure can inspect more of the underlying evidence directly. The code runs automatically. The citations are checked. The key claims are linked to independent replications. Reviewers receive adversarial prompts that ask what would falsify the argument. Institution names are hidden during an early evaluation stage and revealed later only when execution capacity matters. The point of this thought experiment is not that machines have replaced judgment. It is that judgment has more evidence available before it borrows confidence from pedigree.

The same paper, two pedigrees

Imagine a manuscript arriving at a review system with every identifying detail removed. The title is ordinary. The methods are competent. The result is interesting but not sensational. The prose is clear enough. Three reviewers read it and produce three mixed assessments. One sees a valuable contribution. One sees an incremental paper. One doubts whether the result will generalize. Nothing about this situation is unusual.

Now change one thing. Reveal that the paper comes from a globally famous laboratory. The senior author has thirty thousand citations. The institution has produced Nobel laureates. The work is adjacent to a research program already described in keynote talks and high-profile editorials. The manuscript has not changed by a single word.

Would the expected judgment change?

The scientifically pure answer is that it should not. The human answer is that it might. Experiments in peer review have repeatedly investigated whether signals such as author identity and institutional prestige alter evaluation. The evidence is heterogeneous across disciplines and designs, but it is strong enough to reject the comforting assumption that reputation never leaks into judgment. A large experiment in computer science found that single-blind reviewers gave an advantage to papers associated with famous authors and prestigious institutions. Later work comparing single- and double-blind reviewing likewise found that hiding identity could reduce some prestige effects, although not necessarily transform final acceptance outcomes. Recent research on open peer review continues to report associations between institutional prestige and evaluation outcomes. The point is not that reviewers are irrational. It is that prior information is difficult to prevent from becoming part of the evidential frame.

Now reverse the experiment. The manuscript is revealed to come from an unknown laboratory at a small university in a peripheral research system. The authors have few citations. The senior author is not known to the reviewers. Perhaps the reader becomes more attentive to methodological weakness. Perhaps novelty claims are scrutinized more aggressively. Perhaps uncertainty that would have been interpreted charitably in the famous laboratory becomes evidence of immaturity here.

The experiment also has a practical implication for the reader. Throughout the book, whenever a prestigious signal appears, ask two separate questions. First, what legitimate information does this signal provide? Second, what part of the judgment would remain if the signal were removed? The gap between those answers is where borrowed credibility lives. Sometimes the gap will be justified. Sometimes it will reveal a shortcut that has grown too powerful. The objective is not to eliminate the first answer but to make the second answer easier to obtain.

That discipline also prevents a reverse prejudice. An unknown institution is not evidence of superior originality, and an outsider is not epistemically privileged by being excluded. Removing pedigree from one stage of evaluation should expose the work to stricter direct inspection, not to romantic sympathy. The purpose of masking is to reduce irrelevant variance so that evidence, method, and argument can carry more of the decision.

A real submission also makes the thought experiment harder than a philosophical puzzle. Affiliations can contain relevant information about access to equipment, ethics approvals, patient cohorts, or long-term datasets. That is why masking should be staged rather than absolute. The first pass can ask what the work says and whether the evidence coheres; a later pass can ask whether the claimed execution context is credible. Separating these questions preserves legitimate information while preventing it from leaking prematurely into judgments of novelty or correctness. The experiment is therefore not a demand for blindness. It is a demand for timing: reveal a signal when the decision actually requires it.

A particularly vivid field experiment was reported by Huber and colleagues in 2022. More than 3,300 researchers were invited to review the same finance paper, with the visible authorship varied between a Nobel laureate, a relatively unknown early-career researcher, and anonymity. The recommendation pattern changed sharply with the name shown: rejection was much more frequent when the little-known author was visible than when the Nobel laureate was visible. No single experiment settles the size of prestige bias across science, but this design demonstrates why status cannot simply be assumed irrelevant when identity is exposed.

The question behind the experiment

None of these reactions requires malice. A reviewer can reason that a respected laboratory has a history of careful work and is therefore less likely to have committed an elementary error. That inference may even be statistically defensible. The difficulty is that scientific institutions do not only evaluate claims. Their evaluations change the future distribution of resources. A favorable review produces publication; publication produces citations; citations improve funding prospects; funding attracts students; students produce more results; results reinforce institutional prestige; prestige changes the priors applied to the next manuscript. A small epistemic shortcut can become a structural feedback loop.

The paper from nowhere is therefore not simply an ethics puzzle about fairness. It is a systems problem. In a world of limited attention, some method must determine what gets examined first. In a world of limited funding, some method must determine which experiments are run. In a world with more researchers than permanent positions, some method must determine who receives time, laboratories, and careers. The question is not whether selection will happen. The question is what selection mechanisms do to the future evidence environment.

This book begins from one distinction. Credibility is not truth. Credibility is an estimate made under uncertainty about how seriously a claim, person, method, or institution should be taken before full verification. Because full verification is usually unavailable, credibility is indispensable. But precisely because it is indispensable, its allocation deserves the same scientific scrutiny we apply to other consequential systems.

The paper from nowhere has no pedigree. That makes it costly to evaluate. A mature scientific system should not pretend that this cost does not exist. It should make the cost smaller.

A second reason to keep the experiment in mind is that prestige rarely enters a decision under its own name. It appears as a judgment that the problem is important, that the team is likely to execute, that the work is more credible, or that scarce reviewer attention is better spent here than elsewhere. Some of those inferences can be rational. The discipline is to name which inference is being made. Once the purpose of the signal is explicit, one can ask whether the same information could be obtained more directly, whether the signal is being double-counted, and whether it should still matter after the underlying evidence becomes inspectable.

The same discipline applies to anti-establishment signals. A controversial author, a small institution, or a rejected idea can acquire a romantic credibility of its own. That is merely pedigree with the sign reversed. The book's standard remains symmetrical: neither prestige nor marginality should settle an inspectable claim. Both can supply context; neither can substitute for the work of examining evidence. This symmetry is important because institutional reform loses legitimacy if it simply replaces an incumbent hierarchy with a mythology of outsiders.

The same logic applies beyond journals. A grant panel can evaluate scientific merit separately from execution capacity. A hiring committee can distinguish the originality of a candidate's contributions from the prestige of the laboratories in which those contributions were made. A university can assess the reliability of a dataset without translating the reputation of the hosting institution into a verdict about every claim derived from it. These distinctions are administratively less convenient than one global score, but they are epistemically cleaner. They also turn a vague complaint about bias into a design problem with testable alternatives: mask some information, reveal it at a defined stage, record whether judgments changed, and compare whether the resulting procedure predicts later scientific performance better than the status-rich baseline.

A useful way to make this distinction operational is staged disclosure. Early evaluation can hide names, affiliations, previous awards, and other status cues while reviewers examine the claim, method, and internal coherence. Context can then be revealed when it becomes relevant to questions that genuinely depend on it: whether a laboratory has access to a required cohort, whether a team has demonstrated that it can operate a specialized facility, or whether a conflict of interest changes how a result should be interpreted. This is not an argument for universal double-blinding. In many areas anonymity is imperfect, and in some evaluations the identity of the team is part of the object being assessed. The point is to stop exposing information by default merely because a document format happens to contain it.

The deeper issue is not whether every use of pedigree is irrational. When information is sparse, reputation can be a statistically useful prior. The design question is where the prior is allowed to operate and how quickly it must yield to direct evidence. A journal may reasonably use track record to decide whether an expensive data audit is feasible; it has a weaker justification for letting the same track record influence whether a reported effect exists. Treating these as separate decisions is one of the recurring techniques in the chapters ahead.

The story starts before anything has gone wrong. A manuscript enters a world in which no one has enough time. The editor cannot reproduce its experiments. The reviewer cannot inspect every dependency on which it rests. The hiring committee cannot read a decade of work from every candidate. Under those conditions, reputation is not a moral failure. It is an information-compression device. This part follows that device from its legitimate origin to the first moment at which it begins to compound. The aim is to understand why pedigree is useful before asking when it becomes dangerous. If reform begins by denying the scarcity that created reputation, the same hierarchy will simply reappear under a different name.

Chapter 1 - Why Credibility Exists

Elena Varga's manuscript enters a system before anything has gone wrong. The editor is competent, the reviewers are busy, the journal has standards, and the paper contains enough uncertainty that serious evaluation would take days rather than minutes. Yet a decision about scarce attention must be made quickly. The first chapter therefore begins with a constraint rather than an accusation: scientific communities cannot directly verify everything they need to act upon.

From that constraint follows the need for credibility signals. Degrees, laboratories, journals, collaborators, prior grants, and citation histories are not irrational ornaments; they are compression devices. They summarize costly histories into forms that can guide attention. The central distinction is that a signal can be useful without being dispositive. Once that distinction is lost, the infrastructure of trust begins to manufacture the very evidence that future decisions will use.

On a Monday morning, an editor opens an inbox containing more competent manuscripts than the journal can possibly examine in depth. Somewhere else, a funding officer faces hundreds of proposals and a fixed deadline. A department searches for one colleague among two hundred plausible candidates. The ideal of direct evaluation collides with a mundane fact: evaluation is itself a scarce scientific resource. Before asking whether prestige distorts judgment, we need to understand why institutions began to use prestige at all. The answer is not vanity. It is the cost of knowing enough to decide.

Why trust is unavoidable

Trust works because it lets people begin before they have verified everything. A doctoral degree, a known laboratory, or a reputable venue reduces uncertainty enough to allocate attention. The mistake is to treat that first allocation as the final scientific judgment. This distinction between triage and verdict will recur throughout the book: pedigree can rationally determine where inspection starts, but evidence should increasingly determine where confidence ends.

The idealized picture of science begins with evidence. A claim is proposed, evidence is gathered, methods are described, other scientists inspect the work, and confidence changes according to the quality of the argument. The picture is not wrong. It is incomplete because it omits the most constraining variable in real scientific life: attention.

There is more knowledge than any scientist can read, more code than any reviewer can inspect, more data than any committee can audit, and more proposals than any funding agency can evaluate with full depth. The number of papers is not merely large; it makes exhaustive judgment physically impossible. Scientific communities therefore operate through selective inspection. The result is a layered trust architecture in which most claims are accepted provisionally because some other person or institution has already spent attention on them.

This architecture is not an embarrassment. It is one of the reasons large-scale science is possible. A researcher who had to personally verify the construction of every instrument, independently reproduce every cited experiment, and audit every statistical package would never reach the frontier of a field. Intellectual division of labour requires delegated credibility.

The modern university is part of that delegation. A doctoral degree tells strangers that a person survived a particular training process. A faculty position adds another filter. A successful grant indicates that peers considered a proposal credible enough to fund. Publication in a selective venue indicates that editors and reviewers found the work worth entering the formal literature. Citations indicate that other researchers considered it relevant enough to use, discuss, or contest. None of these signals proves truth, yet each can reduce the cost of deciding where to allocate scarce attention.

The first error in reform debates is therefore to imagine a world without proxies. Remove institutional prestige and people will use other signals: publication venues, collaborators, writing style, citation counts, social media visibility, conference invitations, recommendation letters, or informal network reputation. Remove all visible signals and decision-makers will infer hidden ones. Human systems under scarcity do not stop ranking. They change ranking technologies.

The right question is not whether a proxy exists. It is whether the proxy is calibrated to the decision for which it is being used.

A journal's reputation might be a moderately useful filter when a generalist reader chooses what to read on a train. It is a much weaker basis for deciding whether a particular researcher should receive tenure. A university's reputation may contain information about average selectivity. It is a poor substitute for evaluating a candidate's actual work. Citation counts may indicate intellectual influence. They can also reward controversy, fashion, methodological centrality, or membership in a large field. The same signal can be useful at one level and misleading at another.

This problem has become visible enough that major reform initiatives now state it explicitly. The San Francisco Declaration on Research Assessment argues against using journal-based metrics such as the Journal Impact Factor as surrogates for the quality of individual articles or researchers. The Leiden Manifesto similarly insists that quantitative evaluation should support qualitative expert assessment rather than displace it. CoARA, the Coalition for Advancing Research Assessment, has placed related principles into a broader institutional reform program that emphasizes qualitative judgment supported by responsible use of indicators. These are not anti-measurement positions. They are attempts to restore the distinction between signal and object.

The following table is not a ranking of signals. It separates the legitimate information contained in a proxy from the much stronger conclusion that institutions often allow the proxy to imply. Its purpose is to make the boundary visible: a signal can be useful for allocating attention while remaining insufficient for judging a specific claim.

The table below is not a ranking of signals. It is a boundary map. Each row asks what a familiar proxy can legitimately contribute to a decision and where the inference should stop. The purpose is to make a common category error visible: a signal that is useful for allocating attention becomes dangerous when it is silently promoted into evidence for the truth of a particular claim.

Table 1. What a proxy can tell us - and where it stops

A boundary map for using reputational signals without converting them into verdicts.

Each signal contains information, but none should silently expand into a verdict about truth or individual merit.

Signal	Legitimate use and boundary
Institutional pedigree	Useful prior about selection, training environment, and resources. It does not establish the quality or truth of a particular claim.
Journal prestige	Evidence that a manuscript passed a venue's filters and a practical aid for allocating reading attention. It is not article-level verification.

Signal	Legitimate use and boundary
Citation count	Evidence of influence, uptake, controversy, or field position. It does not distinguish truth from popularity without further analysis.
Prior funding	Evidence that a researcher previously passed a competitive process and may have execution capacity. It can also encode cumulative advantage.
Expert reputation	A rational reason to allocate scarce attention. It should lose weight as direct, reproducible evidence becomes available.

Read the second column as a contract with the proxy. Institutional pedigree can justify a prior expectation; journal prestige can justify attention; citations can justify the statement that work influenced a field. None should be allowed to answer a question it was not designed to answer. This discipline will reappear later when the book separates scientific merit, feasibility, replication, and institutional capacity.

Read this table from right to left as well. Whenever an institution uses the left-hand signal to answer a question that belongs in the right-hand boundary, credibility has changed category. The reform proposed in this book is largely an attempt to keep those categories separate for longer.

The scale of modern review makes the scarcity concrete. Peer review is largely uncompensated specialist labor, and estimates have placed its global annual value in the billion-dollar range. This matters because recommendations to 'just read the work carefully' can hide a resource assumption. The scientific system cannot afford exhaustive human verification of every manuscript, dataset, grant, and career. The realistic goal is to reserve expensive judgment for questions that require it while making cheaper forms of checking more automatic, reusable, and transparent.

A proxy is not a verdict

The scale of the problem is concrete. Aczel, Szaszi, and Holcombe estimated that journal reviewers spent more than 100 million hours on peer review in 2020, with a salary-based value above $1.5 billion for US-based reviewers alone. Their estimate was intentionally conservative. The figure helps explain why systems rely on compression. Deep evaluation is expensive enough that institutions inevitably ration it. The reform question is therefore economic as much as ethical: what forms of checking can be made cheaper, reusable, and less dependent on prestige signals?

A useful way to think about the problem is as a compression error. Institutions take multidimensional, uncertain evidence and compress it into manageable labels: accepted, rejected, funded, not funded, excellent, average, top quartile. Compression is necessary because decisions must be made. But information is lost, and the lost information is not random. A binary grant decision can hide the fact that two proposals had nearly identical scores. A journal brand can hide enormous variation in the quality of individual articles. A university ranking can hide strong groups inside otherwise ordinary institutions and weak groups inside elite ones.

The danger increases when compressed labels are fed back into future decisions. If a grant winner is subsequently treated as more credible because they won the grant, then yesterday's lossy compression becomes today's input data. If a publication in a prestigious journal increases the probability of future publication because editors recognize the author, a selection event becomes a reputation event. Over time, the system can confuse the historical record of being selected with the underlying property the selections were meant to detect.

This is why debates about pedigree are often morally misframed. The important question is not whether prestige is deserved. Some prestige is richly deserved. The important question is whether the informational content of prestige is being applied at the right granularity and with an appropriate decay rate.

A mature system would treat pedigree as a prior. It would allow the prior to influence where attention is allocated, because attention is scarce, but it would force the posterior to depend on claim-level evidence. It would also make the prior visible. If a committee gives extra weight to a track record, the weight should be explicit enough to inspect. Hidden reputation effects are more dangerous than acknowledged ones because they cannot be calibrated.

The economics of attention also clarifies why reform has been difficult. Deep review is expensive. Aczel, Szaszi, and Holcombe estimated that journal peer review consumed more than one hundred million reviewer hours globally in 2020, with enormous implicit labour value. Even if the exact estimate is debated, the order of magnitude matters. Science already spends a vast amount of expert attention on evaluation. Asking every institution simply to evaluate everything more carefully is not a reform program. It is an unfunded demand for a resource that is already scarce.

The promising question is therefore operational: how can we reduce the cost of knowing enough to make a good decision? Better data infrastructure can help. Standardized disclosure can help. More transparent review histories can help. Automated checking can help. Shared evaluation artifacts can help so that the same manuscript is not independently re-evaluated from zero by several journals. AI can help if it is used to increase the depth and coverage of inspection rather than to manufacture a new universal score.

The architecture of scientific credibility begins with scarcity. Any reform that ignores scarcity will recreate proxies under another name.

A useful way to discipline proxies is to ask what latent variable they plausibly measure. Journal prestige may partly measure prior selectivity and audience attention. Citation counts may measure use, visibility, controversy, or dependence. A grant award may measure that a proposal survived one competitive process at one time. None maps one-to-one onto truth, rigor, originality, or future importance. DORA, the Leiden Manifesto, the Hong Kong Principles, and CoARA differ in emphasis, but they converge on this basic point: indicators can inform qualitative judgment only when their scope and limitations remain visible.

> The manuscript has now passed the first scarcity filter, and the problem changes. What began as a practical signal about where to spend attention becomes transferable social capital. The affiliation affects the reviewer; the review affects the venue; the venue affects attention; attention affects citations; and those citations later return as evidence about the author. Each step can be locally reasonable while the chain as a whole becomes self-reinforcing. The next sections examine this transfer as a mechanism rather than a moral failing.

The most important property of credibility is that it travels. A university can lend some of its reputation to a newly hired researcher; a famous researcher can lend attention to a paper; a prestigious journal can lend seriousness to a claim; a funded project can lend legitimacy to the next proposal. This transfer is often useful because it lets strangers cooperate. The difficulty is that transferred credibility can outlive the evidence that originally justified it. What begins as a sensible prior can become a form of inherited epistemic capital.

How credibility moves

Borrowed credibility is easiest to see when the carrier changes but the underlying evidence does not. The same result can attract different levels of attention after an author moves institutions, after a famous collaborator joins the paper, or after a journal brand is attached to the manuscript. None of those changes are meaningless. They alter expectations about competence, resources, and prior filtering. They simply do not alter the experiment that was performed. Keeping those two facts separate is the central discipline of this chapter.

Scientific credibility rarely stays where it was earned. It moves.

A university earns prestige through generations of faculty, students, discoveries, wealth, and selective admissions. A newly hired researcher inherits some of that prestige immediately, before producing anything at the institution. A journal develops a reputation through editorial policy and accumulated publication history. A new article inherits part of that reputation the moment it appears under the journal's masthead. A scientist gains authority through a body of work. A new claim made by that scientist begins life with a different social probability of being taken seriously than the same claim made by a stranger.

This transfer is not unique to science. Brands work the same way. Credit markets work the same way. Legal systems rely on institutional standing. The unusual feature of science is that the object ultimately being pursued is not trust but truth. Trust is an instrument used because truth cannot be directly inspected at scale.

We can therefore distinguish several layers of credibility. Institutional credibility attaches to organizations. Personal credibility attaches to researchers. Procedural credibility attaches to methods such as randomized trials, blinded analysis, or registered reports. Venue credibility attaches to journals and conferences. Network credibility attaches to collaborators, mentors, and professional communities. Evidential credibility attaches to the actual support for a claim. The first five can alter our expectations. Only the last directly bears on whether the claim should survive.

In practice the layers are entangled. Consider a hypothetical early-career scientist with an excellent idea but little pedigree. To obtain resources, the scientist may need a senior collaborator. The collaborator adds intellectual value, but also transfers credibility. The proposal is now easier for a panel to interpret as feasible. If funded, the project produces a paper that is easier to place because the laboratory is recognized. If published, the paper gives the junior scientist a stronger track record. Eventually the scientist acquires independent credibility. This can be a healthy mentorship process.

The same mechanism becomes problematic when the transfer overwhelms evidence. A senior name included primarily for status can change the reception of work without changing its epistemic content. An elite affiliation can alter assumptions about methodological competence. A well-known theoretical framework can make an incremental extension appear more important because the field already knows how to narrate it. A new framework from outside the dominant network faces both the burden of proving itself and the burden of teaching reviewers how to value it.

The term borrowed credibility is useful because it avoids a moral judgment. Borrowing is not theft. It can be rational, temporary, and productive. The question is whether the borrowed asset is repaid by evidence.

A healthy system allows credibility transfer at the beginning of evaluation but progressively attenuates it as direct evidence becomes available. At the moment of triage, reputation may help decide which of ten thousand submissions receives immediate attention. Once the methods and data are in front of the evaluator, the influence of reputation should shrink. Once independent replication exists, it should shrink further. In principle, mature evidence should become increasingly independent of pedigree.

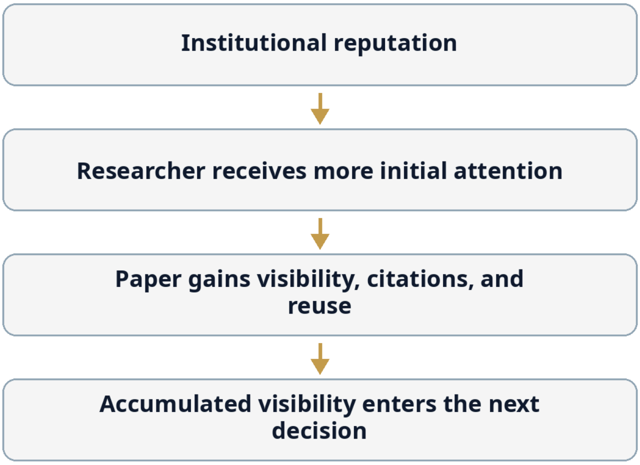
A credibility transfer becomes dangerous when each stage is allowed to certify the next stage without independent inspection.

The simplest way to see the mechanism is as a sequence rather than a tangled network.

Figure 1 compresses the credibility-transfer mechanism into one direction so that the feedback loop is easy to see without a tangled diagram. The final box is not the end of the process: accumulated visibility becomes an input to the next hiring, funding, publication, or collaboration decision.

Figure 1. The credibility-transfer chain

How a reasonable prior can become a cumulative advantage when later decisions reuse the previous signal.



A useful prior can become a self-reinforcing signal.

The important feature is not any single arrow. Each transition is individually plausible. The systemic effect appears because the output of one reasonable shortcut becomes the input to the next. Breaking the loop therefore does not require abolishing reputation; it requires inserting direct evidence at more stages.

Prestige can begin as a prior and then compound through attention and future selection.

The arrows describe a possible pathway, not an inevitable one. At every transition, direct evidence can interrupt the chain. That is precisely why better verification infrastructure matters: it gives later decisions something richer than the residue of earlier decisions.

Status transfer is observable because the scientific product can remain constant while the social wrapper changes. Sun, Barry Danfa, and Teplitskiy studied more than five thousand submissions around ICLR's switch from single- to double-blind review. Scores for the most prestigious authors fell under double-blinding, even though acceptance effects were smaller because many affected papers remained above the acceptance threshold. The result is important less as a universal estimate than as evidence that identity can function as a low-cost signal inside review, exactly where a content-only ideal would prefer it not to.

When attention begins to look like evidence

A useful test is to ask what should happen as direct evidence accumulates. If code is available, the importance of institutional reputation should fall. If an independent team reproduces the result, author fame should fall further. If multiple groups fail to reproduce it, prestige should not be able to hold confidence in place indefinitely. In a well-calibrated system, borrowed credibility is front-loaded: it helps decide what deserves attention, then yields to evidence as evidence becomes available.

Modern academic systems often do the opposite. Reputation can compound. A paper from a prestigious group is more visible, which can generate more citations. More citations strengthen the authors' metrics. Stronger metrics improve funding prospects. Funding supports larger teams and more output. The outputs receive more immediate attention because they come from a visible group. In this circuit, evidence and attention are not independent variables.

The distinction is important because attention can be mistaken for confirmation. A theory with thousands of citations may have been tested thousands of times, or it may simply have become the obligatory language of a field. A highly funded topic may be important, or it may have developed the infrastructure required to keep winning. A large community can generate real epistemic convergence, but it can also create path dependence by training the next generation in the questions, tools, and assumptions the community already recognizes.

Credibility transfer also operates through policy. Governments rely on advisory panels because legislatures cannot reproduce scientific literatures. Panels rely on disciplinary authorities. Authorities rely on journals and consensus documents. By the time a scientific conclusion enters regulation, several layers of trust have been stacked. This is unavoidable, but the stack should be inspectable. A robust system should make it possible to trace a policy claim back through the consensus statement to the studies, methods, data, and unresolved disagreements beneath it.

The same principle applies in reverse when a celebrated claim fails. Retractions and failed replications often update the credibility of a paper more quickly than they update the ecosystem that borrowed from it. Reviews keep citing it. Guidelines may lag. Secondary literature may preserve the claim. Educational material can continue presenting an outdated conclusion years after the evidential base weakened. Credibility is therefore sticky. It transfers quickly and decays slowly.

This asymmetry suggests a design requirement. Scientific systems need explicit mechanisms for credit propagation and credit withdrawal. Citations currently propagate attention but not necessarily evidential status. A future knowledge infrastructure should distinguish supportive citation, neutral citation, methodological dependence, contradiction, failure to replicate, conceptual inheritance, and historical mention. A paper should not remain epistemically young forever simply because the PDF does not change.

The deeper implication is that credibility should be treated as a dynamic relational property rather than a static score. A researcher can be highly credible in one domain and speculative in another. A method can be powerful under one data regime and misleading under another. A journal can publish both excellent and weak papers. A theory can be robust in its core and fragile at its frontier. Compressing these relationships into a single prestige number is convenient, but it is precisely the convenience that creates distortion.

A more reliable science would still have famous institutions and respected scientists. It would simply make fame less transferable than evidence.

Once a paper becomes visible, attention itself produces secondary traces that later evaluators can mistake for independent validation. Citations generate invitations, reviews, keynote mentions, collaborations, and inclusion in training datasets. Each trace can be reasonable in context, yet together they create a path-dependent record. The distinction to preserve is between evidence that a claim has influenced a community and evidence that independent tests have increased confidence in the claim. Those two histories often correlate, but they are not the same history.

Chapter 2 - Compounding Advantage and Institutional Capture

A year later, Elena's paper has become a line in a CV and a source of small reputational gains. She now applies for funding near a competitive threshold. The difference between being ranked just above or just below that threshold is administratively small but materially enormous: one outcome creates staff, data, time, and visibility; the other creates delay. Later evaluators will observe those consequences and may read them as evidence that the initial ranking was more precise than it really was.

This is where pedigree becomes dynamic. Advantage compounds, but it does not compound only because institutions favor famous names. It compounds because earlier decisions change the opportunity set from which later merit is produced. Institutional capture can then emerge without conspiracy when the people, methods, journals, panels, and infrastructures that define quality become increasingly drawn from the same successful pathways.

Imagine two proposals separated by one place in a ranking. The panel sees almost no difference between them, but the budget creates a hard boundary. One team receives staff, data, travel, and time. The other returns to teaching and tries again next year. Eight years later their CVs look radically different. It is tempting to read that difference as proof that the original panel saw something profound. Sometimes it did. Sometimes the system itself manufactured part of the difference. The Matthew effect is the name for that compounding.

The first small advantage

The Matthew effect is not merely a citation phenomenon. Bol, de Vaan, and van de Rijt examined applicants around a funding threshold in the Netherlands and found that winners just above the line accumulated more than twice as much subsequent funding over eight years as near-identical nonwinners just below it. Part of the effect came from participation: losing once made some researchers less likely to apply again. A tiny difference at one moment changed both resources and future willingness to compete.

In 1968, sociologist Robert K. Merton gave scientific inequality one of its most durable names: the Matthew effect. The phrase referred to the tendency for already eminent scientists to receive disproportionate recognition for contributions, while less established researchers received less. Later work generalized the idea into cumulative advantage: favorable positions generate resources that make future favorable positions more likely.

Cumulative advantage is often discussed as if it were a story about unfair recognition. That is too narrow. In science it is a production mechanism.

Suppose two early-career researchers are nearly indistinguishable in ability, training, and proposal quality. One receives a competitive grant by a small margin. The other misses it by the same small margin. The winner hires a doctoral student, collects new data, attends conferences, and gains time protected from other work. The grant itself becomes evidence of excellence in the next evaluation. The nonwinner spends time writing another application, perhaps takes on more teaching, delays the experiment, or leaves the research track. Five years later the two CVs no longer look nearly identical. The initial selection has helped create the difference that future selection will interpret as pre-existing merit.

This is not merely hypothetical. A major study of Dutch research funding examined applicants near a funding threshold and found that those just above the threshold accumulated more than twice as much funding over the following eight years as those just below, despite near-identical initial review scores. Part of the effect arose because early nonwinners became less likely to keep applying. The decision did not simply distribute money. It changed participation in the future tournament.

This is the point at which meritocratic language becomes dangerously ambiguous. A person with a stronger record may genuinely be better positioned to deliver a project. If earlier funding allowed that person to build a stronger laboratory, the difference is real. But the record now mixes at least two causal components: underlying capability and opportunity produced by previous selection. Evaluators usually observe the mixture, not the decomposition.

The Matthew machine operates through multiple channels. Money compounds because equipment and staff produce further results. Network position compounds because successful researchers are invited into collaborations that expose them to early ideas and future opportunities. Visibility compounds because famous work is read sooner. Administrative credibility compounds because institutions are more willing to support researchers who have already attracted external funding. Student quality compounds because ambitious trainees seek laboratories with strong reputations. Time compounds because researchers with secure positions can pursue riskier projects and recover from failure.

Again, none of this implies that successful researchers are undeserving. The mistake is to assume that the observed distribution of success is a direct measurement of the distribution of scientific ability. It is a path-dependent outcome produced by ability interacting with prior selection, institutional resources, timing, networks, field size, luck, and persistence.

CASE IN POINT The Dutch funding-threshold study is valuable because it resembles a natural experiment. Applicants with almost identical review scores were separated by a budget line. Those just above the line later accumulated dramatically more funding than those just below. The finding does not imply that the winners were undeserving. It shows that a decision can be both reasonable at the time and causally important later. Once that is visible, future evaluators should be careful not to read the later resource difference as pure evidence of underlying talent.

The clearest causal evidence for cumulative advantage in funding comes from threshold designs. Bol, de Vaan, and van de Rijt examined early-career applicants in a large Dutch program and compared researchers who scored almost identically on opposite sides of the funding threshold. Over the following eight years, winners accumulated more than twice as much additional funding as near-miss nonwinners. Part of the divergence arose because losing reduced later participation. The study does not imply that funding creates no real productivity; it shows that the institutional decision itself becomes part of the future career signal.

Merit, opportunity, and compounding

This finding complicates the language of merit. Later in the career, the winner may genuinely have a stronger laboratory, a larger team, and a more impressive record. Those are real capabilities. But some of the capability was created by the earlier allocation. Assessment therefore needs to distinguish demonstrated competence from accumulated opportunity. The purpose is not to erase track record. It is to avoid using the consequences of past selection as if they were independent confirmation of the wisdom of that selection.

Funding systems intensify this dynamic when they use past funding as evidence for future fundability. A panel may reasonably prefer a principal investigator who has demonstrated project management.

Yet if every grant program makes the same inference, early exclusion becomes harder to reverse. The system becomes increasingly confident about distinctions it partly manufactured.

One response is to ignore track record. That would discard useful information. A better response is to separate what track record is supposed to measure. Did the researcher complete prior work? Did they share data? Did predictions survive? Did the research create reusable infrastructure? Did the group publish negative findings? Did they correct errors transparently? Did they train researchers well? These are richer variables than cumulative grant income or journal prestige.

Another response is to introduce explicit mechanisms for uncertainty near decision boundaries. Research funding frequently turns continuous, noisy judgments into binary outcomes. Yet peer-review ratings have limited inter-rater reliability, and the difference between the final funded and unfunded proposal may be smaller than the uncertainty of the evaluation procedure. Treating a narrow rank order as precise can produce enormous downstream divergence from weak initial evidence.

This is one reason some funders have experimented with modified lotteries. The idea is not to fund random science. Proposals first pass a quality threshold; among those that are difficult to distinguish reliably, some fraction of allocation is randomized. Recent work has developed increasingly sophisticated taxonomies of such mechanisms, showing that different lottery designs trade epistemic selection against impartiality, concentration, and cost. In New Zealand, a funding program using a modified lottery also created the unusual opportunity to study funding effects with something close to randomized assignment. A 2024 analysis found no clear effect of winning on later publication or citation rates in that sample, a result that should not be overgeneralized but usefully challenges the assumption that panels can always identify projects whose funding will yield superior conventional outputs.

Randomization is uncomfortable because science is culturally attached to the idea that experts should be able to rank quality. But refusing to randomize does not remove randomness. It can hide randomness inside reviewer assignment, panel composition, mood, disciplinary fashion, and small score differences. An explicit lottery can sometimes be more honest than an implicit lottery disguised as precision.

The more general solution is portfolio design. A funding system should not ask only which individual proposal looks best. It should ask what distribution of projects produces a healthy research ecosystem. Some funding should reward demonstrated capability. Some should support new entrants. Some should be reserved for replication. Some should deliberately fund heterodox methods. Some should finance infrastructure. Some should tolerate long horizons and high failure rates. A portfolio can recognize cumulative advantage without allowing cumulative advantage to determine the entire future.

The Matthew machine cannot be eliminated because success changes capability. The objective is to prevent success from becoming a self-validating proof of deservingness.

This makes retrospective merit difficult to interpret. A strong publication record can reflect ability, effort, good judgment, access to resources, favorable collaborators, and previous selection decisions that expanded the feasible set of projects. The appropriate response is not to deny merit but to stop treating observed achievement as if it were produced under equal opportunity sets. Institutions can partially correct this by evaluating selected contributions in context, by distinguishing outputs from resources, and by avoiding criteria that mechanically convert past funding into evidence for future funding.

The paper has now entered institutions. Editors choose reviewers, reviewers interpret norms, grant panels allocate future capacity, and careers begin to depend on decisions made under uncertainty. No conspiracy is needed for these local decisions to become a structure. A field can close around its own methods because those methods are familiar, auditable, and represented on committees. A funding system can prefer reliable incumbents because failure is politically expensive. A journal can favor recognizable work because its readers reward recognizable work. This part examines how reasonable decisions can lock together into an unreasonable system, and why the right response is institutional design rather than moral accusation.

> Compounding advantage does not require a closed conspiracy to become institutional. Once successful approaches supply the reviewers, editors, curricula, datasets, benchmark conventions, and grant-panel expertise of a field, the field begins to reproduce its own definition of maturity. This is a form of endogenous capture: the institution can remain populated by conscientious people while still becoming less sensitive to alternatives that do not pass through its established pathways.

Institutional capture is easiest to recognize when an outsider buys influence. The more subtle form is endogenous. A community becomes so successful at defining legitimate questions, methods, and venues that alternatives must first speak its language to be considered serious. Nothing illegal needs to happen. The system can be populated by conscientious people and still become difficult to challenge. The useful question is therefore not who secretly controls a field, but what feedback loops make a field less able to notice its own blind spots.

Capture from the inside

Endogenous capture often begins as standardization. A field agrees on methods so results can be compared. Journals recruit reviewers who understand those methods. Courses train students in them.

Funders want proposals that can be evaluated reliably, so applicants frame questions in the established vocabulary. Over time, the standard becomes not just a tool for answering questions but a filter on which questions appear answerable. The boundary between quality control and intellectual closure is rarely announced; it emerges gradually.

Institutional capture is usually imagined as an external actor taking control of an organization: an industry influences a regulator, a political party fills a public body with loyalists, or a donor conditions funding on favorable conclusions. These forms matter, but scientific institutions can be captured in a quieter way. A system can become captured by the incentives, categories, and professional interests of the people who legitimately operate it.

Call this endogenous capture.

A field begins with a set of important questions. Researchers build methods to answer them. Journals emerge, conferences form, graduate programs teach the methods, and funders create panels staffed by the people who know the field. This is exactly what institutional maturity looks like. Yet maturity produces assets that must be maintained: laboratories, datasets, professional societies, career ladders, canonical courses, expensive instruments, journal ecosystems, and conceptual vocabularies. What began as a way to investigate a problem can gradually become a community whose survival depends on continuing to define the problem in recognizable terms.

There need not be a meeting where incumbents agree to exclude alternatives. Selection can happen through reasonable local decisions. Reviewers prefer proposals whose feasibility they can judge. Editors prefer manuscripts legible to their audiences. departments hire candidates who can teach existing curricula. Funding agencies seek panels with recognized expertise. Students choose supervisors with placement records. Over time, a paradigm acquires not only intellectual support but infrastructure.

This creates a subtle asymmetry. An established research program is evaluated primarily on whether the next increment is useful. A radically different program is evaluated on whether the entire conceptual reframing is justified. The new approach must often prove more because it fits fewer existing institutional categories.

Thomas Kuhn's account of normal science described how paradigms organize puzzle-solving. One need not accept every implication of Kuhn to see the institutional consequence: research communities require shared assumptions to coordinate. Without some stability, every paper would need to renegotiate first principles. The same coordination efficiency that makes cumulative knowledge possible also raises the entry cost for ideas that cross boundaries.

Capture becomes more consequential when the same network performs several roles. Senior researchers may edit journals, review grants, advise governments, sit on hiring committees, define conference themes, supervise students, and author field-defining reviews. These roles are individually legitimate. In aggregate they can create a closed credibility circuit in which the people best qualified to judge a field are also those most invested in its current structure.

A practical way to detect this kind of closure is to look for decisions that are hard to explain without reference to the field's own history. Are proposals rejected because the question is weak, or because no established panel category knows how to score it? Are methods criticized because they are technically inadequate, or because they violate an inherited convention whose original purpose has been forgotten? These questions should not be answered by outsiders alone. The most useful audits combine insiders, adjacent disciplines, and methodological specialists who can distinguish genuine standards from path dependence. A field that can explain why its boundaries exist is less vulnerable than one that merely reproduces them.

Endogenous capture is best treated as a property of correction channels rather than as an accusation about motives. A field is resilient when strong contrary evidence can find reviewers, venues, resources, and career paths capable of carrying it. It becomes fragile when every path to testing the alternative is controlled by the same assumptions the alternative challenges. This definition makes capture empirically discussable: reviewer concentration, topic concentration, dependency on a small set of datasets or infrastructures, and the treatment of cross-paradigm evidence can all be observed without diagnosing anyone's psychology.

When a field becomes its own gatekeeper

This is why term limits or outsider seats, by themselves, are insufficient. A new committee can reproduce the same ontology as the old one. More robust defenses include methodological diversity, explicit recording of reviewer disagreement, rotating evaluation pools, external challenge processes, and funding streams whose purpose is to explore questions that established panels find difficult to rank. Capture is best resisted by creating alternative paths through the institution, not by searching for perfectly neutral individuals.

The solution cannot be to replace experts with nonexperts. Scientific evaluation requires domain knowledge. Nor should rotation be treated as a universal cure. A panel of constantly changing evaluators can lose memory, become easier to influence administratively, and rely even more heavily on surface proxies. The institutional design problem is to preserve deep expertise while preventing any one epistemic network from becoming the sole gate through which a question must pass.

Several mechanisms can help. Cross-panel review can expose proposals to adjacent disciplines. Minority reports can preserve unresolved disagreement instead of compressing panels into artificial consensus. Reviewer conflict-of-interest rules can extend beyond financial ties to direct competition and close collaboration. Funding agencies can audit whether repeated panel composition predicts repeated institutional concentration. Journals can distinguish methodological review from significance review. Appeals can focus on procedural errors rather than inviting endless relitigation of scientific judgment.

The most important reform may be observability. Institutions often lack the data to know whether they are captured. They can report acceptance rates and demographic statistics yet remain unable to answer more structural questions. How concentrated are reviewer networks? How often do the same institutions review one another? Do new entrants receive systematically different methodological demands? Which topics persist because they produce strong evidence, and which persist because they control large infrastructures? How many proposals fall just below funding thresholds year after year? Which conceptual communities rarely cite or review one another despite studying the same underlying problems?

These questions are measurable. Modern network analysis can reveal structural closure. Natural-language systems can map conceptual overlap that disciplinary labels conceal. AI can compare review language across institutions and detect whether similar weaknesses are treated differently depending on pedigree. None of these analyses proves capture, but they can identify where human audit is warranted.

The language of capture should therefore be used carefully. If every disagreement with a field is called capture, the term becomes a rhetorical weapon. The standard should be institutional: does a system possess sufficiently independent routes for new evidence to alter resource allocation and accepted belief? A field can be strongly consensual without being captured if contrary evidence is genuinely able to change the consensus. A field can be intellectually diverse and still be captured if all viable career paths depend on the same small set of gatekeepers.

Capture is not defined by agreement. It is defined by the cost of correction.

A field can also test itself by creating controlled routes for ideas that do not fit its dominant taxonomy. One route is a cross-panel mechanism in which a proposal rejected as poorly framed by one community can be examined by an adjacent community before it disappears. Another is retrospective analysis of rejected applications: not to prove that panels were wrong, but to ask whether certain methodological families, institutions, or question types repeatedly vanish at the same stage. The useful unit of analysis is the pattern, not the aggrieved individual case. Capture becomes an institutional hypothesis that can be tested against decision records, rather than a rhetorical accusation that can neither be confirmed nor falsified.

The same institutional density that creates closure also creates enormous benefits. Shared methods make results comparable; specialist journals create cumulative conversations; canonical training lets new researchers enter quickly; large infrastructures enable questions no isolated laboratory could answer. Reform should therefore target monopoly over evaluation, not specialization itself. Adjacent-panel review, external replication, minority reports, rotating service roles, and explicit monitoring of network concentration preserve the advantages of expertise while creating additional routes for correction.

Chapter 3 - Gatekeepers, Publication, and the Career Tournament

Elena's next reviewer is also a neighboring competitor. This is ordinary rather than scandalous: the people capable of detecting subtle errors are often the people with the strongest intellectual and professional stakes in the outcome. Scientific institutions therefore face a permanent design problem. They need expert judgment precisely where expert judgment is least socially neutral.

Publication does not end that problem. Acceptance is a useful checkpoint, not a replication event, and the incentives facing the author change immediately after the paper appears. A risky verification study may be valuable to the field but weak for a short-term career; a polished extension may do the reverse. Gatekeeping, verification, and career incentives are therefore one connected mechanism rather than three independent defects.

Scientific evaluation has a structural paradox. The people most qualified to judge a frontier are often the people most affected by how that frontier is defined. A reviewer may be a competitor. A grant panelist may lead a neighboring consortium. An editor may have built a career on one methodological tradition. Expertise and interest are not separable by wishing them apart. The practical task is to use expertise while designing procedures that make conflicts visible, bounded, and less decisive.

Expertise and conflict

The empirical literature on prestige bias is instructive because it refuses a simple story. Tomkins, Zhang, and Heavlin found that single-blind review in a computer-science conference advantaged papers by famous authors and prestigious institutions. Sun, Barry Danfa, and Teplitskiy later found that switching another major conference to double-blind review reduced scores for the most prestigious authors, although acceptance effects were more complicated. Nielsen and colleagues, in contrast, found only weak country- and institution-status effects in a preregistered experiment across six disciplines. The honest conclusion is not that prestige bias is everywhere or nowhere. It is that review design changes what information can influence judgment, and effects vary by context.

Peer review contains a structural conflict that would look strange in many other domains. The people most capable of evaluating a research proposal are often the people closest to competing with it.

This is not an accident. Expertise is local. A researcher who understands the technical risks of a new method is likely to work on nearby problems. A grant panel that excludes everyone with overlapping interests may be left with evaluators too distant to judge feasibility. A journal that avoids reviewers from the same subfield may obtain formally independent but substantively shallow reports.

The phrase conflict of interest is therefore too binary for scientific competition. Direct financial conflicts can often be prohibited. Intellectual competition cannot. Indeed, intellectual competition is part of how science advances.

The institutional question is how to exploit the information possessed by competitors without allowing competition to become invisible governance.

A reviewer may know that a proposed experiment has already failed in several laboratories, information not yet published. That knowledge can save public money. The same reviewer may know that a competing result would weaken their own research program. Both facts can be true at once. Human beings are poor at introspecting exactly how such incentives influence judgment. Requiring declarations helps, but declarations cannot remove subconscious weighting.

Anonymity is one partial response. Double-blind review attempts to reduce identity-based bias by hiding authors from reviewers. Experimental evidence in computer science has found meaningful prestige effects under single-blind review and some reductions when identity is hidden. Other fields show mixed results, and perfect blinding is often impossible because topics, datasets, citations, and writing styles reveal authorship. Still, the principle matters: if identity is not necessary for a particular evaluative task, the burden should be on the system to explain why it is exposed.

A more radical approach is task decomposition. Instead of asking one reviewer for an overall recommendation, the review can be separated into questions with different conflict profiles. One evaluator checks statistical reasoning. Another examines methodological feasibility. Another evaluates novelty through literature comparison. Another tests whether citations support claims. Another assesses whether the research question matters to a particular community. Overall judgment is then assembled from explicit components rather than emerging from one opaque impression.

The conflict becomes especially visible in small fields. If only twenty people can competently evaluate a method, excluding everyone with a collaboration, citation, institutional, or competitive relationship may leave no reviewers at all. The objective therefore cannot be perfect social distance. What matters is procedural visibility. The institution should know which relationships exist, whether a reviewer is evaluating a direct substitute for their own work, and whether several reviewers share the same dependency. A single disclosed conflict can be manageable; a panel whose conflicts all point in the same direction is a structural signal.

Conflict is especially hard to manage in science because intellectual competition is not an exceptional condition. Excluding every close competitor would often exclude the best-informed reviewers. The realistic objective is layered conflict management: disclosure, recusal for direct stakes, diversification of reviewers, decomposition of review tasks, and retrospective audits of patterns. The stronger the discretionary power of a reviewer, the stronger the case for independent evidence about how that discretion is used.

Designing around competitive review

Conflict-of-interest rules should therefore be treated as architecture, not etiquette. Disclosure is necessary but weak if every viable reviewer belongs to the same dense network. A better system can combine recusal thresholds, reviewer diversity, explicit competing-interest statements, independent methodological reviewers, and post hoc audits of unusually closed review networks. The point is not to eliminate competition among experts. Competition is often evidence of an active field. The point is to prevent competition from becoming invisible governance.

This reduces the power of pedigree because some review tasks become claim-local. A citation either supports a statement or it does not. Code either reproduces a reported result under specified conditions or it does not. A sample-size calculation can be inspected independently of whether the principal investigator is famous. Not every aspect of science can be modularized this way, but more can than current practice assumes.

AI increases the feasibility of such decomposition. A model can scan references for mis-citation, compare versions of methods, search for prior art, identify missing baselines, and generate adversarial questions. Yet the output should not be a synthetic accept score. The useful product is a structured dossier that exposes different dimensions of the work to human judgment.

This suggests a general rule for institutions: where gatekeepers are also competitors, reduce the amount of discretionary judgment that must be trusted without evidence. This does not mean rigid bureaucracy. It means turning invisible criteria into inspectable ones.

Consider funding panels. A common process asks reviewers to score excellence, impact, and feasibility, then merges scores into a rank. The apparent precision of the final ranking hides the fact that evaluators may mean different things by the same number. One reviewer uses a 5 only for near-perfect work. Another uses 5 for anything fundable. One penalizes risk. Another rewards it. One views the applicant's prior success as proof of feasibility. Another views it as evidence that scarce money should go elsewhere. The final decimal average can look more objective than the inputs warrant.

A better process could record not only scores but disagreement structure. If evaluators agree that a project is methodologically strong but disagree radically about significance, that is different from a project with mediocre agreement on every dimension. Funding portfolios could deliberately reserve room for projects with polarized expert assessments, because high disagreement can sometimes signal genuine uncertainty at the frontier rather than poor quality.

The same idea applies to hiring. Instead of a committee silently integrating pedigree, letters, metrics, talks, and intuition, institutions can specify which claims about a candidate are being made and what evidence supports each claim. Is the candidate unusually original? Then identify the work that demonstrates originality. Are they independent? Trace intellectual contributions rather than inferring independence from corresponding-author position. Are they reliable collaborators? Use structured evidence rather than prestigious coauthors as a proxy.

Science will never eliminate gatekeeping because resources are finite. The objective is to prevent gatekeeping power from becoming epistemically invisible.

Competitive review also benefits from separating judgments that reviewers routinely collapse. A domain expert may be well placed to judge whether a question matters, a statistician to judge whether an inference is supported, and a program manager to judge whether the team can deliver the promised infrastructure. Asking one reviewer to provide a single global score mixes these roles and makes conflicts harder to interpret. Decomposed review will not remove subjectivity, but it reveals where subjectivity enters. It also makes disagreement informative: two experts can agree that a method is sound while disagreeing about significance, and the committee can preserve that distinction instead of averaging it into an apparently precise number.

One practical improvement is to stop forcing heterogeneous judgments into a single apparently precise score. The same proposal can be methodologically excellent, strategically risky, unusually novel, and difficult to execute. Reviewers may agree on the facts and disagree on values. Preserving the vector of judgments, and sometimes the disagreement itself, gives a funding committee more information than an average decimal. It also makes space for portfolio decisions in which a funder knowingly buys different kinds of risk instead of pretending that every project lies on one universal ladder of excellence.

> The gatekeeping problem becomes sharper after acceptance because publication carries a semantic ambiguity. It means that a manuscript has survived a particular review process, but ordinary language often upgrades that fact into the stronger claim that the result has been independently verified. The distinction matters because peer review and replication

purchase different kinds of confidence and should not be treated as interchangeable institutional services.

Publication has acquired a social meaning stronger than its literal meaning. To many readers, 'peer reviewed' sounds like a quality seal attached to the truth of a claim. In practice, review is a constrained assessment of a manuscript under time pressure. Reviewers rarely rerun the entire pipeline, reproduce the laboratory environment, or verify every citation and transformation. The distinction matters because reform becomes possible only after we stop demanding that peer review perform a job it was never designed to perform.

What peer review actually certifies

The reproducibility debate gives a concrete sense of the gap between publication and verification. In the 2015 Reproducibility Project, 100 psychology studies were replicated and the average replication effect was about half the original effect size; by one commonly discussed criterion, 36 percent of replications were statistically significant compared with 97 percent of the originals. The result was not a declaration that psychology was false. It was evidence that publication filtered for a different property than robust repetition.

A published paper is an event. It is not a verdict.

This distinction is obvious in principle and routinely blurred in practice. Publication means that a manuscript passed a particular editorial process at a particular time under a particular set of constraints. It does not mean that the result was independently reproduced, that every analytical choice was optimal, that every citation was accurate, or that the central claim will survive contact with future evidence. Yet in public discourse, policy documents, and sometimes scientific argument itself, the phrase “peer-reviewed research shows” is often used as though peer review were a certification procedure for truth.

Peer review was never designed to bear that burden. Reviewers usually do not rerun experiments. They rarely inspect raw laboratory records. They may not have access to the full data or code. They work under time pressure, and many reviews are unpaid contributions added to already crowded professional schedules. Their task is closer to informed plausibility screening than forensic verification.

This matters because the incentives around publication are asymmetric. Positive, clean, surprising results have historically been easier to publish than ambiguous or null findings in many fields. Publication bias can therefore change the apparent state of evidence even when every published paper is individually honest. If ten laboratories test an effect and only the two positive studies reach the literature, a later reader encounters a distorted world.

Registered Reports were designed to attack this problem at the mechanism level. In the model, journals evaluate the question and proposed method before the outcome is known. A study that passes Stage 1 can receive in-principle acceptance, after which publication is not contingent on obtaining an exciting result, provided the approved protocol is followed and the work meets quality checks. The design changes the payoff structure: methodological seriousness is rewarded before statistical significance is available as a social signal.

Evidence accumulated over the first decade of Registered Reports is encouraging without justifying triumphalism. Comparative studies in psychology and neuroscience found that Registered Reports were rated higher on methodological and analytical rigor while not clearly sacrificing novelty or creativity. Reviews of the format emphasize that it is not a universal solution. It works best for questions that can be specified before data collection and is less natural for some exploratory, qualitative, theoretical, or rapidly adaptive research. The broader lesson is more important than the format itself: if an institution dislikes an outcome produced by incentives, it should change when the reward is assigned.

This lesson also reframes the reproducibility debate. Calls for researchers to be more careful are valuable but incomplete. A system can make careful behavior expensive. Thorough documentation takes time. Releasing data requires curation. Maintaining software requires labor. Replication is often less prestigious than novelty. Publishing a null result may be harder than publishing a positive one. If career progression depends on a rapid stream of visible outputs, exhorting scientists to perform costly invisible work will have limited effect.

CASE IN POINT The STAP-cell episode offers an unusually clear lesson about the limits of prestige. The work appeared in Nature and involved respected institutions, yet problems with images, methods, and replication emerged rapidly after publication, followed by retraction. Nature's own editorial response emphasized that some fatal faults were not detectable from ordinary manuscript review. The lesson is not that elite journals are unreliable. It is that even elite journals cannot convert a document-review process into experimental verification.

The 2015 Reproducibility Project in psychology made the distinction between publication and later verification unusually visible. One hundred published studies were selected for replication. The replications produced effects, on average, substantially smaller than the originals, and far fewer replications reached conventional statistical significance. The project was not evidence that peer review had failed in every case; it was evidence that a published paper is one observation in an evolving evidential process. Verification happens after publication as well as before it, and institutional interfaces should make that temporal structure explicit.

Replication as a missing market

Cancer biology revealed a second lesson. The Reproducibility Project: Cancer Biology found not only weaker or inconsistent replication results but practical difficulty in reproducing experiments because essential details were missing or difficult to reconstruct. Even the famous 2014 STAP-cell episode illustrates the limit of editorial inspection: Nature's own retrospective noted that referees and editors could not have detected the fatal faults from the submitted material alone. Verification is a separate activity, with its own costs, infrastructure, and incentives. If it is socially important, it needs an explicit market rather than an assumption that peer review has already performed it.

The reproducibility movement has therefore increasingly emphasized infrastructure and incentives. The 2017 manifesto for reproducible science described threats spanning methods, reporting, dissemination, evaluation, and incentives rather than reducing the problem to individual misconduct. This is a healthier model. Most unreliability does not require fraud. It can emerge from researcher degrees of freedom, low statistical power, selective reporting, publication bias, opaque preprocessing, fragile software, undocumented decisions, and ordinary confirmation pressure.

It is useful to distinguish publication, validation, replication, and synthesis. Publication makes a claim visible. Validation checks whether the reported analysis and method are internally credible. Replication tests whether a finding survives a new attempt under sufficiently comparable conditions. Synthesis evaluates how the claim fits the wider evidence base. These stages can occur in different orders and are not equally appropriate for every field, but collapsing them into the single label “peer reviewed” hides too much.

A future scientific infrastructure should allow these states to remain visible. A paper might be published but not computationally reproduced. Another might have its analysis reproduced from shared code but lack independent replication. A third might have multiple successful replications but face a strong theoretical contradiction. A fourth might be superseded by a better dataset. None of these statuses is captured well by citation count or journal name.

The implications for public communication are significant. Scientific institutions often face pressure to present clear conclusions. Policymakers need action, journalists need concise language, and citizens cannot be expected to reconstruct every uncertainty. But clarity does not require pretending that all published evidence has the same maturity. A system can communicate confidence tiers, replication status, unresolved methodological disputes, and the difference between observational association and causal evidence without abandoning expert guidance.

AI may help because many forms of verification are repetitive and expensive. Systems can attempt to execute code, compare reported sample sizes across sections, detect inconsistent statistics, inspect whether a reference supports the proposition for which it is cited, identify duplicated or anomalous images, search for related replications, and maintain a live evidence graph. These tools will make errors and must themselves be validated, but their economics differ from human-only review. A machine can perform the first pass on thousands of papers and direct expert attention toward anomalies.

The proper metaphor is not an artificial judge issuing a sentence. It is instrumentation. A microscope does not decide whether a biological theory is true; it reveals structure that changes what humans can evaluate. AI-assisted verification should aim for the same role.

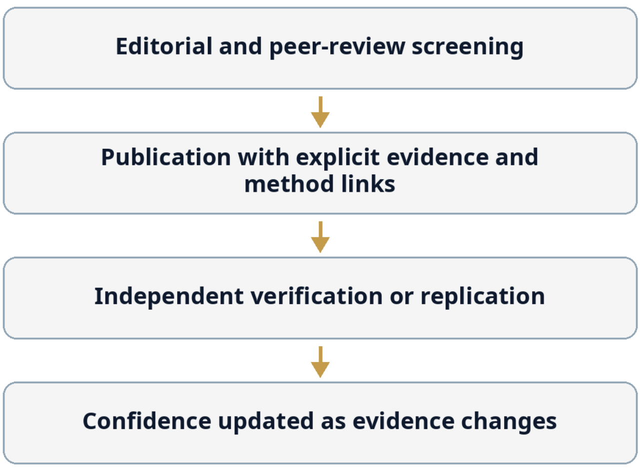
If publication becomes the beginning of a claim's evidential life rather than its ceremonial endpoint, scientific self-correction can become faster and less dependent on prestige.

The correction is architectural: stop asking one process to perform every epistemic function. Publication, verification, replication, and updating can be separate layers that communicate with one another.

Figure 2 replaces the popular mental model of publication as a seal with a sequence of confidence-building layers. Editorial and peer review remain important, but they are followed by evidence linking, independent checking, and later updating.

Figure 2. Publication is one layer, not the final seal

A publication is a checkpoint inside a longer verification process.



Publication is a checkpoint, not a terminal truth state.

This representation makes post-publication correction normal rather than exceptional. A paper can remain historically valid as a record of what was reported while the current confidence state of its claims changes. The institutional object that should be maintained is therefore not only the document but the evolving evidential relationship around it.

A paper can be useful and publishable before every important claim has been independently verified.

This design makes uncertainty visible instead of treating it as embarrassment. A newly published claim can be labeled as promising but not yet independently replicated; an old claim can gain confidence through repeated success; a failed replication can narrow a claim without invalidating every contribution in the paper.

The economics of replication are beginning to change at the policy level. In 2026 the US National Institutes of Health announced an agency-wide initiative to strengthen replication and reproducibility, explicitly describing replication as foundational rather than peripheral. This does not solve the incentive problem, but it signals an important reframing: verification is infrastructure that must be funded, rewarded, and organized. A research system cannot demand reliable knowledge while treating the production of independent checks as volunteer housekeeping.

> Verification is also shaped by the career economy of the people who must perform it. Researchers respond to incentives without needing to believe that metrics are intellectually superior to judgment. When hiring, promotion, and funding rely on visible outputs, rational researchers learn which projects create those outputs reliably. The resulting behavior is not evidence of moral decline; it is evidence that institutional objectives have been translated into measurable proxies with predictable strategic consequences.

A researcher can love truth and still optimize for survival. Salaries, visas, laboratories, doctoral positions, promotion, and the ability to continue asking questions are tied to outputs that institutions can count. That makes academic incentives unusually powerful: they operate not only on income but on access to the activity that gives a scientific career its meaning. The result is not that researchers become cynical. It is that perfectly decent people learn the grammar of what the system rewards.

The researcher as an optimizing agent

The mechanism can be understood without accusing anyone of gaming. If hiring committees request publication counts, candidates will maximize publishable units. If promotion rewards prestigious venues, researchers will spend time optimizing venue fit. If funders ask for predictable milestones, proposals will describe uncertainty as a sequence of controllable tasks. Each response is individually rational. The distortion appears when the measurable representation of science becomes more legible to the institution than the underlying epistemic contribution.

Science is a method for producing knowledge and a labor market for allocating careers. The two systems overlap but do not have identical objectives.

A good scientific institution wants reliable knowledge, productive disagreement, long-term capability, methodological care, and occasional radical breakthroughs. A career tournament must make repeated decisions about hiring, promotion, salary, laboratory space, grants, and status. Those decisions require comparable signals. The result is a persistent tension: activities that improve the epistemic commons are not always the activities that maximize an individual's probability of winning the next evaluation.

Consider replication. A careful replication of an influential result can be extremely valuable to a field. If the result survives, confidence increases. If it fails, future work can redirect. Yet the replicating researcher may receive less prestige than the researcher who introduced the original claim. The social value of verification can exceed its private career return.

The same divergence appears in software maintenance, dataset curation, peer review, mentorship, documentation, negative results, and error correction. These are infrastructure activities. They make everyone else's research better, but traditional assessment systems often struggle to assign individual credit to them. By contrast, papers, citations, grants, and prestigious authorship positions are legible. Careers therefore drift toward outputs that committees can count.

Metrics did not create this problem. They were introduced partly to make evaluation less dependent on private reputation and opaque patronage. Quantitative indicators can reveal productivity that old-boy networks overlook. Citation data can expose influence across institutional boundaries. Bibliometrics can be useful at population scale. The trouble begins when measures designed to inform judgment become targets that organize behavior.

DORA and the Leiden Manifesto emerged from precisely this concern. Their criticism is not that numbers are immoral but that context disappears when indicators migrate between levels of analysis. A journal-level citation average is applied to a paper. A paper's citations are applied to a scientist. A scientist's h-index is applied to a hiring decision in which contributions, field size, career stage, and collaboration structure differ. A measure that contains some information becomes a convenient universal currency.

Universal currencies are powerful because they allow exchange. A dean may not understand a specialized paper but can understand that it appeared in a famous journal. A panelist outside the subfield may not know whether a contribution changed practice but can understand that it has five hundred citations. Once such currencies become institutionally accepted, researchers rationally optimize for them.

This creates what might be called career-facing science: research shaped partly by how it will appear to future evaluators. The effect need not be cynical. Researchers internalize the standards of their field. They learn what counts as a publishable unit, what topics are fundable, what methods signal rigor, how much uncertainty can be admitted without weakening a paper, and which venues matter. Professionalization is partly the acquisition of these tacit rules.

Career incentives also operate through time horizons. A project that needs five quiet years competes badly against one that can generate three papers and a demonstrator in eighteen months. Early-career researchers feel this pressure most sharply because temporary contracts make delay personally expensive. The rational response is to decompose ambitious questions into fundable and publishable units. Sometimes that decomposition improves science. Sometimes it removes the very integration that made the question important. Any assessment reform that ignores employment precarity will therefore address symptoms while leaving a powerful cause untouched.

Research assessment reforms increasingly recognize that integrity-related behaviors consume time and can conflict with output-maximizing incentives. The Hong Kong Principles explicitly argue for rewarding responsible methods, complete reporting, open practices, replication, synthesis, mentoring, and other contributions that strengthen trustworthy research. The point is not to construct a longer checklist. It is to change what a career can rationally optimize for, so that making work easier to verify is not professionally dominated by producing another countable paper.

When the metric becomes the job

Reform movements such as DORA, the Leiden Manifesto, and CoARA try to widen the object of assessment beyond journal-level prestige and crude counts. The European Commission's 2026 baseline assessment of research-assessment reform reports broad but uneven movement from commitment toward implementation. That unevenness is important. Replacing one metric with a narrative CV does not automatically remove status effects; it can merely move them into prose. Better assessment requires clearer criteria, selected contributions, structured evidence, and auditing of whether the new process actually changes outcomes.

The danger is not strategic behavior itself. Every profession has strategy. The danger is when the strategy for succeeding as a scientist becomes weakly coupled to the behavior society actually needs from science.

A healthier assessment system would therefore evaluate a portfolio of contribution types. But this must be done carefully. Simply adding more categories can create more bureaucracy. Researchers may then optimize a dozen indicators rather than five. Narrative CVs can become exercises in polished self-description. Qualitative review can reintroduce pedigree through recommendation networks. Reform must avoid replacing metric gaming with narrative gaming.

One practical approach is evidence-backed narrative assessment. Instead of asking applicants to list every output, institutions can ask them to make a small number of explicit claims about contribution and attach verifiable evidence. A candidate might claim that a software tool became infrastructure for a community, supported by usage data, external dependencies, maintenance history, and independent testimonials. Another might claim unusually careful error correction, supported by documented revisions and open issue histories. A third might claim successful mentorship, supported by trainee outcomes interpreted with appropriate context.

This resembles a scientific argument more than a scorecard. It also makes AI assistance useful. Systems can check claims against public records, normalize across field-specific practices, detect double counting, and identify unsupported assertions. The evaluator remains responsible for value judgment, but the cost of verifying the evidence is reduced.

Economic reform must accompany assessment reform. Short contracts and hypercompetitive grant dependence make long-horizon epistemic behavior difficult. Researchers who must continually fund their own salaries cannot be expected to treat every uncertain, negative, or slow project as equally viable. Institutions that want risky science need financing structures that can absorb failure. Permanent or long-term base funding, protected exploratory budgets, replication allocations, and infrastructure careers are not luxuries; they are mechanism design choices.

The same is true of publishing. If journals depend economically on article volume, prestige competition, or attention, their incentives will shape editorial behavior. If universities depend heavily on rankings that reward particular outputs, internal reform may be punished externally. Scientific institutions are nested inside measurement markets.

This means local virtue has limits. A university cannot fully reform hiring if funders, rankings, and national evaluation systems continue rewarding the old currency. A journal cannot eliminate sensationalism if authors route their most visible work elsewhere. A funder cannot diversify portfolios if political oversight interprets failure as waste. Reform therefore requires coordinated changes at several layers.

The career tournament will never disappear. Scientists need jobs, status matters, and scarce opportunities require selection. The goal is to make winning the tournament more strongly correlated with building reliable knowledge, and to reduce the number of socially valuable scientific behaviors that require personal career sacrifice.

Metrics are most dangerous when they become a common currency across contexts. A citation count may be informative within a field and almost meaningless across fields; the journal impact factor describes a distribution at journal level and is a poor surrogate for the quality of one article. The Leiden Manifesto and DORA both emerged from this mismatch between what indicators can measure and what institutions ask them to decide. Responsible assessment is therefore less about finding a better master metric than about refusing the idea that one master metric should exist.

Published science leaves records. Missing science does not. That asymmetry makes the system easier to criticize for what it produced than for what it discouraged. Yet the larger loss may live in rejected proposals, abandoned replications, unfashionable questions, peripheral institutions, and results that were perfectly informative but insufficiently exciting. Here the book turns toward absence. The problem is not to romanticize outsiders or dissent. Most rejected ideas are rejected for good reasons. The problem is that a mature system should be able to ask what kinds of error its selection machinery systematically makes, including errors that leave no publication behind.

Chapter 4 - The Knowledge We Do Not Produce

The most consequential failures of a research system are not always visible in its literature. Some appear as empty shelves: replications that were never funded, negative results that were never written, questions that did not fit a panel, and researchers whose work never acquired enough visibility to become part of the searchable conversation. Elena encounters this when a project that would test a fashionable result is judged less attractive than a project that extends it.

The same mechanism operates at larger scales through consensus and geography. Consensus is indispensable when no individual can inspect all evidence, but it can become sticky when the pathways into the conversation are uneven. Scientific location, language, institutional networks, and accumulated visibility affect which claims are encountered and therefore which claims later appear to be central. The missing science is thus partly a problem of incentives and partly a problem of what the system makes legible.

Libraries display what science produced. They do not display the replications that were never funded, the null results left in a drawer, the proposals made safer after reviewer advice, or the researcher who left after three narrowly missed calls. This missing archive matters because selection changes the future distribution of questions. A scientific system can be accurate about the work it admits and still be systematically narrow about the work it allows to exist.

The empty shelves

Negative results illustrate the economics of invisibility. A study that carefully shows an effect is absent can save other laboratories years of effort, yet it is often harder to publish, cite, or use for career progression than a novel positive result. Replications suffer a similar problem: their social value can be high while their novelty value is low. When reward systems conflate novelty with contribution, the scientific record becomes systematically less informative about what failed.

Science policy usually evaluates what exists. We count papers, grants, patents, datasets, trials, citations, startups, and discoveries. This creates a severe observational bias. The most consequential effects of a selection system may lie in the work that was never attempted.

A rejected proposal leaves a record, but an idea abandoned before submission may not. A young researcher who observes the funding landscape and chooses a safer topic generates no counterfactual document. A laboratory that cannot afford a replication produces no failed grant. A scientist who leaves academia after repeated near misses becomes a career statistic, not a missing research program. A field that never forms because its concepts fall between established disciplines leaves almost no evidence at all.

Call this the science that never happened.

The concept matters because selection systems shape the distribution of questions before they shape the distribution of answers. If a funding system consistently rewards projects with clear preliminary data, investigators rationally generate preliminary data before asking for major support. This can improve feasibility, but it also favors questions for which preliminary work is affordable. If high-impact journals reward strong narratives, researchers may design projects likely to produce narratable results.

If promotion depends on publication frequency, long projects become expensive career bets. The epistemic landscape is edited upstream.

The invisible effect is particularly strong for interdisciplinary work. A proposal that combines two mature fields may be legible to neither panel. Each discipline can identify what the proposal lacks according to its own standards, while no evaluator owns the integrated question. The project is not necessarily radical; it is administratively homeless.

Researchers learn this. They adapt language to calls, restructure ideas around available funding instruments, emphasize dimensions recognizable to panels, and postpone questions that lack institutional homes. Such adaptation is not evidence of dishonesty. It is what competent people do inside incentive systems.

The opportunity cost can be large even when the funded portfolio is excellent. Suppose every selected project is defensible. It does not follow that the selection mechanism produced the best portfolio. A system can choose many good projects while systematically omitting categories of good projects. This is why evaluating grant decisions solely by asking whether funded work succeeded is insufficient. The relevant comparison includes plausible alternatives that were excluded.

Counterfactual evaluation is difficult but not impossible. Funding agencies can track near-threshold applicants longitudinally. They can compare panels with different selection mechanisms. They can run partial lotteries among similarly rated proposals. They can deliberately fund a sample of projects rejected by conventional procedures to estimate false-negative rates. They can compare outcomes across reviewer disagreement patterns. They can study what happens when identity or institutional information is masked.

The file-drawer problem is not limited to statistical null results. Software that failed to scale, datasets that proved too noisy, attempted replications that exposed undocumented dependencies, and theoretical approaches that reached a clear dead end can all contain information that would save others time. Yet the scholarly record is optimized for completed stories. A richer system would allow these outcomes to be registered in lightweight, citable forms without pretending that every failed attempt deserves a full journal article. The goal is not to publish everything. It is to preserve enough negative information that the community does not repeatedly pay for the same avoidable ignorance.

Selective publication can materially distort what a field appears to know. Turner and colleagues compared FDA records for 74 antidepressant trials with the published literature. Nearly a third of the registered studies were not published, and the published record made 94 percent of trials appear positive, compared with 51 percent in the FDA assessments. The apparent effect size was also inflated. This case is unusually measurable because a regulator preserved a less selective record. In many fields, the missing shelf cannot be reconstructed so cleanly.

The cost of unasked questions

The same logic applies before data collection. Panels tend to reward proposals whose value can be narrated clearly in advance. Frontier work often has the opposite property: the uncertainty is the reason to do it. One practical response is to create small portfolios specifically for high-disagreement proposals, replication, exploratory work, and neglected questions. The institution then acknowledges that it cannot rank every form of scientific value on one scale. The aim is not to fund contrarianism. It is to preserve option value where uncertainty is irreducible.

These experiments should not be framed as attacks on peer review. They are tests of a decision technology. Medicine does not insult physicians by evaluating diagnostic procedures. Aviation does not insult pilots by studying cockpit design. Science should be willing to experimentally evaluate the institutions through which science is allocated.

A deeper difficulty is defining success. If alternative funding is judged only by citations, the experiment may reproduce the same prestige dynamics it is meant to test. If it is judged only by publication count, risky long-horizon projects will appear inefficient. Outcome measurement must include field-appropriate epistemic value: whether results changed beliefs, produced reusable tools, falsified important claims, opened new methods, trained durable capabilities, or generated subsequent independent work.

AI can make the invisible somewhat more visible. Large-scale text analysis can map abandoned or repeatedly rejected concepts across proposals if privacy and governance rules permit. It can identify clusters of ideas that receive strong reviews on some dimensions but repeatedly fail overall. It can compare funded and unfunded proposals for conceptual diversity. It can estimate whether entire regions of a research landscape are underexplored relative to adjacent evidence or social need.

Such systems must be governed carefully. Grant proposals contain confidential ideas and personal information. Automated analysis could itself become a surveillance layer that discourages speculative thinking. Data should therefore be used primarily for aggregate institutional learning, with strong access controls, retention limits, and separation between program evaluation and individual researcher scoring.

There is also a philosophical limit. We can never know the full space of discoveries that did not occur. The purpose of the concept is not to calculate a definitive lost-science number. It is to force institutional humility. A portfolio that looks successful from the inside may still be narrow because the selection system has trained the population to submit only what it knows how to recognize.

One practical reform is to create protected channels for epistemic variance. Some grants can use conventional expert ranking. Some can fund replications. Some can support tool-building. Some can select from quality-thresholded proposals by lottery. Some can be awarded to high-disagreement proposals. Some can support early-stage questions with deliberately weak requirements for preliminary results. The mechanisms themselves should be diverse because every mechanism creates its own blind spots.

A research ecosystem should not only maximize expected success. It should preserve the option to discover that its current map of promising science is wrong.

The opportunity cost of selection can be estimated indirectly. Funders can sample proposals just below a threshold, track whether similar ideas later succeed elsewhere, and compare neglected topics with changes in technology or social need. Journals can examine whether repeatedly rejected methods later become standard. None of these exercises reconstructs the counterfactual world perfectly, but they make absence less invisible. The purpose is not to embarrass past panels with hindsight. It is to learn which categories of uncertainty a selection process handles badly, then create a complementary channel for them. A system that measures only the performance of what it funded will systematically overestimate the completeness of its own search.

The opportunity cost is larger than unpublished null results. Funding and career structures influence which questions are proposed at all. A risky project can disappear one step earlier, when an applicant anticipates panel preferences and rewrites it into a safer form. This is difficult to observe because no rejected manuscript exists. Institutions can nevertheless create protected spaces for exploratory, cross-disciplinary, replication, and long-horizon work, then evaluate those mechanisms as portfolios rather than demanding that each project mimic the risk profile of conventional funding.

> The absence of research is difficult to observe, so institutions often rely on consensus as a map of where reliable knowledge already exists. That is usually rational. Consensus aggregates distributed expertise and protects non-specialists from having to adjudicate every controversy themselves. Yet the same social mechanisms that make consensus informative can make it path-dependent. The problem is not consensus itself but the conditions under which dissenting evidence can enter, survive, and eventually update it.

No modern society can operate by asking every citizen to personally verify every technical claim. Consensus is therefore not an embarrassment but an essential bridge between specialization and collective action. The problem appears when we forget that consensus is an inference about a community's current best judgment, not a metaphysical property of nature. A system should be able to rely on consensus strongly while still preserving the machinery by which consensus can be revised.

Why consensus is rational

Consensus is strongest when multiple independent lines of evidence converge and when the community contains genuine routes for criticism. It is weaker when many apparent confirmations descend from the same dataset, method, laboratory, or theoretical assumption. The number of papers alone therefore tells us less than the diversity of the evidence graph. Ten dependent replications can be less informative than two genuinely independent ones.

Scientific consensus is one of civilization's most valuable epistemic technologies. It is also one of its most misunderstood.

A nonspecialist facing a complex question cannot reconstruct the relevant literature. Even specialists cannot independently reproduce every result in their own field. When many qualified researchers, using different methods and data, converge on a conclusion, that convergence is meaningful evidence. Ignoring it because “experts can be wrong” is irrational. Experts can be wrong, but a world in which every individual treats personal intuition as equivalent to accumulated specialist evidence is not a world of intellectual freedom. It is a world with no scalable epistemology.

The problem begins when consensus changes grammatical role. “Most relevant experts currently judge X to be the best-supported interpretation” is an epistemic summary. “X is true because experts agree” is an argument from authority. The two often collapse in public discussion because the first sentence is cumbersome and the second is socially efficient.

Consensus has several possible sources. Researchers can converge because independent evidence points in the same direction. They can converge because methods have standardized and produce comparable observations. They can converge because training establishes shared conceptual priors. They can converge because journals and funding select compatible work. They can converge because dissenting alternatives repeatedly fail. They can also converge because alternatives never became institutionally viable. Usually several mechanisms coexist.

This does not make consensus suspect. It makes the genealogy of consensus relevant.

A mature evidence system should therefore represent not only that a consensus exists, but why. How many independent lines of evidence support the conclusion? How dependent are they on shared datasets, instruments, or assumptions? Have serious contrary hypotheses been tested? How many apparent replications originate from the same research network? Does the consensus persist across methodological communities or mainly within one? Which claims are central and which remain contested at the edges?

These questions matter because apparent numerical agreement can conceal correlated evidence. Ten studies that reuse the same dataset are not equivalent to ten independent datasets. Twenty papers that depend on the same software pipeline may share a hidden error. Multiple laboratories trained in the same methodological tradition may reproduce the same bias. The effective number of independent evidential paths can be far smaller than the publication count suggests.

This is where pedigree and consensus intersect. Prestigious groups can coordinate attention. Once a framework becomes associated with leading institutions, more students enter it, more methods are developed for it, more datasets are collected in its vocabulary, and more review expertise accumulates around it. The framework may deserve this success. Yet the resulting consensus contains both empirical convergence and infrastructural investment.

Consensus also has a topology. A thousand researchers can appear to agree while relying on a small number of shared measurements, benchmark datasets, software libraries, or foundational assumptions. Conversely, a smaller community can have unusually strong consensus if independent methods repeatedly converge. This is why counting statements of agreement is a poor substitute for mapping evidential dependence. The distinction becomes crucial in public controversy: confidence should rise when diverse lines of evidence converge, not merely when the same evidential lineage is repeated by many voices.

For nonspecialists, expert consensus remains one of the strongest available guides because it aggregates training, exposure to evidence, and repeated professional scrutiny. The error is to treat consensus as if it were independent of the structure that produced it. A robust consensus is stronger when its evidence comes from independent methods, datasets, institutions, and research traditions. An apparently large consensus built on one measurement pipeline or one genealogical cluster can be more fragile than its citation count suggests.

How consensus becomes sticky

This suggests a better public language. Institutions should be able to say not only that a consensus exists but what kind of consensus it is: how broad the evidence base is, where uncertainty remains, which assumptions are shared, and what observations would materially change confidence. Such communication is more demanding than a binary fact-check label, but it is also more resistant to both cynicism and dogmatism. The goal is not to weaken expert consensus; it is to explain why confidence is warranted and how revision remains possible.

A strong scientific culture should make it possible to say this without giving comfort to arbitrary contrarianism. The existence of institutional effects does not grant equal standing to unsupported alternatives. A dissenting claim does not become credible merely because consensus can be socially reinforced. The evidential burden on dissent remains. What changes is the obligation of institutions to preserve channels through which strong contrary evidence can actually be heard.

There is a useful analogy with fault-tolerant computing. Redundancy only protects a system if failures are sufficiently independent. Three identical servers with the same software bug are not three independent guarantees. Scientific consensus gains robustness when it emerges from diverse instruments, methods, datasets, theoretical traditions, and research groups. Diversity is epistemically valuable not because every perspective is equally correct, but because correlated failure is dangerous.

Institutional policy can therefore treat methodological diversity as resilience. Funding portfolios can support alternative measurement approaches. Review panels can include adjacent expertise without turning every panel into a demographic or disciplinary checklist. Large consensus reports can explicitly model dependency among evidence sources. Journals can publish well-designed adversarial replications. Scientific societies can maintain living maps of unresolved questions rather than presenting consensus documents as static monuments.

AI can help with evidence dependency analysis. A system can trace dataset reuse, code lineage, citation inheritance, common authorship, shared instruments, and methodological templates. It can reveal that fifty papers ultimately depend on three primary measurements. It can distinguish direct empirical support from chains of secondary citation. It can identify when a claim is repeatedly cited through a review article rather than independently examined.

This is not an argument for an automated consensus score. The point is to expose structure. A human expert panel that can see the dependency graph is in a better position to judge robustness than one looking only at paper counts.

The public communication of consensus also needs reform. Confidence should not be portrayed as unanimity. Scientific disagreement can coexist with strong practical consensus. Climate science, medicine, engineering, economics, and public health routinely require action before uncertainty is eliminated. Institutions lose credibility when they imply that uncertainty is absent and later appear to reverse themselves when new evidence arrives.

The better model is calibrated authority. Experts should be able to say: this conclusion is strongly supported; these mechanisms remain debated; this subgroup is uncertain; this policy judgment adds values and trade-offs beyond the science; these observations would materially change our view.

Consensus is strongest when it includes a map of its own possible failure.

Sticky consensus is most dangerous when changing one's mind carries social costs that are independent of evidence. A laboratory may have students, grants, software, and public commitments built around a framework; a journal may have developed an audience around a particular debate; a regulator may have encoded an estimate into standards. Updating then becomes institutionally expensive even when it is intellectually warranted. This does not imply bad faith. It means that evidence enters a system with sunk costs. A resilient consensus process therefore needs explicit update points, protected space for adversarial synthesis, and a public vocabulary in which narrowing or revising a claim counts as scientific progress rather than as institutional humiliation.

Stickiness can arise because changing belief has coordination costs. Textbooks, guidelines, datasets, grant calls, and software encode prior assumptions, so a correction must propagate through many artifacts before the practical system catches up. This is one reason post-publication correction deserves infrastructural treatment. A retraction or erratum is not complete merely because a journal page changes; downstream reviews, models, datasets, clinical guidance, and automated knowledge bases may still carry the older claim.

> Consensus is also produced inside a geography of unequal visibility. Research systems differ in language, funding, infrastructure, prestige, conference access, editorial representation, and network centrality. These differences do not imply that peripheral work is systematically better; they imply that otherwise similar work may face different probabilities of being noticed, cited, invited, and reused. The credibility gradient therefore has a spatial and network dimension in addition to an institutional one.

The paper from nowhere is not equally 'nowhere' everywhere. A result written in fluent academic English, produced inside a dense collaboration network, and associated with a university that reviewers recognize begins its journey with lower transaction costs. Another result can be equally careful and still require more explanation before it is granted the same attention. Geography therefore enters scientific credibility not only through prejudice but through infrastructure, language, networks, hiring pipelines, and accumulated visibility.

Where science is done matters

Prestige is also embedded in career networks. Wapman and colleagues analyzed US tenure-track hiring from 2011 to 2020 and found that roughly 80 percent of domestically trained faculty came from about 20 percent of PhD-granting universities, with steep prestige hierarchies across fields. This is not proof that hiring committees simply worship logos. Elite institutions genuinely concentrate resources, training, and networks. But the concentration means that institutional pedigree can reproduce itself through the very labor market that later treats pedigree as evidence.

Scientific merit is distributed more broadly than scientific opportunity.

This statement is easy to endorse abstractly and difficult to operationalize because opportunity itself affects visible merit. A researcher with access to elite collaborators, expensive equipment, strong grant offices, experienced mentors, large datasets, international conferences, and a recognized institutional address can produce a record that is genuinely stronger than that of an equally capable researcher without those resources. By the time an evaluator sees the two CVs, structural difference has become empirical difference.

The pedigree gradient therefore has at least three components. Selection matters: prestigious institutions often attract unusually strong researchers. Resources matter: institutions change what researchers are able to accomplish. Perception matters: the same output can be interpreted differently depending on where it comes from. Reform becomes confused when these components are treated as though only one can be true.

Experiments on status signals illustrate the perception component. A preregistered study involving more than four thousand scientists across six disciplines varied the country and institution attached to research abstracts. The authors reported weak evidence of status-related bias overall, an important reminder that prestige effects are neither universal nor necessarily large in every context. In other review settings, including a controlled computer-science experiment, stronger prestige effects have been detected. The correct conclusion is not that bias is everywhere or nowhere. It is that the size and form of the effect is an empirical property of a review system and should be measured rather than assumed.

Geography enters through more than reviewer psychology. Research systems differ in salaries, administrative burden, infrastructure, teaching loads, language, visa access, publication fees, procurement, legal environments, and the density of nearby collaborators. A scientist working in a lower-resource environment may spend more time solving problems that never appear in the final paper. These hidden costs reduce output and then reappear in evaluation as apparent productivity differences.

Network structure compounds the effect. Collaboration is not only a social luxury. It is a channel for tacit knowledge. Researchers learn which questions are becoming important before they are visible in journals. They hear why an experiment failed. They receive invitations to datasets and workshops. They learn how panels interpret a call. They encounter future collaborators. Dense networks therefore generate informational advantage.

The danger is to respond with a simplistic anti-elite policy. Strong research centers create real economies of scale. Concentrating instruments and expertise can be scientifically efficient. Exceptional laboratories can train generations of researchers. Deliberately dispersing all resources equally would destroy useful specialization.

CASE IN POINT Hiring networks show how pedigree becomes infrastructure rather than merely opinion. In a 2022 Nature analysis of US tenure-track faculty, a small minority of universities produced a large majority of faculty, and upward movement in the prestige hierarchy was uncommon across many fields. Such patterns can arise partly from real concentration of training quality and resources. They also mean that the labor market continually reproduces the signal that future committees then use as evidence. Pedigree is both an input to selection and an output of previous selection.

Global disparities appear not only in resources but in attention. Gomez, Herman, and Parigi analyzed nearly twenty million papers and compared citation flows with textual similarity. They found that research from highly active countries increasingly received more citations than similar work from peripheral countries, a pattern they call citational lensing. The finding does not reduce citations to national prejudice; it shows that visibility and reputation operate at country level strongly enough to leave a measurable trace after accounting for topical similarity.

Language, networks, and global visibility

Global inequality adds further layers. English-language fluency reduces friction in review and collaboration. Travel budgets affect conference visibility. Article-processing charges can make open access easier for wealthy institutions than for poor ones. Large laboratories can absorb compliance and data-management costs that small groups cannot. UNESCO's Open Science Recommendation treats openness, infrastructure, and equitable participation as connected rather than separate goals. That is the right frame:

reducing pedigree effects requires lowering the practical cost of participating in the shared evidence system.

The more defensible objective is permeability. A concentrated center is less problematic if researchers, ideas, data, and resources can enter and leave it through open channels. A prestigious network becomes dangerous when it is also a closed network.

Permeability can be increased through shared infrastructure, portable grants, open data, interoperable research platforms, visiting programs, cross-institutional doctoral supervision, remote access to specialized equipment, and funding mechanisms that reward collaboration without requiring peripheral partners to become decorative subcontractors. Evaluation can also normalize for opportunity where appropriate, although this must be done transparently to avoid replacing one opaque judgment with another.

A particularly important reform is to separate affiliation from early-stage idea evaluation whenever identity is not needed. Double-blind review is imperfect, but it can serve as an ablation test on pedigree. Funding agencies could run parallel masked evaluations on subsets of proposals and compare ranking changes. Journals can periodically audit whether identity exposure alters reviewer severity. Hiring committees can evaluate selected work samples before reading full CVs. These experiments do not require institutions to believe that pedigree bias is severe; they allow institutions to find out.

AI can make such auditing much richer. Language models can redact identifying clues more effectively than simple name removal, although perfect anonymization remains difficult. They can produce matched representations of proposals with institution-dependent information removed. They can compare review rationales to detect whether the same methodological concern is described as fatal for one group and manageable for another. They can simulate counterfactual presentation, not to accuse individual reviewers, but to estimate system-level sensitivity to status signals.

There is an additional risk: AI systems themselves may encode geography and pedigree. A model trained on the scientific web will have seen prestigious institutions mentioned more often, associated with awards, influential papers, and positive language. If asked to rank researchers or proposals without careful design, it can reproduce the existing hierarchy while appearing neutral because the decision is computational.

This makes transparency essential. Any AI used in consequential research assessment should be tested for counterfactual stability under changes to names, affiliations, country, gender-coded cues where legally and ethically appropriate, and other irrelevant attributes. The model should also be evaluated for field and language effects. An automated evaluator that performs well on mainstream English-language machine-learning papers may be unreliable in a small humanities field or on work written by non-native English speakers.

Pedigree cannot simply be deleted from science. Institutions are real, resources are real, and track records contain information. But pedigree should not be allowed to masquerade as a property of the claim itself.

English-language publication, conference travel, editorial networks, preprint visibility, and access to high-resource collaborations all affect how quickly work enters the central conversation. These mechanisms are partly cumulative: being cited creates visibility, visibility creates collaborators, and collaborators create further visibility. A better system cannot equalize every structural difference, but it can lower avoidable barriers by widening discovery, using content-based retrieval, diversifying reviewer pools, and auditing whether institutional and national signals influence decisions beyond the tasks where they are genuinely relevant.

Diagnosis is cheap compared with institutional redesign. Every reform changes incentives, creates new paperwork, and can be gamed. The task is therefore not to invent a perfect metric or a pure committee. It is to make credibility less dependent on a single channel. This part moves from the paper and the career to the architecture around them: how claims could be evaluated separately from pedigree, how money could reward correction as well as novelty, what law can regulate without legislating truth, and why scientific governance needs the equivalent of checks and balances.

Chapter 5 - Separating Evidence from Status

By this point Elena's paper is no longer merely a document. It has citations, associated code, follow-up claims, attempted replications, and an institutional history. Yet most evaluation systems still compress this evolving evidential object back into a few labels: journal, author, grant, institution, citation count. The chapter asks what happens if the unit of evaluation changes from the prestige-bearing paper to the inspectable claim and its evidence.

That shift also changes economics. Reliability work remains scarce because institutions reward it inconsistently. Replication, adversarial checking, maintenance, negative results, and long-horizon risk need explicit economic homes. The goal is not to abolish selection but to stop forcing every scientific value through the same tournament for novelty.

A journal article is a convenient container, not the natural unit of scientific truth. One paper can contain a strong measurement, a fragile causal claim, a useful dataset, a speculative interpretation, and a figure that later turns out to be wrong. Yet assessment often assigns one reputation to the whole object. Digital science makes a more granular architecture possible: claims, evidence, methods, code, data, and replications can be represented separately and linked. Once that happens, pedigree can become context rather than destiny.

From papers to claims

The shift from document-level to claim-level evaluation is more than a software feature. It changes what can be credited and corrected. A dataset can remain valuable after one interpretation fails. A method can be reused even if the headline conclusion is revised. A specific claim can acquire replication evidence without forcing the reader to decide whether the entire paper is 'good' or 'bad.' Granularity reduces the social pressure to defend whole intellectual identities when only one component is under dispute.

The unit of scientific communication is still, to a surprising degree, the paper. A paper bundles a question, literature review, method, data description, results, interpretation, rhetoric, authorship, institutional affiliation, and a set of references into one document. The bundle was historically sensible. It remains useful for human reading. But it is a poor data structure for a system that wants to know what, exactly, is supported by what.

A paper can contain a robust measurement, a fragile causal interpretation, a speculative theoretical section, and a mistaken citation at the same time. Yet evaluation often assigns the entire object a single social status. It is accepted or rejected. It appears in a prestigious venue or it does not. It accumulates citations as a whole. Later authors may cite it for claims that were peripheral to the original study. A complex epistemic object is compressed into a binary publication event and a scalar citation count.

A more mature architecture would separate claims from containers.

Consider a simple scientific sentence: “Method A improves outcome B under condition C.” That claim can be represented as an object connected to the experiment that generated it, the dataset used, the code that implements the analysis, the assumptions required for the inference, the prior work it extends, subsequent replications, contradictory findings, and later refinements. The paper remains the narrative in which the authors explain the work, but the claim gains a life that can be updated independently of the document.

This changes the role of pedigree. An author identity can remain attached for credit and accountability, but it no longer needs to carry the evidential burden. A reader can ask whether the claim has independent support, whether the replication uses the same data source, whether the effect survives alternative analysis, and whether a later study narrows the conditions under which it holds.

Such a system would not create certainty. It would create provenance.

Provenance is one of the underdeveloped concepts in scientific reform. We often ask whether a claim is cited, but not how the claim changed as it moved through the literature. We ask whether a dataset is open, but not whether downstream papers used the same version or preprocessing decisions. We ask whether a result replicated, but not whether the replication was genuinely independent of the original code and assumptions. Provenance turns the history of a claim into inspectable structure.

This is technically feasible in pieces today. Persistent identifiers can link papers, datasets, software, researchers, and grants. Version control can record code evolution. Structured reporting can expose methods. Knowledge graphs can represent relationships among claims. AI systems can extract candidate claims from natural language and map them to supporting passages, references, and data artifacts. What is missing is not a single technology but an institutional decision to treat these connections as first-class scientific outputs.

The same granularity can improve credit. Current authorship often bundles conceptual work, software, data curation, experimental execution, analysis, and writing into one ordered list. Contribution taxonomies help, but a claim graph could go further by connecting specific contributions to specific scientific objects. A dataset maintainer can receive durable credit when dozens of later claims depend on that dataset. A replication team can be visible at the point where confidence changes. This makes the scientific record more faithful to how modern collaborative work is actually produced.

The proposal to move toward claim-level infrastructure should be modest about ontology. Science does not need a universal machine-readable theory of meaning before useful structure becomes possible. Stable claim identifiers, provenance links, declared methods, dataset versions, software versions, and relationships to replications or contradictions already allow important questions to be answered. The paper can remain the human narrative. The claim graph is an additional layer for navigation, audit, and updating.

Pedigree as prior, evidence as update

A claim-oriented system also allows pedigree to be localized. Reputation might legitimately affect confidence that a laboratory followed a difficult protocol, but it should have little bearing on whether a mathematical transformation is correct when the code is executable. Different claims invite different evidence. The architecture therefore becomes epistemically plural: human expertise, formal checks, replication, provenance, and institutional history can each contribute where they are informative rather than being collapsed into one prestige-weighted judgment.

The benefits would extend beyond reliability. Credit could become more precise. A researcher who created a dataset could receive visible downstream credit without needing honorary authorship on every paper. A software maintainer could be recognized through dependency. A scientist who falsified a popular claim could be connected to the correction rather than appearing merely as another citation. Research objects could accumulate reputational histories appropriate to what they actually contributed.

This matters because authorship currently performs too many functions. It signals intellectual contribution, responsibility, status, mentorship, laboratory membership, and sometimes political negotiation. The more tasks a single convention performs, the easier it is for credibility to leak between them. Claim-level provenance allows responsibility and credit to be decomposed.

The same decomposition can improve review. Instead of asking a reviewer whether the paper is “strong,” systems can ask which claims are central, what evidence each claim requires, which dependencies are shared, and where uncertainty enters. An empirical claim may be well supported even if the theoretical interpretation is weak. A method may be useful even if the headline comparison is overclaimed. Review can become less like judging a performance and more like inspecting an evidential structure.

AI is particularly suited to the clerical layer of this work. Models can propose claim extraction, align citations to propositions, identify unsupported leaps, and generate machine-readable provenance records. But automated extraction should remain contestable. Authors and reviewers need to correct mappings. Systems should preserve original source spans. Confidence should be attached to extracted relations. A generated graph must never become more authoritative than the text and data from which it was derived.

The legal and institutional implications are significant. Funding agencies could require machine-readable provenance for publicly funded projects in proportion to project size and field norms. Journals could publish claim maps alongside articles without making them a condition for every discipline immediately. Universities could evaluate contribution graphs rather than relying primarily on journal brands. Research infrastructures could maintain public version histories for important claims.

The transition should be gradual because standardization can itself become a gatekeeping technology. A highly formal representation that fits experimental sciences may distort qualitative or theoretical research. The objective is not to force every discipline into one ontology. It is to make evidence relationships more explicit where doing so improves accountability.

The underlying principle is simple. A scientific claim should become progressively less dependent on who said it as the evidence around it becomes richer.

That is what it means to separate the claim from the pedigree.

Treating pedigree as a prior also clarifies when it should disappear from the decision. Before an experiment is performed, a team's history may be relevant to whether an ambitious protocol is feasible. Once data, code, and procedure are available for inspection, those direct objects should carry progressively more weight. This is a temporal rule as much as an epistemic one: use reputation when information is scarce, then deliberately retire it as better information arrives. Institutions rarely specify this handoff, so the initial prior quietly survives into later judgments. A claim-centric architecture can encode the transition explicitly and ask evaluators to justify any remaining dependence on status after the relevant evidence is visible.

A Bayesian metaphor is useful here only at the conceptual level: reputation can influence the initial expectation, but direct evidence must be capable of moving the judgment substantially. No equation is required. The institutional implication is concrete. Early triage may legitimately use coarse contextual signals; later high-stakes evaluation should expose more of the underlying record. As verification becomes cheaper, the stage at which pedigree remains decisive should shrink.

> Separating claims from pedigree is technically useful only if institutions also pay for the activities that keep claims reliable. A system can publish perfect provenance and still undersupply replication if careers and grants reward novelty almost exclusively. The next sections therefore move from informational architecture to economic design: how to fund correction, informative failure, protected risk, and competing

selection mechanisms without creating a new bureaucracy around every good intention.

The easiest scientific activity to finance is new production. A proposal promises discovery, a project creates outputs, and a press release can describe progress. Correction has weaker political theater. Replication may confirm what was already believed; maintenance prevents failure rather than producing novelty; careful curation looks like overhead. If institutions want a self-correcting science, however, they must build an economy in which doubt is not merely praised rhetorically but given time, status, and money.

Paying for correction

Economic reform should start with a simple observation: self-correction is a public good, and public goods are often underproduced when rewards attach mainly to private or visible outputs. A laboratory that replicates another group's work creates value for the whole field but may receive little career advantage. A maintainer who keeps a dataset usable can be indispensable while appearing unproductive under publication metrics. Funding rules can correct this mismatch by making verification, curation, software, and negative evidence eligible first-class outputs.

Science celebrates skepticism in theory and often underfunds it in practice.

Novel claims generate projects. Confirmation can extend them. Contradiction can terminate them. From the perspective of the knowledge system, all three outcomes are useful. From the perspective of careers, journals, and institutions, they are not equally rewarded. The asymmetry produces an economy in which generating claims can be more profitable than checking them.

The solution is not to ask scientists to become less ambitious. It is to create markets and budgets for epistemic maintenance.

Replication is the clearest example. A finding that influences clinical practice, regulation, a major theoretical literature, or billions in subsequent research should attract verification resources proportional to its downstream importance. Today, replication often depends on the interests of individual laboratories or special initiatives. A more systematic model would treat verification as insurance. The more social or scientific capital accumulates around a claim, the stronger the incentive to test whether the foundation is stable.

This leads to a simple funding principle: verification budgets should scale with epistemic exposure.

If a result becomes a critical dependency for a field, a small percentage of the resources that depend on it could be reserved for independent replication, robustness checks, data audits, or alternative measurement. This is not punishment for successful researchers. It is the scientific equivalent of stress-testing infrastructure after it becomes important.

Negative results require another adjustment. A well-designed study that fails to find an expected effect can save others from repeating the same path. Yet its value is diffuse. The originating laboratory bears the cost while the community receives the benefit. This is a classic public-goods problem. Registered Reports partially solve it by decoupling publication from outcome. Dedicated repositories and journals can help, but prestige and discoverability matter as much as storage. A null result that nobody finds has little systemic value.

Funding can therefore recognize informational value rather than positive direction. Grants can require pre-specified dissemination of informative null results where appropriate. Evaluation can credit rigorous disconfirmation. Synthesis systems can actively retrieve negative evidence rather than relying on citation popularity, which tends to privilege visible positive narratives.

Funding allocation itself should also acknowledge uncertainty. Conventional panels often create rank orders that imply a degree of precision the underlying judgments do not support. Research on grant peer review has documented limited agreement among evaluators, especially near decision boundaries. Modified lotteries are one institutional response. After proposals cross a quality threshold, random selection among the difficult-to-distinguish set can reduce arbitrary differentiation and potentially broaden access.

There is also a useful analogy with safety engineering. No serious infrastructure budget assumes that all money should go to new features while inspection, maintenance, and incident response are financed only when something fails publicly. Scientific systems often do exactly that. Verification budgets appear ad hoc after controversy instead of being treated as normal operating cost. A small protected allocation for replication and maintenance would not need to dominate research spending to change behavior. Its existence would signal that correction is expected work rather than an accusation against the original researchers.

Several real mechanisms already acknowledge uncertainty in research selection. The Swiss National Science Foundation, for example, allows a lottery when the funding line cuts through proposals judged to be of exactly the same scientific quality. This is a narrow use of randomization, not a rejection of expert review. Its importance is conceptual: the institution refuses to pretend that a ranking is more precise than the judgment that produced it. Similar logic can be applied to replication reserves, negative-result publication, and protected budgets for high-uncertainty work.

Portfolio funding and protected risk

The second economic principle is to stop pretending that fine-grained rankings are more precise than the evidence permits. Thresholded lotteries are one response: expert review identifies proposals that are clearly fundable, while random selection is used only within a band where ranking noise is high. The point is not random science. It is honest uncertainty. Combined with portfolio funding, long-horizon support, and dedicated replication reserves, such mechanisms can reduce the extent to which a single narrow decision becomes a lifelong cumulative advantage.

The important word is modified. A pure lottery over all proposals would throw away expert information. A thresholded lottery uses expertise where it is relatively robust — excluding clearly unsuitable work and identifying a fundable range — while declining to pretend that experts can always discriminate finely inside that range. Recent philosophical and simulation work on funding lotteries makes the trade-off explicit: stronger randomness can improve some forms of fairness and reduce concentration while sacrificing some ability to prioritize proposals assumed to be highest in expected epistemic value. There is no universal optimum because the objectives differ.

A diversified funding system can use several mechanisms simultaneously. High-confidence infrastructure projects may merit conventional selection. Exploratory grants may use broad thresholds and lotteries. High-risk programs may deliberately select proposals with strong reviewer disagreement. Replication funds can prioritize downstream dependency. Early-career programs can limit the weight of prior grant success. Long-horizon fellowships can support people rather than projects where the goal is intellectual independence.

This is not administrative complexity for its own sake. It is portfolio theory applied to knowledge production. A financial institution would not place every asset into the same risk class. A national research system should not use one evaluation logic for every epistemic function.

Economic reform also requires changing how institutions absorb failure. A project portfolio designed for real novelty will include failures. If auditors, politicians, university boards, or media treat every failed project as evidence of waste, agencies will rationally choose safe work. Society cannot simultaneously demand breakthrough research and zero visible failure.

Accountability should therefore distinguish negligence from informative failure. A project that followed a sound method and discovered that the hypothesis was wrong may be successful science. A project that failed because basic controls were ignored is different. Current reporting often compresses both into whether deliverables were achieved.

AI could support a richer accountability model by reconstructing project histories. Did the team follow the planned protocol? Were deviations documented? Did the result reduce uncertainty? Were data and tools preserved? Did the project produce evidence that changed future decisions? Automated systems can help extract these signals, but the value judgment must remain human and policy-driven.

There is also a case for paying directly for review and verification. The existing peer-review system relies heavily on invisible labor. Aczel and colleagues' estimate of more than one hundred million annual reviewer hours illustrates the scale. Compensation alone will not solve quality problems, and professionalizing review could create its own bureaucracy, but strategic payment can make deeper methods review, statistical audit, code execution, or replication planning economically viable.

A scientific economy that pays only for production will overproduce claims. A scientific economy that pays for doubt creates a market for correction.

Economic reform works best when mechanisms are matched to distinct failures. No single funding instrument can simultaneously maximize novelty, reliability, equality of opportunity, and long-term capacity. The table below therefore treats funding as a portfolio of functions rather than a contest for one universal notion of excellence.

Table 2 groups funding mechanisms by the behavior they are intended to purchase. The common theme is that reliability work must have an economic home. If every reward is attached to novelty, institutions should not be surprised when replication, negative results, maintenance, and adversarial checking are undersupplied.

Table 2. Mechanisms that pay for correction, not only novelty

Institutional mechanisms that reward verification, uncertainty, and informative failure.

The first column names a funding mechanism; the second explains the institutional problem it is meant to address.

Mechanism	Institutional purpose
Replication reserve	Direct a defined share of resources toward independent checks of claims with high downstream importance or uncertainty.
Registered funding	Evaluate important parts of the research question and design before outcomes are known, reducing pressure for attractive results.
Thresholded lottery	Use expert review to define a clearly fundable set, then acknowledge ranking uncertainty within the near-tie region.

Mechanism	Institutional purpose
High-disagreement portfolio	Preserve some proposals that divide experts, treating disagreement as possible frontier information rather than automatic weakness.
Infrastructure careers	Reward software, datasets, standards, curation, and verification as durable scientific contributions rather than invisible support labor.
Long-horizon support	Protect a limited share of funding from annual output pressure so difficult questions can survive periods of low publishability.

These mechanisms are complements rather than a menu from which one universal replacement should be chosen. A replication reserve solves a different problem from a thresholded lottery; long-horizon funding solves a different problem from Registered Reports. The design task is to match the mechanism to the uncertainty the institution is actually facing.

These mechanisms should be piloted, compared, and retired when they fail. Their value is not ideological. It lies in producing different error profiles. A research system becomes more resilient when one committee's preference does not determine every downstream opportunity.

There is a complementary reason to diversify funding instruments. A system made entirely of large competitive grants favors applicants who can absorb long application cycles, assemble recognizable consortia, and survive repeated rejection. Small rapid grants, replication funds, infrastructure awards, long-horizon fellowships, and threshold lotteries serve different epistemic functions. Their value is not that one mechanism is fair in every context. It is that a heterogeneous funding ecology prevents one noisy ranking procedure from becoming the universal route through which scientific possibility must pass. The portfolio is therefore institutional as well as financial: it distributes not only money, but also different ways of being allowed to try.

Protected risk also requires protected time. A grant that nominally funds exploratory work but subjects investigators to the same short reporting cycles and output expectations as an incremental project will often converge toward incremental behavior. One practical reform is to separate fiduciary milestones from epistemic milestones. Spending, ethics, data stewardship, and safety can be audited on schedule, while the scientific direction is allowed to change when early evidence undermines the original plan. This preserves accountability without rewarding investigators for pretending that uncertainty disappeared when the proposal was accepted.

Portfolio design also changes the meaning of failure. If every funded project is expected to succeed, reviewers will rationally prefer predictable work and investigators will rationally promise predictable outcomes. A portfolio can instead define several legitimate risk classes and judge performance at the program level. Some projects should fail if the program genuinely explores uncertain territory. The accountability question then becomes whether the failures were informative, competently pursued, and appropriately distributed, not whether every grant produced the forecast result.

Chapter 6 - Governing Scientific Institutions

Reform becomes more difficult when scientific organizations exercise public power: they allocate public money, control careers, determine access to infrastructure, and shape the expert record used in policy. Elena's experience now moves from the journal and grant panel to the rules governing those decisions. The relevant question is not whether law should decide scientific truth. It should not. The question is which procedures public institutions may legitimately be required to make visible, contestable, and auditable.

A constitutional view of science follows from that boundary. No single mechanism should dominate credibility. Expert review, replication, open evidence, plural funding instruments, appeals, conflict management, and independent audit should constrain one another. The purpose of legal and institutional design is to preserve epistemic competition while preventing procedural power from hardening into permanent authority.

Calls for legal reform can become dangerous quickly. Legislatures should not decide which statistical method is correct, which theory deserves consensus, or which scientific conclusion is politically acceptable. Yet law already structures science through employment, funding, procurement, intellectual property, public records, whistleblower protection, and due process. The relevant question is narrower: which procedural rights and transparency duties can make scientific institutions harder to capture without turning governments into arbiters of truth?

What law can legitimately change

Law is most useful when it protects process. Publicly funded institutions can be required to disclose material conflicts of interest, preserve decision records, provide reasoned explanations at an appropriate level, protect good-faith reporting of misconduct, and make research outputs accessible when privacy, security, or legitimate commercial interests do not justify restriction. The 2021 UNESCO Recommendation on Open Science provides an international policy frame for open publications, data, software, and infrastructures. European whistleblower rules offer another procedural analogy: protection is aimed at enabling reporting and enforcement, not at deciding the substantive truth of every allegation.

Law should not determine scientific truth. It can determine the conditions under which scientific institutions exercise power.

This boundary is crucial. A legal regime that prescribes acceptable theories would destroy scientific autonomy. A legal regime that treats universities, journals, and funders as entirely private epistemic worlds ignores the public resources and public consequences that flow through them. Modern science occupies an intermediate position: it requires autonomy in substantive judgment and accountability in procedure.

The most defensible legal reforms therefore target due process, transparency, conflicts, access, and institutional concentration rather than conclusions.

Public funding provides the clearest case. When taxpayers finance research, there is a strong argument that resulting publications should be accessible, subject to legitimate privacy, security, and intellectual-property constraints. The same logic can extend to underlying data and code when disclosure is ethically and legally possible. Open access is not merely a distribution policy. It reduces pedigree dependence by allowing people outside wealthy institutions to inspect the actual evidence.

Transparency should also apply to evaluation rules. Funding agencies need not reveal every confidential reviewer identity, but they can publish criteria, conflict policies, scoring calibration, panel composition principles, aggregate concentration measures, appeal procedures, and retrospective evaluations of the selection mechanism. If a public body uses an algorithmic tool, the existence and role of that tool should be disclosed.

Conflicts of interest deserve a broader vocabulary. Financial conflicts are only one category. Direct scientific competition, close collaboration, supervisory relationships, institutional dependence, and adversarial litigation can all matter. The law should not attempt to enumerate every intellectual conflict, but agencies can be required to maintain and enforce proportionate conflict frameworks.

Whistleblower protection is another foundational issue. Science depends on the ability to challenge data, methods, and institutional conduct. Yet researchers may be professionally vulnerable when allegations concern powerful laboratories or institutions. Legal protection should distinguish good-faith reporting from reckless accusation while ensuring that employment dependence does not make serious concerns impossible to raise.

The hardest area is appeals. Scientific judgment cannot be converted into ordinary administrative litigation every time an applicant dislikes a decision. That would paralyze evaluation and encourage strategic legal pressure. Appeals should therefore focus primarily on procedure: undisclosed conflict, factual error about eligibility, failure to follow published rules, discriminatory treatment, or misuse of automated systems. The purpose is not to give courts jurisdiction over whether a theory deserved funding.

The distinction between procedure and substance can be tested with a simple question: could the same legal rule protect two scientists who disagree about the underlying theory? A rule requiring conflicts to be disclosed can. A rule requiring public grants to preserve an auditable decision record can. A rule protecting a researcher from retaliation for reporting fabricated data can. A rule declaring one contested scientific interpretation legally correct cannot. This symmetry test is a useful safeguard against reforms that quietly convert governance into ideological control.

Law is strongest when it regulates procedure, transparency, rights, and conflicts rather than scientific conclusions. Public funders can be required to publish assessment criteria, document material conflicts, preserve records, provide reasoned decisions where administrative law requires them, and maintain routes for appeal on procedural grounds. None of this gives a court competence to decide whether a theory is true. It gives participants a way to challenge arbitrary process without turning scientific disagreement into litigation.

Transparency, appeals, and protection

A useful legal boundary is that the state should regulate the conditions under which scientific power is exercised rather than prescribe scientific conclusions. Due process can govern a grant appeal; it cannot determine whether a hypothesis is true. Competition law can examine market concentration in scholarly infrastructure; it should not decide which journal is intellectually excellent. Employment law can protect researchers who report serious integrity concerns; it should not immunize every scientific disagreement from ordinary criticism. Procedural humility is what prevents reform from becoming politicized epistemic control.

Research assessment also raises data-protection concerns. An AI system that builds a detailed credibility profile of every scientist could become a powerful surveillance infrastructure. It might infer ideological tendencies, collaboration networks, health information from biomedical work, or sensitive career patterns. Europe in particular already has strong legal traditions around data minimization, purpose limitation, and automated decision-making. Research institutions should extend these principles to internal analytics rather than assuming that scientific goals automatically justify comprehensive profiling.

Automated evaluation should be legally constrained by function. Low-stakes tools that check formatting or missing citations require different governance from systems that influence hiring or grants. The higher the consequence, the stronger the requirements for validation, auditability, human review, contestability, and documentation. A model that ranks proposals should never be treated as a neutral calculator. Its training data, prompt design, retrieval sources, and scoring rubric encode policy choices.

There is also a competition-law dimension worth exploring. Academic publishing, data infrastructure, analytics, and research platforms can become concentrated. If a small number of commercial or institutional actors control publication, citation data, researcher profiles, evaluation metrics, and AI assessment tools, credibility infrastructure itself becomes a strategic bottleneck. Open standards and data portability may therefore be as important for epistemic pluralism as antitrust is for economic competition.

Law can also create safe spaces for experimentation. Public funders could be authorized to run controlled trials of alternative review mechanisms. Universities could receive regulatory support for portable benefits and cross-institutional appointments. Data-sharing frameworks could reduce legal uncertainty around reproducibility while preserving privacy. Procurement rules could require interoperability for publicly funded research infrastructure.

A useful legal doctrine would be the protection of epistemic competition. The state should not decide which scientific view wins, but it can ensure that public institutions do not make correction needlessly difficult. This resembles constitutional design more than substantive regulation.

Such a doctrine would ask whether entry channels exist for new researchers, whether evidence can challenge incumbent programs, whether evaluation criteria are knowable, whether conflicts are managed, whether publicly financed claims are inspectable, whether automated systems can be contested, and whether institutional decisions leave auditable traces.

The law cannot make science true. It can make scientific power easier to inspect and harder to monopolize.

Legal reform is easiest to misunderstand because law is powerful. The appropriate targets are procedural conditions around scientific authority, not substantive conclusions. The table distinguishes reforms that can plausibly improve accountability from the epistemic limits that should constrain them.

Table 3 deliberately limits the role of law. Each lever addresses procedure, transparency, concentration of power, or protection of participants; none authorizes legislators to choose scientific winners.

Table 3. Legal levers and their epistemic purpose

Procedural legal tools that strengthen scientific correction without legislating scientific conclusions.

Law can protect openness, due process, and contestability without deciding which scientific position is correct.

Legal lever	Purpose and boundary
Conflict disclosure	Require material relationships and roles to be visible where they can affect public decisions; disclosure does not by itself establish bias.
Reasoned decisions	For high-stakes public funding, preserve structured reasons and relevant review records so procedures can be audited without exposing every confidential deliberation.
Whistleblower protection	Protect good-faith reporting of serious integrity or legal violations; do not convert ordinary scientific disagreement into protected accusation.
Open outputs	Attach proportionate access and provenance duties to publicly funded work while preserving justified privacy, security, and legitimate IP exceptions.
Appeal and recusal rules	Provide limited review of procedural errors and conflicts; appeals should not become a second scientific panel for every rejected proposal.
Infrastructure competition	Use procurement and competition policy to avoid unnecessary dependence on a single closed assessment or publishing infrastructure.

The boundary matters because legal remedies can easily overshoot. A system that makes every peer-review disagreement appealable in court would paralyze expertise. A system with no procedural rights can conceal arbitrary or conflicted decisions. The workable middle is auditable procedure with restrained substantive intervention.

The recurring theme is proportionality. A procedure can be transparent enough to audit without becoming performative bureaucracy. A review can be contestable without making every judgment litigable. Institutional design improves when rights and duties are specific about the failure they are intended to prevent.

Transparency must also be proportionate. Fully public review can expose junior reviewers to retaliation, reveal confidential proposals, or disclose sensitive data. Conversely, total opacity makes pattern auditing impossible. Good design separates levels: public rules and aggregate statistics, protected but auditable review records, confidential conflict declarations, and narrowly defined procedures for independent investigation. Whistleblower protection belongs in the same architecture because the cost of reporting institutional failure should not be professional extinction.

> Procedural law can prevent some abuses, but the deeper design principle is pluralism. Scientific institutions should avoid allowing one committee, one metric, one model, or one prestige hierarchy to become the sole path to legitimacy. Checks and balances are useful in science for the same reason they are useful elsewhere: different mechanisms fail differently. A robust system combines them so that the error of one mechanism can be exposed by another.

Liberal constitutional systems begin from a pessimistic insight: good intentions are not enough, so power should be divided. Scientific governance can use the same insight without pretending that a university is a state. When the same small network controls hiring, funding, publication, awards, and disciplinary definition, errors can reinforce one another. The answer is not permanent opposition. It is institutional plurality: different mechanisms should fail differently and be able to correct one another.

Checks and balances for knowledge

Checks and balances in science need not mimic government branches. They can be built from heterogeneous evaluation paths. A proposal can be scientifically reviewed while implementation capacity is assessed separately. A paper can be published while verification status remains open. A hiring committee can evaluate selected contributions before seeing broad metrics. A field can preserve both mainstream funding and a smaller mechanism for high-disagreement work. The principle is that no single proxy should control every gate.

The phrase constitutional science does not imply a written constitution for every laboratory. It describes a design principle: no single mechanism should be trusted to allocate all scientific credibility.

Political constitutionalism begins from a sober observation. Even well-intentioned actors can accumulate power, information can be incomplete, and institutions can drift. The response is not to find perfectly virtuous rulers but to distribute authority, create checks, preserve appeals, and make important actions visible. Scientific institutions face analogous problems.

Peer review is powerful but noisy. Metrics are scalable but reductive. Replication is informative but expensive. Markets can aggregate dispersed beliefs but can be gamed and are unsuitable for many questions. Public participation can reveal neglected values but cannot replace specialist competence. AI can inspect enormous volumes of material but can hallucinate, inherit bias, and optimize the wrong objective. Reputation carries real information but compounds historical advantage.

The obvious conclusion is not to choose the best mechanism. It is to combine mechanisms whose failure modes differ.

A constitutional architecture for science would separate epistemic functions. One system decides whether a project meets minimum methodological standards. Another allocates some exploratory funding. Another tracks replication status. Another manages conflict disclosures. Another audits concentration and network closure. Another evaluates downstream social or policy relevance. These functions can interact without collapsing into one score.

This is important because universal scores create universal attack surfaces. If an h-index becomes the dominant career currency, people optimize h-index. If a national evaluation formula determines university funding, universities reorganize behavior around the formula. If an AI credibility score becomes decisive, researchers will learn to optimize the model. Goodhart's law is not an argument against measurement. It is an argument against monoculture.

Constitutional design also preserves minority judgments. Review panels often seek consensus because a final decision is required. Yet disagreement contains information. A proposal rated transformative by two experts and implausible by three is not equivalent to a proposal everyone finds moderately good. The same mean score can conceal radically different epistemic situations.

Institutions should therefore store and sometimes publish disagreement structure in appropriate form. Consensus reports can include minority analyses. Funding portfolios can allocate a small share to high-variance proposals. Major scientific assessments can distinguish disputed assumptions from disputed observations. This is not procedural indulgence. It is error preservation. If the majority is wrong, the system retains a trace of the alternative path.

Separation of functions is particularly important because academic institutions frequently reuse the same signals. Journal prestige affects hiring; hiring affects access to grants; grants affect laboratory capacity; laboratory capacity affects publication volume; publication volume feeds rankings; rankings affect the next hiring decision. A constitutional design tries to break some of these correlations. It does not require every decision to be independent, which would be impossible. It requires enough partially independent channels that one early judgment cannot automatically become evidence in every later judgment.

Scientific checks and balances are functional rather than political metaphors. Journals, funders, universities, replication teams, data repositories, professional societies, and post-publication critics should not all depend on the same decision path. When independent institutions can challenge one another, error has more routes to become visible. The challenge is to preserve interoperability without creating a central authority that can make one mistaken classification propagate everywhere at once.

Plural selection mechanisms

Plural mechanisms also create diagnostic information. If masked and unmasked assessments diverge sharply, status signals deserve inspection. If expert ranking and a prediction market disagree, the disagreement itself can be informative. If different reviewer pools consistently favor different methods, the institution has learned something about its internal structure. Diversity of evaluation is therefore not only a fairness device. It is a sensor for hidden assumptions.

Rotation is another constitutional tool, but it should be used selectively. Term limits for certain editorial, panel, and governance roles can reduce entrenched networks. Yet rapid rotation can destroy expertise and institutional memory. A better approach may combine stable professional staff with rotating scientific judgment, transparent appointment criteria, and limits on repeated concentration of decision power among the same networks.

Independence matters as well. An institutional credibility audit should not be controlled entirely by the body being audited. Universities can maintain internal research-integrity offices, but national or cross-institutional mechanisms may be needed for serious cases. Journals can correct papers, but independent replication databases can provide an external record. Funders can evaluate their own programs, but selected mechanism studies should be conducted by independent metascience groups.

The architecture should also include exit and interoperability. Researchers should be able to move data, software, identity records, and contribution histories between platforms. Journals should not be the only durable home for scientific objects. Funding records should use open identifiers. Review artifacts, when sharing is ethically appropriate, should be portable enough to prevent repeated waste.

AI fits constitutional science best when multiple systems check one another. A literature-mapping agent should not be the same component that assigns a funding score. A citation-verification model can provide evidence to reviewers without knowing the applicant's identity. A bias-audit system can test the evaluator using counterfactual inputs. A reproducibility agent can execute code in a sandbox independent of the narrative-review model. Separation of computational powers matters for the same reason separation of institutional powers does: correlated failure is easier to detect when components have different roles.

The system must also preserve human responsibility. “The algorithm recommended it” is not an acceptable constitutional position. Every consequential automated output should have an institutional owner who can explain what role it played. Human oversight should not be ceremonial; reviewers need authority to reject machine output and must be given enough information to do so intelligently.

Scientific constitutionalism ultimately rejects two fantasies. The first is the fantasy of perfectly objective institutions. The second is the fantasy that abolishing institutions produces epistemic freedom. Large-scale knowledge requires organized trust. The task is to organize trust so that no single pathway becomes impossible to challenge.

Plurality should extend to assessment itself. CoARA's reform agenda emphasizes qualitative judgment supported, rather than dominated, by responsible indicators. DORA rejects journal-based metrics as surrogates for individual work. Registered Reports move part of publication judgment before results are known. Thresholded lotteries acknowledge near-ties. None is a complete constitution. Together they illustrate a more robust principle: different mechanisms should fail differently, so that one systematic bias does not determine every career and every research direction.

For centuries, the central constraint has been the price of attention. That constraint is changing. AI can read at a scale no committee can match, compare claims across literatures, inspect citations, execute code, and generate structured criticism. This does not make AI an oracle. It changes a cost curve. Tasks that were previously too expensive to perform routinely can become ordinary. The opportunity is large precisely because the danger is large: an automated system can also reproduce prestige bias, invent evidence, reward superficial fluency, or turn a flawed assessment rule into industrial-scale exclusion. The question is not whether AI will participate in evaluation. It is what role it should be allowed to occupy.

Chapter 7 - AI and the Falling Cost of Evaluation

The economics underlying the previous chapters changes when machines can inspect more material than a human reviewer could read. AI can retrieve citations, compare claims, execute bounded code checks, identify statistical inconsistencies, summarize methodological dependencies, and generate adversarial questions. None of these actions makes a model a scientific sovereign. Together, however, they reduce the cost of due diligence that previously made pedigree so attractive as a shortcut.

The opportunity comes with a symmetrical risk. A model trained on the record of science can inherit the status gradients, publication biases, and methodological fashions of that record and then reproduce them at greater speed. The relevant design question is therefore not whether AI should enter evaluation; it already is. It is how to use automation to expose evidence while preventing opaque model judgments from becoming a new form of borrowed credibility.

Many institutions of credibility were designed for a world in which careful evaluation was slow, expensive, and human. That world is not disappearing, but its economics are changing. A machine can now compare thousands of papers before lunch, search for missing citations, execute parts of an analysis, and identify disagreements that no individual reviewer would have time to map. The result is not automatic truth. It is an expansion of what can be checked routinely. That expansion can reduce the need to borrow credibility from prestige.

The collapse in evaluation cost

The critical economic change introduced by AI is not that machines become wise. It is that some forms of scrutiny become cheap enough to repeat. Citation checking can be continuous rather than occasional. Statistical assumptions can be flagged automatically. Similar claims can be compared across thousands of papers. Code and data links can be tested at submission and again years later. Even imperfect tools can be valuable when used to allocate scarce human attention toward anomalies that deserve deeper inspection.

Scientific reputation partly exists because verification is expensive.

If every reader could instantly inspect raw data, rerun analysis, trace every citation, compare the result against the complete literature, identify hidden dependencies, and test alternative explanations, journal brands would matter less. They would still matter for curation and community, but less credibility would need to be borrowed from them.

For most of scientific history, such inspection was impossible. Even a conscientious expert could deeply review only a tiny fraction of the literature. The information problem forced social shortcuts.

Artificial intelligence changes the marginal cost of several evaluative tasks. It does not make judgment free, and it certainly does not make truth automatic. But it can make first-pass inspection abundant.

A language model connected to scholarly databases can compare a manuscript's novelty claim against thousands of papers. A code agent can attempt to execute an analysis in a sandbox. A vision model can inspect images for suspicious duplication or manipulation. A citation agent can retrieve cited sources and ask whether the relevant passage actually supports the sentence. A statistical checker can recalculate reported quantities. A provenance system can identify whether supposedly independent studies share authors, datasets, code, or preprocessing pipelines. A replication monitor can update the status of influential claims as new work appears.

Each capability is imperfect. Together they change the economics.

The important distinction is between generation and evaluation. Generative AI makes producing plausible scientific text much cheaper. Systems such as the AI Scientist demonstrated that models can automate substantial parts of the machine-learning research cycle, including idea generation, code writing, experiments, and manuscript production in constrained settings. In 2025, Sakana AI reported that an end-to-end AI-generated paper passed peer review at an ICLR workshop under a disclosed experimental arrangement. Whatever one thinks of the scientific depth of such systems, the directional implication is clear: the cost of producing research-like artifacts is falling.

If production becomes cheap while evaluation remains expensive, credibility systems become overloaded. Journals receive more submissions. Reviewers face more text. Low-quality work can consume expert attention simply by looking polished. The old scarcity relationship reverses: writing is abundant, checking is scarce.

AI must therefore be used on both sides of the equation. Automating research without automating verification creates an epistemic denial-of-service problem.

The first gains are likely to come from narrow tasks with objective failure conditions. A citation either supports the proposition attributed to it or it does not. A referenced dataset is accessible or it is not. Code can often be executed, versions can be compared, duplicated text or images can be flagged, and numerical inconsistencies can be surfaced. These checks do not settle the scientific question, but they remove avoidable noise from human review. As the infrastructure matures, more sophisticated tasks can be attempted without pretending that all of scientific judgment is one generic language-model problem.

The new abundance of machine-assisted inspection changes which compromises remain defensible. If a system can automatically test whether code links resolve, whether reported tables can be regenerated, whether citations entail the claims attributed to them, and whether a manuscript overlaps substantially with prior work, it becomes harder to justify leaving all such checks to a small number of unpaid reviewers. The human bottleneck can move upward toward interpretation, importance, ethics, and genuinely novel methodological judgment.

From scarce review to continuous review

This creates what we might call evaluation elasticity. When the marginal cost of a check falls, institutions can demand more direct evidence before resorting to prestige proxies. But elasticity cuts both ways. AI also reduces the cost of generating papers, reviews, grant applications, and plausible nonsense. The volume problem may therefore grow faster than the evaluation savings. The winning architecture will not be a single model reading everything; it will be a layered system that uses automation to collect evidence, formal tools where possible to verify it, and humans to make accountable judgments under explicit uncertainty.

The most promising near-term role is not autonomous acceptance but triage and augmentation. AI can perform broad, reproducible checks before humans spend scarce expertise. It can tell a reviewer that three citations appear not to support the associated claims, that the reported sample count differs between methods and results, that a baseline has a newer implementation, or that an apparently novel method resembles prior work under different terminology. The human can then decide whether these observations matter.

The 2026 randomized study of an LLM feedback agent at ICLR 2025 offers an instructive example. The system did not replace reviewers. It gave feedback on the reviews themselves, flagging vague statements, possible misunderstandings, and unprofessional comments. More than twenty thousand reviews were included; reviewers who received automated feedback sometimes revised their reports, and blinded evaluation found the revised reviews more informative. The intervention improved a human process by adding a second layer of inspection.

This pattern is likely to generalize. AI can review the reviewer, audit the auditor, and compare parallel judgments. The value comes from redundancy rather than sovereignty.

Cheap evaluation also makes new institutional experiments possible. A funding agency could run masked AI-assisted analyses on every proposal to identify overlooked prior art or methodological risk, while prohibiting the system from generating an overall score. A journal could provide authors with a pre-submission verification report. A university could audit whether hiring dossiers make claims unsupported by the underlying record. A research council could map whether funded projects occupy an unusually narrow conceptual region.

As evaluation becomes cheap, however, surveillance becomes cheap too. The same infrastructure that checks claims can score people continuously. Universities may be tempted to build real-time productivity dashboards. Funders may use AI to predict grant success. Governments may profile controversial research. Publishers may create proprietary credibility rankings that become unavoidable career currencies.

The danger is not hypothetical because optimization pressure follows available metrics. Once a machine can calculate a number cheaply, institutions tend to ask whether they can use it.

The correct design principle is therefore asymmetrical: automate evidence gathering aggressively; automate consequential judgment cautiously.

Evidence gathering is usually decomposable and contestable. A system can show which passage supports its claim that a citation is mismatched. A global credibility score is harder to audit because it embeds weights and values. The more a machine output compresses heterogeneous evidence, the more it begins to recreate the pedigree problem in algorithmic form.

AI's greatest contribution to scientific credibility may be precisely the opposite of scoring. It can make scores less necessary by making the underlying evidence easier to inspect.

Continuous review should not mean continuous surveillance of researchers. The object being monitored is the scientific artifact and its evidential neighborhood. A high-impact claim can accumulate new replications, citations, corrections, dataset revisions, and contradictory measurements without requiring the authors to be permanently examined. This distinction matters for both legitimacy and workload. The system can allocate more automated checking to claims with large downstream dependency and less to low-impact material. Human attention is then triggered by meaningful state changes rather than by a universal demand to rereview everything. Evaluation becomes event-driven: a new replication, correction, or dependency can reopen a question when there is a reason to do so.

Continuous review should not mean continuous surveillance of researchers. It means that evidential objects can be rechecked when their dependencies change. A dataset version can trigger alerts to downstream analyses; a retraction can propagate to literature maps; a replication can update the visible status of a claim; software can be re-executed when environments change. This is closer to maintenance in engineering than to policing. The system watches the state of knowledge artifacts, not the private behavior of scientists.

> Lower evaluation cost does not imply that judgment should be delegated wholesale. The history of metrics already shows what happens when a useful instrument becomes a target. An AI score can acquire institutional authority even faster because it appears precise, scalable, and technically sophisticated. The productive role for AI is narrower and stronger: perform inspectable evidence work, generate challenges, reveal uncertainty, and leave consequential interpretation open to accountable human judgment.

The most tempting use of AI is also the most dangerous: ask a model for a score and call the score objectivity. A better design begins with a less glamorous role. AI can help reviewers notice omitted controls, trace claims to sources, compare manuscripts with prior art, and rewrite vague criticism into something actionable. A 2026 randomized study at ICLR 2025 showed that LLM-generated feedback caused a substantial share of reviewers to revise their reports and made revised reviews more informative. That is a useful model of augmentation: improve the human review process rather than silently replace its authority.

AI as assistant, not sovereign

The ICLR 2025 experiment reported by Thakkar and colleagues is useful because the model did not decide acceptance. It gave reviewers feedback on their reviews. More than 20,000 reviews were included; 27 percent of reviewers receiving feedback changed their reports, incorporating thousands of suggestions, and blinded evaluation found the revised reviews more informative. This is an example of AI improving the quality of an institution without claiming epistemic sovereignty.

The phrase “AI reviewer” suggests a familiar object: a model reads a paper and emits a review. That is the least interesting version of the idea.

A human review is already a compressed product. The reviewer reads selectively, remembers some parts of the literature, notices certain methodological issues, interprets novelty through personal experience, and writes a narrative summary. Asking one language model to imitate that entire process simply creates another opaque generalist reviewer, with different but equally consequential biases.

The better architecture is plural and specialized.

One system checks whether central factual claims are supported by citations. Another looks for obvious statistical inconsistencies. Another compares the methods with reporting standards relevant to the field. Another searches for prior art. Another tests whether the paper's conclusions outrun its reported evidence. Another attempts to reproduce computational results. Another generates the strongest plausible alternative explanation. Another estimates how dependent the evidence is on a small set of datasets or assumptions. A final component does not decide acceptance; it assembles an auditable dossier for human reviewers.

This approach has several advantages. It localizes error. If the novelty checker performs poorly in a niche field, its output can be discounted without discarding the statistical audit. It allows different tools to use different models and retrieval systems. It creates natural opportunities for disagreement. Most importantly, it prevents one model's latent notion of “good paper” from becoming a hidden constitution for science.

Recent studies show both the promise and the fragility of LLM-based review. The large ICLR experiment published in Nature Machine Intelligence in 2026 found that machine feedback could improve the specificity and informativeness of human reviews. Other work has found that LLM editorial recommendations can be highly sensitive to prompt formulation and can show only fair agreement between models even when many individual critique statements are grounded in manuscript text. This distinction matters. A model may be useful at identifying local issues while remaining unreliable at the global accept-or-reject decision.

Scientific judgment is unusually difficult for language models because quality is not fully contained in text. A method that looks elegant on paper may depend on laboratory realities the model cannot see. A benchmark improvement may be unimportant because practitioners know the benchmark is saturated. A theoretical result may be mathematically correct but conceptually sterile. A qualitative paper may derive strength from interpretive context that standard checklists flatten. Review is partly an assessment of evidence and partly a judgment about what a community should spend attention on next.

The second component is inherently normative. AI can assist, but pretending it is merely technical hides the value choices.

CASE IN POINT AI-assisted review is already moving from speculation to institutional experiment. The ICLR 2025 study did not ask models to accept or reject papers. Instead, an automated feedback agent criticized the reviewers' own reports for vagueness, misunderstanding, and unprofessional phrasing. Reviewers remained accountable for the final text. This narrow design is revealing: AI may produce the largest early gains not by replacing experts, but by making expert reasoning more explicit, consistent, and inspectable.

The strongest 2026 evidence for a useful intermediate role comes from ICLR 2025. Thakkar and colleagues ran a randomized study covering more than twenty thousand reviews. Reviewers who received LLM-generated feedback changed 27 percent of their reviews and incorporated more than twelve thousand suggestions; blinded evaluation found the revised reviews more informative.

Crucially, the model was improving the review rather than replacing the reviewer. That division of labor is a better starting point for institutional adoption than autonomous accept-reject scoring.

The new failure modes

The failure modes are equally important. LLMs can hallucinate sources, imitate consensus language, overvalue polished prose, and inherit prestige correlations from training data. Recent work has also emphasized that standard accuracy incentives can encourage guessing rather than abstention. A scientific reviewer should therefore be rewarded for calibrated uncertainty, source-grounded criticism, and explicit refusal when evidence is insufficient. High-stakes systems should expose provenance and intermediate checks rather than return an unexplained score that merely looks numerical and objective.

Another problem is hallucination. A reviewer that confidently invents a related paper, misstates a theorem, or attributes an unsupported result to a citation can damage authors while sounding authoritative. Work published in Nature in 2026 framed hallucination partly as an incentive-design problem: evaluation systems that reward accuracy without sufficient penalty for incorrect guessing can encourage models to answer when they should abstain. The implication for scientific review is direct. An artificial reviewer should be rewarded for calibrated refusal.

“I cannot verify this claim from the available sources” is often a better scientific output than a fluent paragraph.

This suggests that AI reviewers need explicit epistemic state. They should distinguish retrieved evidence from model inference, identify source spans, expose uncertainty, and abstain when relevant material is unavailable. Where the system proposes a criticism, it should make the basis inspectable. Where it predicts significance or novelty, it should label that judgment as subjective and model-dependent rather than factual.

Training data is another source of concern. Historical reviews contain historical norms. If senior authors historically received more charitable interpretations, a model trained to imitate those reviews may learn prestige bias. If certain countries or institutions are underrepresented, writing styles associated with them may be treated as lower quality. If fashionable topics receive higher scores, automated systems can strengthen fashion by learning popularity as a quality signal.

Counterfactual testing should therefore be mandatory for consequential deployments. The same manuscript can be evaluated with affiliations changed, names masked, rhetorical style simplified, citation prestige perturbed, and field labels altered. If irrelevant changes strongly alter the output, the system is not ready to influence real decisions.

There is also a strategic problem. Authors will learn the reviewer. If the model family, rubric, or tool becomes predictable, manuscripts can be optimized for it. This is not necessarily bad; authors already write to human norms. But machine optimization can become more extreme because feedback is cheap and repeatable. A paper may be iteratively tuned until the evaluator predicts acceptance while the underlying contribution barely changes.

The answer is not secrecy. Secret algorithms create accountability problems. The answer is plural evaluation and periodic adversarial testing. Institutions can use multiple independent systems, vary nonessential implementation details, and audit whether optimizing one evaluator transfers to genuine human judgments and downstream reliability.

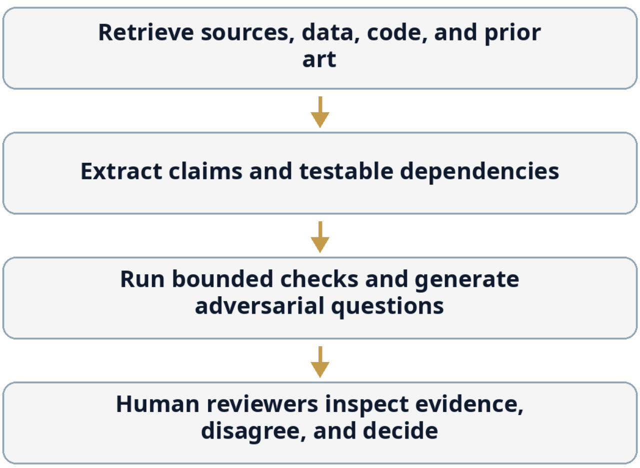
An artificial reviewer should therefore be conceived less like a colleague and more like a laboratory of evaluative instruments. Its purpose is to expand what can be inspected, not to replace the social process by which a community decides what matters.

A safe AI review architecture can be represented as a pipeline of bounded responsibilities. Each stage produces evidence for the next rather than a hidden final score.

Figure 3 shows the division of labor advocated in the AI chapters. The machine performs scalable evidence work and generates questions; humans retain the authority to interpret context, tolerate justified uncertainty, and make consequential decisions.

Figure 3. AI-assisted review without an AI sovereign

A safe default division of labor between scalable machine checks and accountable human judgment.



Automation expands inspection; humans retain consequential judgment.

This architecture also improves accountability. When the system surfaces a questionable citation or a failed code execution, the evidence can be inspected directly. When a model instead produces one opaque score, reviewers and authors are forced to argue with an authority whose internal weighting may be impossible to reconstruct.

Automation gathers and tests evidence; accountable humans remain responsible for consequential decisions.

The value of this design is inspectability. If a source is wrong, the retrieval stage can be corrected. If a statistical check is inappropriate, the methodological reviewer can reject it. If the final decision is controversial, the institution can show what evidence informed it. A monolithic score offers none of these repair points.

Automation introduces its own pedigree problem. A model trained on historical reviews can learn historical preferences for institutions, methods, topics, writing styles, and canonical citations. A model connected to bibliometric databases can reproduce visibility bias at machine speed. Hallucinated criticism can impose asymmetric costs on authors who must rebut it. The safeguards therefore need to be architectural: source-grounded outputs, explicit uncertainty, reproducible tool versions, counterfactual bias tests, human responsibility, and the ability to inspect what evidence produced a flag.

Chapter 8 - Auditing Bias and Institutional Health

Once evaluation becomes partly computational, a new possibility appears: the system itself can be subjected to experiments. Elena's manuscript can be presented with different affiliations, author names, career histories, or geographic contexts while the scientific content remains fixed. Changes in judgment become measurable status sensitivity rather than an argument about motives.

The same logic scales from individual decisions to institutional health. An epistemic guardian need not decide who is right. It can monitor concentration of grants, closed review networks, unusually persistent unreplicated claims, correction latency, citation dependence, and disparities between pedigree signals and later evidential performance. Observability is valuable precisely because it separates detection of fragility from accusation of misconduct.

Bias is difficult to discuss when every case is unique. Counterfactual testing makes the problem more concrete. Take the same manuscript and change only the identity signals: famous laboratory versus unknown laboratory, central institution versus peripheral institution, prestigious senior author versus early-career author. If the assessment changes, the evaluator is responding to something other than the scientific content. The pedigree ablation test turns a moral accusation into an auditable experiment.

Counterfactual review

Counterfactual auditing can be applied to people as well as models. A grant agency can periodically rerun historical proposals with institution and identity signals masked. A journal can compare score distributions before and after anonymization. A hiring committee can examine whether candidate rankings change when venue names and institutional logos are hidden during an initial contribution-level assessment. These experiments do not prove discrimination in any individual case. They reveal sensitivity of the procedure to signals that may or may not be appropriate for the decision.

Bias is often discussed through intentions. Did a reviewer consciously favor a famous laboratory? Did a committee discriminate against a lesser-known institution? Did an editor recognize the author and become more charitable? These questions matter ethically, but they are difficult to answer and often produce defensiveness.

A systems approach asks a different question: how sensitive is the decision to information that should not matter?

This is the pedigree ablation test.

Take the same research object and remove or alter status-bearing signals. Hide the institution. Mask the senior author's name. Replace the affiliation with a matched institution of different prestige. Remove journal branding from a paper used in a promotion dossier. Reformat the manuscript so that visual cues no longer reveal the venue. Ask independent evaluators to judge the variants. The difference between judgments becomes an estimate of sensitivity to pedigree.

Such tests already have precedents in studies of blind review, but AI makes them much cheaper and more scalable. Thousands of counterfactual versions can be generated while preserving substantive content. Review language can be compared not only for final scores but for severity, benefit of the doubt, requests for additional experiments, claims of novelty, and interpretation of ambiguity.

The test should be used diagnostically, not punitively. Individual reviewers work with limited time and cannot be expected to be free of every cognitive prior. Publishing a leaderboard of “most biased reviewers” would create fear, strategic language, and legal conflict while teaching institutions little. Aggregate analysis is more useful. Which disciplines show strong sensitivity? Which review stages amplify it? Does masking improve agreement or simply change the form of disagreement? Are certain criteria more pedigree-sensitive than others?

Ablation can extend beyond prestige. Where ethically and legally appropriate, systems can test sensitivity to gender-coded names, country, language style, seniority, institutional sector, and collaboration patterns. But the purpose must remain tightly bounded. These experiments process sensitive characteristics and can themselves become harmful profiling tools. Governance should specify who may run them, how results are aggregated, and whether raw individual-level data are retained.

The experiment can also be inverted. Instead of hiding prestigious information, institutions can deliberately add a prestigious signal to a weakly known submission and observe whether the score rises. Repeated over many historical cases, the distribution of changes becomes a calibration curve for status sensitivity. This approach is more informative than asking reviewers whether they are biased, because most status effects are not conscious. They often appear as reasonable changes in perceived feasibility, importance, or risk. Counterfactual review measures the effect without requiring a theory about motive. A second use is calibration over time. If status sensitivity falls after a review reform, the institution has evidence that the procedure changed behavior rather than merely changing its language. If it rises again as reviewers learn to infer identities from topic or citation patterns, masking alone may be insufficient. Periodic counterfactual audits make reform measurable and reversible instead of ceremonial.

Counterfactual review works best as an audit rather than as a permanent theater of artificial submissions. Institutions can sample decisions, vary only the signals they want to test, and estimate how much recommendations move. The same method can probe journal names in hiring files, institutional affiliations in grants, author seniority in manuscript review, or country labels in abstract assessment. The experiment must be designed carefully because masking can remove legitimately relevant information. Its value is that it converts a vague suspicion into a falsifiable institutional question.

Measuring status sensitivity

The design challenge is to distinguish legitimate context from irrelevant status. Some evaluations genuinely need information about resources, team capability, prior delivery, or access to specialized facilities. The answer is staged disclosure. First assess the scientific object under minimal identity information; then reveal execution-relevant context in a separate phase. The difference between the two assessments becomes data. Instead of arguing abstractly about whether pedigree matters, institutions can measure where and how much it changes their own decisions.

Pedigree ablation is also useful for AI systems. Before an automated evaluator is deployed, developers can create controlled test suites in which only irrelevant identity features change. A robust system should show limited variance when evaluating claim-local properties. If a citation checker changes its conclusion because the corresponding author moves from a famous university to an unknown one, something is clearly wrong. If an “impact” estimator changes, the interpretation is more complex because institutional resources may genuinely affect future reach. The test forces designers to separate properties that should be invariant from properties that legitimately depend on context.

This distinction is essential. Not every influence of pedigree is bias. If a project requires a unique facility available only at one institution, affiliation is relevant to feasibility. If a team has previously delivered a large international infrastructure project, track record may matter to execution risk. The correct goal is not pedigree blindness. It is pedigree locality: institutional information should influence only the decision dimensions for which it is causally relevant.

A funding system could formalize this through staged review. Stage one evaluates the scientific question, proposed method, and evidence of novelty under maximum feasible masking. Stage two evaluates execution capacity, resources, ethics, team composition, and project management with identity visible. The final decision preserves both scientific and operational information without allowing execution pedigree to silently redefine intellectual merit.

Hiring can use a similar structure. Committees might evaluate a small set of selected contributions before seeing citation metrics or recommendation letters. Later stages can incorporate the broader record. This does not remove reputation; it delays it until direct evidence has had a chance to shape the evaluator's first model.

The idea has a larger philosophical implication. Institutions often debate fairness through principles that are hard to falsify. Ablation turns fairness claims into experiments. If an institution believes its decisions are not strongly driven by prestige, it can test the belief.

Science should be unusually comfortable with such a proposition.

The results should be interpreted carefully. If affiliation changes a score for access to a unique telescope, clinical cohort, or supercomputer, that may be relevant to execution. If it changes a novelty judgment about an already completed experiment, the justification is weaker. An audit therefore needs decision-specific categories rather than one global bias number. The institution can then ask whether the amount of status sensitivity in each category matches an explicit policy. This turns an abstract fairness debate into a governance question with measurable tolerances. Over time, an institution can compare this status-sensitivity profile with outcomes. If masked and unmasked judgments diverge, which version better predicts later reproducibility, project completion, or expert reassessment? The answer may differ by decision type. Pedigree could improve predictions of execution in equipment-intensive grants while adding little to the assessment of a completed theoretical argument. This empirical comparison turns a normative dispute into a design problem. Instead of asking whether status should matter in the abstract, the institution asks where it contributes information, where it duplicates information already available, and where it introduces variance that has no defensible relationship to the decision.

Status sensitivity is not synonymous with discrimination. A team with a specialized facility may genuinely be more capable of executing a difficult experiment; a long record of careful data stewardship may legitimately affect a feasibility judgment. The audit should therefore be task-specific. If identity changes a methodological-quality score for identical methods, concern is stronger than if it changes a feasibility score that explicitly depends on infrastructure. This task decomposition prevents fairness auditing from collapsing into the unrealistic demand that context never matter.

> Counterfactual testing identifies bias in particular decisions; institutions also need a view of their own long-term epistemic behavior. The analogy is observability in complex technical systems. Engineers do not infer the health of a distributed service from one request. They monitor latency, concentration, failure patterns, dependency chains, and recovery. Scientific institutions can do something similar without turning research governance into surveillance of individual belief.

Modern engineering systems are observable because operators can inspect latency, errors, resource concentration, dependencies, and failure trends before the whole service collapses. Research institutions have far weaker observability. They can count publications and money, but they often cannot answer how concentrated review networks are, which important claims remain unreplicated, whether the same experts repeatedly evaluate one another, or how often corrections propagate downstream. An epistemic guardian is best understood as observability infrastructure, not as a machine that declares truth.

Observability for science

An epistemic guardian should be boring in the best engineering sense. It should count dependencies, surface anomalies, preserve provenance, and make institutional patterns inspectable. It might show that a small cluster of reviewers handles a disproportionate share of proposals in one niche, that a highly cited claim has never been independently replicated, or that corrections to a dataset have not propagated to downstream papers. These are operational facts, not machine verdicts about truth.

Modern institutions have observability systems for almost everything except their own epistemic behavior.

Companies monitor cash flow, security incidents, latency, customer churn, and operational risk. Cloud platforms expose dashboards showing failures before users complain. Financial institutions run stress tests. Aircraft contain redundant sensors and recorders. By contrast, a research institution may know how many grants it won without knowing whether its evaluation processes are becoming conceptually narrower, more concentrated, or more dependent on pedigree.

An epistemic guardian is a monitoring system for this neglected layer.

The name should not be confused with an authority that decides truth. A guardian does not determine which theory is correct. It observes whether the machinery through which claims receive attention and resources is behaving in ways the institution has declared undesirable.

At a funding agency, a guardian might measure concentration of awards by institution, repeated reviewer-applicant network proximity, the stability of scores under panel composition, the rate at which near-threshold applicants return, the conceptual diversity of funded portfolios, and the proportion of high-impact claims later replicated. At a journal, it might track review duration by author status, differences between masked and unmasked evaluation, citation-support errors, retraction and correction response times, and reviewer network closure. At a university, it might examine whether promotion decisions depend excessively on journal brands despite stated policy, whether internal seed funding reaches new entrants, and whether research-integrity concerns are resolved differently across status levels.

AI makes this possible because much of the relevant data is unstructured. Review text, proposal narratives, recommendation letters, methodological descriptions, correction notices, and scientific claims cannot be analyzed adequately with simple spreadsheets. Language models and graph systems can extract patterns, provided privacy and uncertainty are handled responsibly.

The guardian should operate according to three constraints.

First, it should be evidence-generating rather than decision-making. It flags patterns and produces auditable analyses. It does not automatically reject a grant because a network metric looks suspicious.

Second, it should be institutionally independent enough to criticize the processes it observes. An internal dashboard controlled by the same leadership whose performance is measured can become decorative compliance. Independence may come from mixed governance, external audit, protected reporting lines, or cross-institutional operation.

A minimal observability layer could begin with data institutions already possess: reviewer assignments, funding decisions, conflicts, retractions, corrections, citation dependencies, and replication links. The technical challenge is less severe than the governance challenge. Metrics must be defined before anomalies are interpreted, and normal variation must not be pathologized. A small field will naturally have more repeated reviewers than a large one. A capital-intensive domain will naturally concentrate resources. Observability is useful when it supplies context and historical baselines, not when every deviation from an average is treated as misconduct. Institutions can also compare the observability layer with outcomes it was not designed to optimize. If a warning about reviewer concentration predicts no later problem, the threshold may be too sensitive. If repeated corrections fail to trigger alerts, it is too weak. The guardian should itself be evaluated as a scientific instrument, with false positives, false negatives, calibration studies, and documented revisions.

Engineering observability is useful as a metaphor because it separates monitoring from control. A dashboard showing reviewer concentration does not decide that concentration is corrupt. A graph showing a highly cited but unreplicated claim does not decide the claim is false. It identifies a state that merits attention. The scientific equivalent should privilege descriptive, source-linked indicators and make thresholds contestable, rather than quietly converting anomalies into allegations.

The epistemic guardian

Governance matters because observability can itself become surveillance. Researchers should know what is measured, which data are retained, how alerts are interpreted, and how false positives can be contested. The system should prefer aggregated structural signals over intrusive personality scoring. Its role is closer to an auditor or safety monitor than to a disciplinary authority. If the guardian begins ranking people by a single opaque credibility score, the cure has reproduced the disease.

Third, it should itself be falsifiable. Every detector will generate false positives. A concentrated field may be concentrated because only a few institutions possess the necessary equipment. A cluster of mutual reviewing may reflect a genuinely tiny specialty. A strong correlation between prestige and scores may reflect real average quality differences. The system must expose data and alternative explanations rather than labeling anomalies as misconduct.

This last requirement separates a guardian from a moralizing algorithm. It asks questions.

Why did institutional concentration increase after a new funding program? Why do proposals from emerging groups receive more requests for preliminary evidence? Why are citations to a famous method predominantly ceremonial rather than evidential? Why did a claim remain in guidelines after multiple contradictory studies? Why are reviewers in one subfield unusually likely to recommend rejection for work using an adjacent methodology?

Some questions will have innocent answers. The value lies in making the institution able to see itself.

Guardians can also monitor AI. As automated tools enter review and research administration, institutions need systems that test drift, prompt sensitivity, counterfactual bias, hallucination rates, source coverage, and human override patterns. If humans always follow the model, nominal human oversight may be meaningless. If humans reject the model only when it recommends outsiders, automation may amplify rather than reduce pedigree effects.

There is a danger of bureaucratic metastasis. Every new accountability mechanism can create reports, committees, and compliance work that distract from research. Guardians should therefore be designed to reduce manual reporting. They should reuse operational traces, automate routine checks, and produce a small number of high-value signals. The success criterion is not the amount of governance activity but the reduction of epistemic blind spots.

Privacy must be built in. Most guardian functions can be performed at aggregate level. Individual-level investigation should require a higher threshold and appropriate due process. Raw review text and proposal content should not be retained indefinitely simply because storage is cheap. Monitoring systems should be constrained by explicit purpose.

The concept becomes particularly valuable across institutions. A national epistemic observatory could publish aggregate indicators of research-system health without ranking individual scientists. It could measure concentration, mobility, replication investment, openness, funding-mechanism diversity, correction speed, and evidence dependency. Such an observatory would complement conventional output metrics with indicators of self-correction capacity.

The health of science is not only how much it produces. It is how efficiently it notices when it may be wrong.

An epistemic guardian should observe the health of a system rather than rank the moral or intellectual worth of researchers. The table below translates that principle into measurable institutional questions. Its purpose is to keep monitoring close to operational risk.

Table 4 turns the idea of an epistemic guardian into an observability specification. The rows describe system-level signals that can be monitored without deciding in advance that an individual researcher has done anything wrong.

Table 4. What an epistemic guardian should observe

System-level signals for observing epistemic fragility before it becomes institutional failure.

The system watches structural conditions that affect self-correction and flags them for human interpretation.

Observed signal	Why it matters
Review concentration	A small closed reviewer pool can create blind spots, conflicts, and excessive dependence on one methodological culture.
Unreplicated high-impact claims	Downstream dependence can grow faster than direct verification, increasing systemic fragility.
Correction propagation	A correction is useful only if dependent datasets, reviews, and claims are updated or at least alerted.
Pedigree sensitivity	Counterfactual changes in affiliation or author status reveal how strongly non-content signals influence evaluation.
Methodological monoculture	Shrinking diversity in methods or assumptions can reduce the system's ability to detect shared failure modes.
Resource concentration	Funding and infrastructure concentration can be efficient, but extreme concentration increases path dependence and entry barriers.

The distinction between anomaly and allegation is essential. Concentrated review may be unavoidable in a tiny specialty; an unreplicated claim may simply be difficult to reproduce; slow correction propagation may reflect technical barriers. Monitoring creates questions for governance. It should not create guilt by dashboard.

None of these signals is a verdict. Concentration can reflect real excellence; an unreplicated claim may simply be too expensive to repeat. The guardian's job is to make these conditions visible early enough that humans can ask whether they are justified.

An epistemic guardian should itself be plural and auditable. Different institutions may run different detectors; external researchers should be able to study their false-positive patterns; no single proprietary credibility score should become mandatory infrastructure. The guardian's strongest outputs are questions: Which claims carry unusually large downstream dependence? Which review networks are unusually closed? Where did a correction fail to propagate? Those questions return responsibility to humans while making institutional blind spots cheaper to inspect.

The final part asks what a scientific system would look like if direct evidence became easier to inspect than it is today. Reputation would not disappear. It would lose some of its monopoly. A paper could become a graph of claims and dependencies rather than a sealed PDF. Confidence could change as replications and contradictions arrive. Institutional prestige could remain a useful prior without becoming a permanent verdict. The destination is not a reputation-free science. It is a science in which reputation is easier to earn, easier to question, harder to transfer blindly, and easier to update.

Chapter 9 - From Reputation Networks to Evidence Networks

Years after publication, Elena's paper is still represented by a stable citation while the evidence around it has changed. Some analyses have been reproduced, a boundary condition has been discovered, and one interpretation has weakened. A static bibliographic record cannot express this trajectory without forcing readers to reconstruct it manually.

Evidence networks offer a different abstraction. Claims receive stable identities and remain linked to methods, data, code, replications, contradictions, and corrections. Credibility then becomes an updateable state with provenance rather than a permanent property inherited from the original venue. Reputation still matters, but it becomes one input among several and should decay when it is no longer supported by current evidence.

Scientific communication is still organized around documents that imitate paper. The PDF is portable, stable, and culturally successful, but it hides dependencies inside prose. A reader sees a conclusion without automatically seeing every dataset, transformation, assumption, competing result, or replication on which confidence should depend. An evidence network changes the unit of navigation. The paper remains readable narrative; beneath it sits a machine-readable graph of what is being claimed and why.

The claim graph

A claim graph can begin modestly. Each major claim receives a stable identifier and links to the evidence that directly supports it, the methods used to produce that evidence, and known replications or contradictions. The system does not need a universal ontology of science on day one. It needs enough structure to answer practical questions: What is this claim based on? Which parts are independent? What changed after a correction? Which downstream conclusions depend on the disputed component?

Today's scholarly graph is primarily a graph of documents and people. Authors write papers. Papers cite papers. Institutions employ authors. Funders support projects. Bibliometric systems count and rank these relationships.

Tomorrow's scientific graph could be organized around evidence.

This sounds like a technical change but is fundamentally institutional. A reputation network asks who is connected to whom and what was cited. An evidence network asks which claims depend on which observations, methods, assumptions, datasets, software, replications, and counterarguments.

The difference can be illustrated with a common citation pattern. A paper states that a certain effect is well established and cites a review. The review cites several original studies. Several of those studies reuse related data. One relies on a method later criticized. Another measured a slightly different phenomenon. A fifth is frequently cited but did not actually support the precise claim. A conventional citation graph shows a dense and reassuring network. An evidence graph reveals that support may rest on fewer independent foundations than the citation count suggests.

Constructing such graphs manually would be prohibitively expensive at scale. AI makes partial automation plausible. Models can extract claims, classify citation roles, align methods, detect dataset reuse, and suggest contradiction relations. Human experts can then validate high-impact nodes. The goal is not perfect machine understanding of science. It is a better index into the evidential structure than keyword search and raw citation counts provide.

Evidence networks would change literature review. Instead of asking for the most cited papers on a topic, a researcher could ask for the strongest independent evidential paths supporting a claim, the most serious contradictory evidence, or the assumptions shared by apparently independent studies. A policymaker could ask which parts of a recommendation rely on mature evidence and which rely on expert extrapolation. A student could see how a textbook conclusion emerged historically and where uncertainty remains.

The graph should also preserve the difference between evidence and commentary. A replication result, a methodological critique, a retraction notice, and an expert opinion are all relevant but not interchangeable. Each needs a typed relationship and provenance. This is where neuro-symbolic or rule-governed AI can be useful: language models can extract candidate relations from prose, while explicit schemas and validation rules prevent the extraction layer from silently inventing dependencies. The graph becomes an auditable intermediate representation rather than an opaque summary produced by a model. The first implementation should therefore resist ambitious claims of semantic completeness. It is enough to represent a handful of high-value relations accurately: supports, contradicts, replicates, corrects, depends on, and reuses. More elaborate semantics can be added only when they improve real decisions. A smaller trustworthy graph is more useful than a universal knowledge representation whose edges cannot be audited. A further design choice concerns granularity. Claims should be split only where a distinct evidential dependency or correction path exists. If every sentence becomes a node, the graph turns into an unreadable transcription of prose; if only headline conclusions become nodes, important assumptions remain hidden. The practical compromise is to identify claims that can fail independently, that support consequential downstream work, or that require different kinds of evidence. This keeps the representation small enough to audit while preserving the places where disagreement and correction actually matter.

Evidence networks also make independence visible. Ten papers can cite a claim while relying on the same dataset, the same software implementation, or the same underlying cohort. A graph can represent those shared dependencies so that a reader does not mistake bibliographic multiplicity for evidential independence. This is especially important for automated synthesis, where systems that count supporting documents without tracing common ancestry can manufacture confidence from duplicated evidence.

Portable evidence and provenance

Open standards matter because proprietary evidence graphs can create a new form of institutional capture. The durable layer should use portable identifiers, open metadata, and exportable provenance. Commercial services can compete on search, visualization, review assistance, and inference without owning the epistemic history itself. UNESCO's open-science framework is relevant here: infrastructure is part of scientific governance, not merely technical plumbing. Whoever controls the metadata architecture can influence what becomes visible.

They would also change credit. Scientific contribution could be recognized through the structure of dependency. A dataset that supports hundreds of findings becomes visibly central. A replication that materially changes confidence in an important claim gains importance even if it is not highly cited. A methodological critique that invalidates a class of analyses becomes part of the evidential history of downstream claims. Credit becomes linked to what the work did to the knowledge graph rather than only where it was published.

This does not imply a universal numerical score. Indeed, the richest advantage of the graph is that it resists scalar compression. A claim can be strongly replicated but narrow. A method can be widely used but controversial. A scientist can contribute excellent infrastructure without producing fashionable theories. Different evaluators can inspect the same graph for different purposes.

Open standards are critical. If evidence graphs are proprietary, the institution controlling them gains enormous power over scientific visibility. Publicly funded research should favor interoperable identifiers, exportable provenance, and transparent relation types. Commercial services can add value, but the underlying scholarly record should not become hostage to one platform.

There are difficult ontology problems. What counts as the “same” claim across different wording? When does a conceptual replication count as support rather than merely related evidence? How should qualitative evidence be represented? What is the unit of contradiction when two studies differ in population or operational definition? These are not reasons to abandon the project. They are reasons to preserve uncertainty and plural representations.

A practical design can allow multiple interpretations. One system may classify two findings as replications; another may classify them as partial conceptual overlaps. The disagreement can itself be represented. Evidence infrastructure should not pretend that semantic mapping is objective when it is not.

The transition from reputation networks to evidence networks would also alter the social meaning of prestige. Prestigious researchers may remain highly connected because they genuinely produce central work. The difference is that their authority becomes more decomposable. A new claim from a famous scientist can begin with attention but must build its own evidential neighborhood.

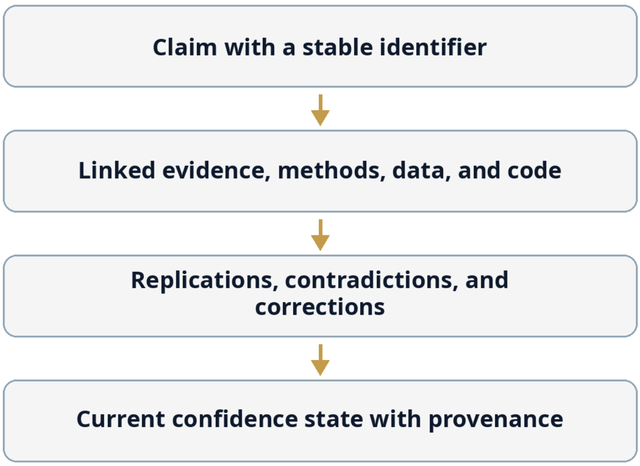
In the long run, this may be the most realistic way to reduce pedigree dependence. Moral appeals cannot eliminate human reliance on reputation. Better evidence infrastructure can make reputation less necessary for tasks where direct evidence becomes searchable.

The evidence-network idea can be implemented incrementally. The core is a simple chain from claim to support to challenge to update, with stable identifiers connecting the pieces.

Figure 4 illustrates the minimal evidence-network architecture proposed here. It is intentionally linear. A claim is connected to its evidential basis, then to later tests and corrections, and finally to a current state that records why confidence changed.

Figure 4. From a paper-centric record to an evidence network

A minimal path from document-centered publishing to living, provenance-aware evidence.



The readable paper remains; the evidential state becomes inspectable.

Real scientific graphs will branch, merge, and contain disagreement, but beginning with a simple dependency chain is preferable to designing a universal ontology that no discipline can adopt. The engineering principle is incremental structure: add enough machine-readable provenance to answer useful questions, then expand only where practice demonstrates value.

Narrative remains human-readable while the supporting structure becomes machine-readable and updateable.

Such a graph does not solve epistemology. It solves a more modest and important problem: it lets disagreement attach to the exact object under dispute. That makes correction cheaper, credibility more granular, and AI assistance safer because the machine has explicit provenance to inspect.

Portability matters because scientific objects outlive platforms. A claim identifier that works only inside one publisher, a provenance record locked to one vendor, or a credibility score that cannot be recomputed creates a new form of institutional capture. Open identifiers, versioned metadata, exportable evidence bundles, and transparent schemas allow universities, funders, journals, and independent tools to disagree while still referring to the same underlying objects. Interoperability is therefore an epistemic safeguard as much as a technical convenience.

> An evidence network becomes genuinely different from a reputation network only when it can forget. Prestige in academic systems often accumulates much more easily than it decays, even when the evidential environment has changed. A living credibility system should preserve historical reputation while updating its relevance to current claims. This is not punishment for past error; it is ordinary temporal calibration in a domain where methods, data, and knowledge continue to evolve.

Prestige has an unusual temporal property: it can persist long after the event that created it. A celebrated paper remains highly cited after serious qualifications appear. A famous laboratory can carry a reputation formed under previous leadership. A metric accumulated in one research era may continue to influence decisions in another. In finance we expect prices to update. In software we expect dependencies to age. Scientific credibility should also be capable of temporal revision without requiring public disgrace or institutional collapse.

Why reputation should expire

Credibility decay should not mean arbitrary punishment for age. Some results become more credible over time because they survive repeated use. The point is to prevent attention from freezing. Confidence should be refreshed by new evidence: successful replication, failed replication, correction, changed measurement standards, improved theory, or discovery of dependence among supposedly independent studies. A claim with no new evidence may remain stable; it simply should not gain authority merely because its citations continue to accumulate.

Prestige is remarkably durable. Evidence should not be.

A famous paper can remain famous for decades even after its central claim is narrowed, revised, or quietly abandoned. A textbook can preserve a simplified conclusion long after specialists have become cautious. A method can remain canonical because an entire field built infrastructure around it. Citations continue accumulating because later authors cite the historical landmark, the standard reference, or the review everyone else cites.

This creates a temporal asymmetry. Credibility is often accumulated through discrete successes but updated weakly when the evidential environment changes.

Financial systems have depreciation. Security credentials expire. Medical guidelines are revised. Software dependencies receive vulnerability notices. Scientific claims usually have no equivalent life-cycle status. The PDF remains fixed while the world around it moves.

Credibility decay does not mean that old findings should automatically become less true. It means that confidence should depend on the current state of checking rather than the age of the original prestige event.

A claim that has survived repeated independent tests can become more credible over time. A claim that became influential but was never independently examined should not inherit permanent confidence merely because it is old and familiar. A claim based on a dataset later found problematic should update downward. A method superseded by better measurement should be marked as historically important without remaining epistemically default.

This suggests dynamic evidence states. A claim can be new, independently checked, multiply replicated, contested, narrowed, superseded, or unresolved. These states are not final labels. They change as evidence arrives.

AI systems could maintain such states by monitoring new literature and proposing updates. If a replication appears, the system links it. If a correction changes a key parameter, downstream claims are flagged. If a retraction removes a foundation, dependent papers become candidates for review rather than automatically invalidated. If a new meta-analysis changes confidence, the relevant graph updates.

Aging matters differently across domains. A theorem does not become false because it is old, but its proof may acquire a simpler verification or reveal a hidden assumption. A clinical effect can become less relevant when treatment standards change. A machine-learning benchmark can lose meaning when its dataset becomes saturated or contaminated. A social-science estimate can depend on institutions that no longer exist in the same form. Credibility updates therefore need domain-specific triggers rather than a universal half-life. The principle is temporal awareness, not automatic depreciation. Updating should also distinguish a claim from the prestige of its originator. A researcher's broad reputation may remain high even when one result is revised, and one failed result need not contaminate unrelated work. Conversely, a celebrated career should not prevent a particular claim from losing confidence. Temporal updating works best when credibility is granular enough that correction does not require either total vindication or total reputational collapse.

Decay is best understood as a requirement to refresh, not as a fixed half-life. Some claims deserve increasing confidence because they are repeatedly used, independently replicated, and integrated with other evidence. Others should lose practical authority when methods become obsolete or when no independent check appears despite large downstream dependence. The update rule should be visible and domain-sensitive. What must decay is not old knowledge itself but the privilege of avoiding re-examination indefinitely.

Living confidence, not permanent authority

This dynamic view also makes correction less humiliating. If the scientific record is expected to update, a revised claim is not necessarily a scandal. It is the normal behavior of a living evidence system. Researchers gain incentives to attach corrections to the exact claims affected rather than defend an entire paper's identity. Institutions gain a way to recognize scholars who improve the record, including when improvement consists of narrowing, qualifying, or retiring their own earlier conclusions.

The challenge is avoiding automated contagion. A retracted paper does not necessarily invalidate every citation to it. A failed replication does not always refute the original effect. Context matters. Dynamic credibility therefore requires relation-aware propagation, not a simplistic score that drops whenever a negative event occurs.

The same principle should apply to people. Track record matters, but evaluation should weight relevance and recency. A researcher who made an important contribution twenty years ago may still be excellent, but the contribution should not function as unlimited credit for unrelated future claims. Conversely, a scientist who made a well-documented error and corrected it should not carry permanent stigma if later work is strong. Credibility systems should allow recovery as well as decay.

Institutions also need decay. University prestige is historically sticky. Rankings and reputation surveys can preserve status across generations. Some persistence is justified because institutional capacity changes slowly. But decision-makers should distinguish current capability from inherited brand. Funding and hiring systems can use more direct evidence about present laboratories, infrastructure, mentorship, and delivery rather than treating century-old prestige as a universal proxy.

Dynamic credibility also changes science communication. Instead of a binary distinction between “scientifically proven” and “debunked,” public interfaces could show how a claim's status evolved. This would make scientific self-correction look less like contradiction and more like the normal operation of a learning system.

The cultural challenge is significant. Institutions often fear that exposing uncertainty weakens trust. In the short term it can. In the long term, trust built on static certainty is brittle. When guidance changes, citizens may interpret revision as evidence that experts never knew anything. A system that publicly represents confidence as updateable has a better chance of making revision legible.

Credibility should therefore behave less like a medal and more like a maintained state.

Institutions can make updating routine by attaching review dates or evidence states to high-impact claims, much as standards bodies periodically revisit technical standards. Nothing needs to happen if the evidential situation is unchanged. The value is that staleness becomes visible. When a claim is heavily reused in policy, medicine, or downstream research, the cost of a periodic evidence check may be small relative to the cost of allowing an outdated assumption to propagate for another decade. This kind of review can be triggered by impact rather than age alone. A minor claim that few people use may not justify repeated attention, while a result embedded in clinical guidelines, regulatory standards, or hundreds of downstream models may deserve active monitoring. The principle is to spend verification effort where dependency makes error expensive. Maintaining confidence states also requires institutional ownership. Someone must be responsible for deciding when a correction is material, when two results are genuinely comparable, and when a downstream dependency should be alerted. AI can suggest these relationships, but governance cannot be delegated to an unattended model. Journals, repositories, funders, and disciplinary infrastructures can share the task through open protocols. The goal is not one central arbiter of confidence but a traceable record of who updated what, using which evidence and under which rule.

A living confidence state should preserve disagreement rather than compressing it into a single number. One replication may fail because the phenomenon is context-sensitive; one correction may narrow rather than overturn a result. Readers need to see what changed and why. A concise status such as 'supported in three settings, failed in one population, original effect revised downward after data correction' is often more scientifically informative than a score of 0.72 that hides the path by which the system arrived there.

Chapter 10 - The Right to Be Wrong

No architecture described in this book can guarantee truth. Elena can follow exemplary methods and still be wrong; a prestigious consensus can survive for years and later fail; an outsider can occasionally see what a mature field missed. The objective of reform is therefore not to engineer error out of science. It is to make error easier to discover, cheaper to admit, and less dependent on the social rank of the person who points to it.

The book ends where it began, with the paper from nowhere. A better system does not ask readers to ignore pedigree, nor does it romanticize obscurity. It asks them to know exactly what kind of information pedigree supplies and to demand a path by which direct evidence can defeat it. Scientific institutions earn trust not by being infallible, but by making correction structurally possible.

A perfectly designed scientific constitution would still produce wrong answers. That is not a defect unique to institutions; it follows from the fact that inquiry reaches beyond what is already known. A system that eliminates error by eliminating risk would eliminate discovery as well. The final design criterion is therefore paradoxical: institutions should become more capable of detecting error while preserving room for intelligent people to be wrong in interesting ways. Science needs discipline, but it also needs permission to surprise itself.

No constitution can guarantee truth

Any architecture that rewards only reproducibility will become conservative. Any architecture that rewards only novelty will become noisy. Any architecture that trusts only experts will become closed; one that distrusts experts will become incompetent. The design problem is therefore irreducibly plural. Science needs strong priors and real opportunities to defeat them. It needs stable institutions and alternative paths around them. It needs standards and protected spaces in which standards can be challenged.

A book about scientific credibility can easily end in the wrong place. After diagnosing bias, cumulative advantage, institutional capture, publication incentives, noisy review, and algorithmic risks, the temptation is to design a system so careful that error becomes institutionally unacceptable.

That would be a catastrophe.

Science requires the right to be wrong.

A speculative theory should be allowed to fail without destroying the career of the person who proposed it. A risky grant should be allowed to produce a null result. A young researcher should be allowed to challenge a famous laboratory and turn out to have misunderstood the problem. A consensus should be allowed to change. An institution should be able to admit that a policy built on reasonable evidence is no longer optimal. A model should be able to abstain.

The purpose of reform is not to eliminate error. It is to improve the relationship between error and correction.

Systems that punish visible failure too strongly produce hidden failure. Researchers avoid risky hypotheses. Institutions conceal uncertainty. Managers select projects with predictable deliverables. Reviewers reward safe methods. Scientists become reluctant to retract or correct because correction carries reputational cost. The resulting literature can look clean while the underlying exploration becomes shallow.

This is why epistemic reliability and psychological safety are connected. Error correction requires environments in which admitting error is less damaging than defending it indefinitely. Scientific prestige should therefore reward transparent correction. Journals should make corrections easy to issue without treating every change as scandal. Funders should distinguish informative failure from negligence. Universities should recognize researchers who identify problems in their own work.

At the same time, tolerance for error cannot become tolerance for carelessness. The distinction lies in process. Did the researcher state assumptions honestly? Were methods appropriate to the question? Were deviations disclosed? Was uncertainty represented? Were data preserved? Did the team respond constructively when problems appeared? A failed prediction can be excellent science. Fabricated data cannot.

This balance resembles exploration and exploitation in other complex systems. Exploitation concentrates resources on approaches that currently look reliable; exploration preserves the possibility that the current map is incomplete. Too much exploitation creates intellectual monoculture. Too much exploration dissipates resources across weak ideas. Scientific institutions already make this trade-off implicitly. A better architecture makes it explicit by maintaining different funding and publication channels for mature, verification-heavy work and for bounded, high-uncertainty exploration. Different institutions can specialize in different parts of this trade-off. Some laboratories and funders can prioritize reliable incremental work; others can protect exploratory programs whose failure rate is intentionally high. Diversity at the institutional level prevents every researcher from being forced into the same risk profile. It also makes the system easier to evaluate because one can compare which portfolios produce which kinds of knowledge. The constitution must finally protect exit. If a journal, funder, database, or assessment provider becomes systematically unreliable, researchers need alternative venues and interoperable records that allow scientific work to move. Exit is an epistemic safeguard because monopoly over infrastructure can turn procedural mistakes into field-wide constraints. Competition alone does not guarantee quality, but the credible possibility of switching reduces the cost of challenging a captured institution. Scientific pluralism therefore depends partly on technical standards that make evidence and reputation portable across organizational boundaries.

Every reform in this book can be gamed, misunderstood, or overextended. Open data can encourage superficial reanalysis. Blinding can hide legitimate feasibility information. Lotteries can be used where expert discrimination is actually strong. AI checks can produce false confidence. The appropriate response is not institutional pessimism but meta-corrigibility: reforms should be piloted, measured, compared, and reversible. A scientific institution should be able to learn that its mechanism for learning is itself defective.

Preserving the right to surprise

The right to be wrong is not a right to be careless. It is the right to test a serious idea without first proving that the idea will succeed. That right becomes credible only when paired with traceability, correction, and accountability. A mature scientific system should make it easier to take epistemic risks because failures are legible and recoverable. The final objective is not a world in which institutions always choose correctly. It is a world in which their mistakes are cheaper to discover than to defend.

AI systems need the same distinction. We should not demand that scientific AI always answer. Reliability improves when systems can expose uncertainty, request more evidence, or decline to judge. The 2026 work on hallucination incentives makes the general mechanism clear: if scoring rewards guesses more than calibrated abstention, models learn the wrong behavior. Scientific institutions can make the same mistake with humans. If careers reward confident positive stories more than careful uncertainty, people learn the wrong behavior.

The right to be wrong therefore has an institutional counterpart: the duty to remain corrigible.

Corrigibility means that a claim can be updated, a decision can be revisited, an institution can be audited, an algorithm can be challenged, and prestige can be overridden by evidence. It is the opposite of both dogmatism and nihilism. Dogmatism says authority has settled the question. Nihilism says no authority or evidence deserves more weight than any other. Corrigibility says some beliefs are strongly justified, but every pathway must remain connected to conditions under which it would change.

This principle also clarifies the role of outsiders. Science needs channels for unconventional ideas, but it does not owe every idea equal attention. Attention is scarce. A claim that contradicts mature evidence should carry a high burden. The fairness requirement is not equal probability of acceptance. It is access to a route by which strong enough evidence could, in principle, overcome pedigree and consensus.

A reliable institution is one that can be surprised.

That may be the best test of all the reforms proposed in this book. Can an unknown researcher with unusually strong evidence change the system's belief? Can a replication overturn a celebrated finding? Can an audit show that a funding mechanism is biased and lead to redesign? Can an AI tool report uncertainty instead of manufacturing a ranking? Can a famous scientist lose an argument without losing human dignity? Can a field preserve enough diversity that an alternative path exists when its dominant assumptions fail?

If the answer is yes, pedigree can remain useful without becoming destiny.

The deeper cultural change is to detach error from humiliation. Serious mistakes still require correction, and misconduct still requires sanction, but ordinary scientific error should be recoverable. When careers depend on never publicly retreating, researchers gain incentives to defend claims beyond the point justified by evidence. When updating is normal and credited, intellectual flexibility becomes compatible with status. A system that wants faster self-correction must make correction a respectable career event rather than a ritual of reputational destruction. This is where economic and cultural reform meet. A funding scheme that tolerates failed exploration is meaningless if promotion committees punish the same failure. A journal that welcomes corrections changes little if institutions interpret correction as evidence of incompetence. The right to be wrong therefore has to be supported across the chain of credibility, from funding and publication to hiring, promotion, and public communication.

The right to be wrong is paired with a right to be heard under a sufficiently strong evidential burden. Science does not owe equal attention to every claim, and mature evidence should produce strong priors. What it owes is a route by which new evidence can change the prior. The outsider should not need pedigree to make decisive evidence legible; the insider should not be protected from decisive evidence by accumulated status. That asymmetry is the core of an open epistemic order.

> The final movement returns to Elena's manuscript after its long institutional journey. It has accumulated a venue, citations, critics, replications, corrections, and a history of decisions made around it. The question is no longer whether the paper deserves trust in the abstract. The question is which parts deserve which degree of confidence, for what reason, and under what conditions that confidence should change. That is the more precise use of trust the redesigned system is meant to support.

The paper returns

Return to the manuscript from the prologue. Nothing about its data has changed, but the institution around it has. Its major claims have stable identifiers. The code can be executed in a controlled environment. Reviewers can see which citations support which statements. The first assessment hides pedigree; a later stage reveals team capacity where capacity actually matters. Replication status remains visible after publication. The paper is not guaranteed to be right. It is simply harder for reputation to substitute for inspection.

Modern science does not suffer from a single credibility crisis. It suffers from a mismatch between the complexity of knowledge and the simplicity of the signals used to manage it.

Pedigree, journals, citations, grants, awards, rankings, and expert consensus are not fake institutions placed on top of real science. They are part of the machinery that made large-scale science possible. They compress information, allocate attention, coordinate communities, and allow people to act before complete verification is available.

But compression creates error. The same mechanisms that reduce cognitive cost can accumulate power. The same reputation that helps us find expertise can become an inherited prior. The same grant that enables good science can create future evidence of deservingness. The same journal that filters literature can become a surrogate for article quality. The same consensus that rationally guides nonspecialists can obscure dependency among its evidential sources.

The solution is not to abolish credibility. It is to make credibility more granular, more dynamic, and more inspectable.

The most consequential opportunity created by AI is not artificial genius. It is a fall in the cost of epistemic due diligence. If machines can cheaply trace claims, check references, rerun code, map dependencies, compare counterfactual reviews, and monitor institutional patterns, then some of the trust previously borrowed from pedigree can be replaced with evidence.

This replacement should be selective. There will always be judgments that cannot be automated cleanly: which questions matter, which risks society should fund, how much uncertainty is acceptable in policy, which forms of knowledge deserve preservation, and which scientific capabilities a community wants to build. These are not bugs left over after better computation. They are value-laden governance decisions.

The architecture proposed here therefore preserves institutions while reducing their monopoly over interpretation. It preserves experts while making expertise easier to audit. It preserves reputation while constraining transfer. It preserves peer review while decomposing it. It preserves metrics while refusing to make them universal currency. It introduces AI as instrumentation, not sovereignty.

The guiding question changes from “Who deserves to be believed?” to “What would we need to inspect in order to believe this claim for the right reasons?”

That is a harder question. It is also a more scientific one.

The purpose of a scientific institution is not to identify a class of people whose statements can safely be accepted. It is to make the cost of discovering what deserves to be believed progressively lower.

The redesigned path also changes what happens when the paper fails. A later replication can weaken one claim without erasing the software, dataset, or methodological contribution that remains useful. A correction can propagate to downstream readers instead of surviving only as a notice attached to an old PDF. The authors can update their position without asking the community to choose between total vindication and reputational ruin. That possibility is central to the book's argument: credibility becomes more robust when institutions can localize error. A system that can say exactly what changed has less need to protect prestige as an all-or-nothing asset.

The difference in the redesigned system is not that Elena's paper is trusted less. In many cases it can be trusted more, because the reasons for confidence are decomposed. Readers can distinguish editorial screening from replication, reputation from execution capacity, citation from independent support, and AI-generated checks from human judgment. Trust becomes more specific, and therefore easier to update without collapsing into either blind deference or generalized suspicion.

A better use of trust

Trust remains. A scientist still relies on colleagues, institutions, instruments, and traditions. The difference is architectural: trust is more local, more explicit about its basis, and easier to update. The deepest reform is therefore not the abolition of pedigree but a reduction in how much epistemic work pedigree is forced to perform. Reputation should help us decide where to look. Evidence should help us decide what to believe.

A better use of trust would therefore feel less dramatic than a revolution. Editors would still edit, experts would still review, universities would still build reputations, and successful researchers would still attract collaborators. What changes is the amount of hidden inference carried by those facts. A prestigious affiliation would mean what it actually means: evidence about an environment and a history, not a certificate attached to every new claim. A citation count would mean attention and influence, not automatic truth. A grant would mean that a proposal won a selection process, not that the future has already validated the panel's judgment.

The institutional benefit is resilience. When credibility is distributed across claims, evidence, methods, replications, and transparent decisions, a failure in one component does not require society to choose between blind loyalty and wholesale distrust. A famous paper can be corrected without discrediting an entire field. A peripheral laboratory can earn confidence through inspectable evidence without first acquiring decades of prestige. An AI system can assist evaluation without being granted an unreviewable authority of its own. Trust becomes less brittle because it is backed by more visible reasons.

That is ultimately the argument of the book. Scientific institutions should not aspire to make reputation irrelevant. They should aspire to make direct examination progressively cheaper, so that borrowed credibility is needed for less time and for fewer decisions. The closer a judgment gets to a consequential claim, the more its confidence should rest on inspectable evidence rather than on the prestige accumulated around the people who carry it.

The same logic scales beyond academia. Regulatory science, journalism, courts, venture capital, and public policy all face decisions in which direct verification is expensive and reputational shortcuts are tempting. Science is simply a particularly revealing case because it explicitly claims to organize belief around evidence. Improving its credibility infrastructure would therefore have value beyond universities: it offers a model for institutions that need expertise without turning expertise into unchallengeable authority.

This also gives institutional trust a more defensible public meaning. Citizens and policymakers cannot personally reproduce most of the science on which modern societies depend. They will continue to rely on expert institutions. The strongest answer to skepticism is therefore not a demand for deference but an architecture that can show how disagreement was handled, what evidence supports a claim, where uncertainty remains, and how correction will be propagated. Trust becomes compatible with scrutiny. Experts can remain genuinely authoritative without being treated as infallible, because their authority rests increasingly on procedures that make challenge and updating normal rather than exceptional.

The practical standard is not that every reader should be able to verify every result. That would recreate the impossible burden from which reputation arose. The standard is that verification paths should exist, evidence states should be intelligible to the relevant specialist, and consequential institutions should be able to explain which shortcuts they used and why. A trustworthy system is not one in which trust is absent. It is one in which trust can be traced to reasons, challenged at the appropriate layer, and revised without requiring the collapse of the entire institution that previously carried it.

The mature alternative to pedigree is not universal verification. It is better allocation of verification. High-impact, high-uncertainty, safety-critical, or heavily reused claims deserve more scrutiny; routine low-stakes claims can rely more on ordinary trust. AI can help identify where additional checking has the highest expected value, while institutions decide the values and risk tolerances that define 'high'. The result is a system that spends its scarce human attention more deliberately.

Appendix A - Concepts for the General Reader

This appendix explains terms that are useful in the argument but should not be assumed knowledge for a general reader. The definitions are deliberately operational: they explain how each concept functions in the book rather than attempting to replace a textbook treatment. The linked Wikipedia pages provide accessible orientation and additional references; they are not used as evidentiary authorities for the substantive claims of the book.

The table is arranged as a vocabulary map. The first column contains the term, kept narrow so the concept remains visually distinct. The second column explains why the term matters, what it does and does not mean, and where a reader can continue. Because several terms have different technical meanings across disciplines, the explanations emphasize the usage adopted in this book.

Appendix Table A1. Key terms and accessible background

A general-reader glossary with direct links to Wikipedia for orientation, not as primary evidence.

Concept	Explanation and accessible link
Authority bias	The tendency to give more weight to a statement because it comes from a person or institution perceived as authoritative. Authority can be informative when expertise is real, but it is not direct evidence that a particular claim is correct. In this book the concept helps explain why pedigree can influence evaluation before the underlying evidence has been examined. Wikipedia

Concept	Explanation and accessible link
Bayesian prior	A prior is the belief or expectation held before new evidence is examined. In ordinary scientific judgment, reputation can function like an informal prior: a careful laboratory may initially be expected to produce more reliable work. The important rule is that new evidence must be allowed to update and, when strong enough, overturn that expectation. Wikipedia
Bibliometrics	The quantitative study of publications and citations. Bibliometric indicators can reveal patterns of influence, collaboration, or attention across large literatures, but they do not directly measure truth or research quality. Problems arise when convenient indicators such as citation counts are used as substitutes for richer evaluation of individual work. Wikipedia
Citation cartel	A coordinated or mutually reinforcing pattern in which authors, journals, or groups cite one another excessively in order to increase citation-based indicators. Not every dense citation network is a cartel; specialists naturally cite one another. The term matters because citation metrics can become strategic targets rather than passive records of intellectual dependence. Wikipedia

Concept	Explanation and accessible link
Conflict of interest	A situation in which a person responsible for a judgment also has interests that could be affected by that judgment. In research, reviewers and experts often work close to the topics they evaluate, so some conflicts are structurally unavoidable. Good governance therefore manages and discloses conflicts rather than pretending that expertise is interest-free. Wikipedia
Counterfactual test	A test that asks what would happen if one feature of a situation changed while the substantive content remained the same. A pedigree ablation test is counterfactual: the same paper can be evaluated under different author names or affiliations. If judgments change substantially, the difference estimates sensitivity to status rather than to scientific content. Wikipedia
Cumulative advantage	A process in which an early advantage creates resources or opportunities that make later advantages more likely. In science, a small funding or prestige difference can lead to better equipment, stronger collaborators, more data, and greater visibility. Later evaluators may then mistake the consequences of earlier opportunity for an independent confirmation of merit. Wikipedia

Concept	Explanation and accessible link
Double-blind peer review	A review arrangement in which authors do not know reviewer identities and reviewers are not shown author identities. The aim is to reduce some status and identity effects. Blinding is imperfect because topics, citations, datasets, or writing styles can reveal identity, and some evaluations legitimately require information about execution capacity. Wikipedia
Epistemology	The branch of philosophy concerned with knowledge: what it means to know something, what counts as justification, and how beliefs should respond to evidence. This book uses the adjective epistemic for properties related to the production, evaluation, correction, and transmission of knowledge rather than for questions of morality or administrative efficiency alone. Wikipedia
External validity	The degree to which a result can be expected to hold beyond the specific conditions in which it was obtained. A finding can be internally sound yet fail when applied to other populations, settings, periods, or measurement systems. Replication across varied contexts helps distinguish a robust regularity from a result that depends on narrow conditions. Wikipedia
Falsifiability	The idea that a scientific claim should expose itself to possible evidence that could show it to be wrong. Falsifiability is not a complete definition of science, but it captures an important institutional virtue: claims should not be protected from revision by making every possible observation compatible with them. Wikipedia

Concept	Explanation and accessible link
Goodhart's law	A warning that when a measure becomes a target, people adapt behavior to improve the measure, potentially weakening its relationship to the underlying goal. Publication counts, impact factors, citation scores, and ranking indicators can all become targets. The problem is not measurement itself but the assumption that a proxy remains neutral after rewards depend on it. Wikipedia
HARKing	An abbreviation for hypothesizing after the results are known. It refers to presenting a hypothesis as though it had been specified before seeing the data when it was actually formulated afterward. Exploratory discovery is valuable; the problem is disguising exploration as prior prediction, because that changes how strongly the result should count as evidence. Wikipedia
Incentive compatibility	A property of a system in which participants can pursue their own reasonable interests without being pushed toward behavior that undermines the system goal. Research institutions are more incentive-compatible when careful verification, data stewardship, correction, and informative negative results can advance a career rather than merely consume time needed for rewarded outputs. Wikipedia

Concept	Explanation and accessible link
Institutional capture	A condition in which an institution increasingly serves the interests, assumptions, or self-reproduction of the groups that dominate its procedures rather than the broader purpose for which it exists. In this book capture usually means endogenous narrowing rather than bribery or conspiracy: normal gatekeeping can gradually make alternative methods and questions harder to recognize. Wikipedia
Knowledge graph	A structured representation in which entities or claims are connected by explicit relationships. In scientific infrastructure, a claim graph can link a claim to data, methods, code, supporting studies, replications, contradictions, and corrections. The goal is not to replace readable papers but to make important evidential dependencies machine-inspectable. Wikipedia
Mechanism design	The study of how rules and institutions can be designed so that individual behavior produces desirable collective outcomes. The book uses this perspective when discussing grants, peer review, replication funding, appeals, and AI governance. Instead of asking researchers to become morally perfect, mechanism design asks how rules change the incentives and information around ordinary decisions. Wikipedia

Concept	Explanation and accessible link
Metascience	The scientific study of science itself. Metascience examines topics such as reproducibility, peer review, publication bias, research incentives, funding mechanisms, and evaluation practices using empirical methods. Many reforms discussed in this book should be treated as metascientific hypotheses: pilot them, measure consequences, compare alternatives, and revise them when they fail. Wikipedia
Open science	A broad movement toward making research processes, data, code, methods, publications, and evaluation more accessible and transparent where ethical and legal constraints permit. Openness can reduce verification costs, but it is not sufficient by itself: badly designed incentives can remain even when every file is public. Wikipedia
P-hacking	The use of flexible analyses, repeated testing, selective reporting, or other researcher choices in ways that increase the chance of obtaining apparently significant results. The term does not imply that every flexible analysis is misconduct. It highlights how analytic freedom can create misleading certainty when the full search process is not visible. Wikipedia

Concept	Explanation and accessible link
Peer review	Evaluation of scholarly work by people with relevant expertise. Peer review can detect errors, improve exposition, and filter obviously weak work, but it is not the same as independent replication. Reviewers usually inspect a manuscript under time constraints and rarely rerun every experiment or reconstruct every transformation from raw data. Wikipedia
Preregistration	A time-stamped record of a study plan made before the relevant results are known. Preregistration can clarify which analyses were planned and which were exploratory, reducing some forms of selective reporting. It is a transparency tool rather than a guarantee of quality; a preregistered study can still have a weak design or be poorly executed. Wikipedia
Principal-agent problem	A problem that arises when one party delegates work or decisions to another party whose incentives or information differ. Research funders, universities, journals, reviewers, and researchers repeatedly occupy principal and agent roles. Monitoring and incentives must therefore be designed with the possibility that locally rational behavior diverges from the institution's stated goal. Wikipedia

Concept	Explanation and accessible link
Provenance	Information about where a result, dataset, statement, or artifact came from and how it was transformed. Scientific provenance can record sources, versions, analysis steps, software, decisions, and later corrections. Good provenance makes it easier to inspect why a claim is believed without requiring readers to trust the reputation of the people who transmitted it. Wikipedia
Publication bias	The tendency for studies with statistically significant, positive, striking, or otherwise attractive results to be published more often than studies with null or inconvenient results. If the visible literature is a selected subset of all completed work, readers can overestimate how consistent or strong an effect really is. Wikipedia
Registered Report	A publishing format in which a study question and method are peer reviewed before the results are known. Journals can offer in-principle acceptance based on the importance of the question and quality of the method, reducing the incentive to make publication depend on a surprising outcome. Results are still scrutinized after the study is completed. Wikipedia

Concept	Explanation and accessible link
Replication crisis	A broad term for evidence that many published findings, particularly in some fields, have been difficult to reproduce or replicate reliably. The phrase should not be read as proof that science generally fails. It describes a set of empirical and institutional problems that motivated stronger methods, transparency practices, replication projects, and metascientific research. Wikipedia
Reproducibility	The ability to obtain consistent results using the same data, code, or analytical procedure, although terminology varies by discipline. Replicability often refers to obtaining compatible findings in a new study or new data. The distinction matters because a perfectly reproducible computation can still be based on a weak experiment, while a sound effect may vary across contexts. Wikipedia
Research integrity	The norms and practices that support trustworthy research, including honest reporting, responsible data management, appropriate authorship, conflict management, correction of errors, and avoidance of fabrication or falsification. Integrity is broader than misconduct rules because reliable science depends on many routine practices that never become disciplinary cases. Wikipedia

Concept	Explanation and accessible link
Selection bias	Systematic distortion caused by the way observations, people, studies, or records enter an analysis. In the context of this book, selection can occur before statistics begin: prestigious work may be more likely to be read, cited, funded, or retained in the visible scientific record, changing the evidence later observers encounter. Wikipedia
Scientific consensus	A broad convergence of expert judgment within a scientific community based on the available evidence. Consensus is often the best practical guide for non-specialists, especially when evidence is complex. It remains corrigible: a healthy consensus has identifiable evidential foundations and mechanisms through which sufficiently strong new evidence can change it. Wikipedia
Statistical significance	A conventional way of assessing whether observed data would be unusual under a specified statistical model, commonly summarized by a p-value threshold. Statistical significance is not the same as scientific importance, practical importance, truth, or replicability. Treating a threshold as a verdict is another example of converting a limited proxy into a stronger claim. Wikipedia

Concept	Explanation and accessible link
Version control	A system for recording changes to files or structured artifacts over time so that earlier states can be recovered and differences inspected. Scientific evidence networks can borrow this idea: claims, data, code, corrections, and confidence states should have identifiable versions rather than silently overwriting their history. Wikipedia
Whistleblower	A person who reports suspected wrongdoing or serious institutional failure, often at personal or professional risk. Effective research governance protects good-faith reporting while also providing fair procedures for those accused. Whistleblower protection matters epistemically because institutions cannot correct problems that participants are structurally afraid to make visible. Wikipedia

The concepts above are connected rather than independent. Goodhart's law explains why a metric can change after it becomes a target; mechanism design asks how rules can reduce that distortion; metascience tests whether the proposed rule actually works; provenance and version control make later checking feasible; and concepts such as selection bias, publication bias, authority bias, and cumulative advantage describe different ways in which the visible scientific record can diverge from a neutral sample of all possible evidence.

A useful reading habit is to return to this appendix whenever a term begins to sound like a verdict. None of these concepts should be used rhetorically to dismiss an institution, a field, or a result. They are diagnostic tools. Their value lies in helping readers ask more precise questions about evidence, incentives, uncertainty, and procedure.

Appendix B - A Practical Institutional Blueprint

A staged implementation path

The reforms in this book are intentionally plural because no single institution can implement the entire architecture at once. A practical transition can begin with a small number of separations that reduce the most damaging forms of credibility transfer without paralyzing research.

The first separation is between scientific merit and execution capacity. Grant programs can review the intellectual core of a proposal under maximum feasible masking before exposing institution, team reputation, and infrastructure. Identity becomes relevant when feasibility is assessed, not before the scientific question has been evaluated on its own terms. The two assessments remain visible rather than collapsing into a single score.

The second separation is between publication and verification. Journals can publish manuscripts while attaching explicit verification states: data available or restricted, code executed or not executed, key statistics checked or unchecked, independent replication absent or present, major post-publication critique unresolved or addressed. These states should be descriptive rather than punitive. They make evidential maturity legible.

The third separation is between evidence gathering and consequential judgment in AI systems. Automated tools can search, compare, reproduce, and flag. Their raw findings should be accessible to humans. Overall accept, hire, promote, or fund recommendations should either be prohibited or treated as experimental advisory outputs until strong evidence demonstrates reliability across contexts.

The fourth separation is between individual assessment and system audit. Epistemic guardians should primarily monitor aggregate processes. Their mandate is to identify concentration, instability, network closure, correction latency, or bias sensitivity. Individual investigation should occur only when specific evidence and due process justify it.

The fifth separation is between scientific disagreement and procedural misconduct. Appeals should not become a second scientific peer-review system. They should address conflicts, rule violations, factual administrative errors, discriminatory treatment, or demonstrable misuse of automated tools. This preserves autonomy while giving participants procedural rights.

A credible implementation program should itself be experimental. Institutions should establish baseline measurements, introduce reforms in limited domains, evaluate unintended effects, and publish results. A double-blind stage may reduce prestige bias but increase reviewer disagreement. A lottery may reduce concentration but create political discomfort. AI verification may catch citation errors but burden authors with false positives. Every reform should be treated as a hypothesis about institutional behavior.

The most important implementation principle is reversibility. Institutions should avoid reforms that create new irreversible monopolies. Open standards are safer than proprietary credibility scores. Multiple review mechanisms are safer than one universal model. Auditable tools are safer than black-box rankings. Pilot programs are safer than immediate system-wide mandates.

Reform should reduce the cost of correction, including correction of the reform itself.

A sensible pilot would begin with one bounded decision process, such as a fellowship call or internal seed-grant program. The institution could separate masked contribution assessment from execution assessment, preserve reviewer disagreement, publish the criteria in advance, and run a retrospective pedigree-sensitivity audit. AI could be used only for evidence collection and review-quality assistance. The pilot should be evaluated against ordinary outcomes such as reviewer workload, applicant trust, diversity of funded approaches, later project performance, and frequency of procedural appeals. Reform becomes credible when it is falsifiable.

The pilot should be instrumented from the beginning. Baseline data from the old process matters because a new mechanism can feel fairer while quietly increasing another error. A masked stage may reduce pedigree effects but make it harder to evaluate genuinely relevant infrastructure constraints. A new appeal process may correct conflicts but consume so much time that decisions become unusable. The point of staging is to expose such trade-offs while the system is still small enough to change. Successful pilots should publish not only their preferred metrics but the failure cases that caused the design to be revised.

A useful implementation sequence is therefore conservative. First collect baseline evidence about the existing process. Then add one separable intervention, such as masked first-stage review or structured conflict metadata. Only after that intervention is understood should AI assistance, claim graphs, or richer audit mechanisms be layered on top. Otherwise a failed pilot will be impossible to diagnose because too many variables changed at once. Institutional reform deserves the same experimental discipline that the book asks of science: controlled changes, explicit hypotheses, recorded failure modes, and willingness to reverse an intervention when its costs exceed its benefits.

A staged transition should begin where evidence about effects can be collected. A funder can pilot masked scientific assessment in one program, retain visible feasibility review, and compare concentration and reviewer disagreement with previous rounds. A journal can add structured evidence states without changing acceptance criteria. A university can evaluate selected contributions before exposing journal metrics. Each pilot should publish what improved, what became worse, and what remains uncertain, turning institutional reform into an object of metascience rather than a declaration of virtue.

What not to centralize

The blueprint should explicitly prohibit a universal researcher credibility score. Different scientific tasks require different evidence, and a single score would recreate the transfer problem in computational form. Institutions should also resist making every open-science practice mandatory regardless of domain; legitimate privacy, security, indigenous knowledge, and commercial constraints exist. The objective is not maximal visibility. It is proportionate inspectability where public or institutional decisions depend on claims that can reasonably be checked.

A second prohibition concerns hidden composite scoring. It is tempting to combine publication quality, replication history, open-science practices, conflicts, funding success, and AI-generated evaluations into one convenient number. The convenience is precisely the danger. Composite scores conceal trade-offs and encourage optimization against the metric. A better interface shows separate evidence dimensions and requires decision-makers to state which dimensions matter for the specific task.

Nor should institutions centralize every review artifact indefinitely. Transparency has costs: candid deliberation can be chilled, commercially sensitive proposals can be exposed, and privacy risks can accumulate. Retention schedules should therefore be proportionate to purpose. Some evidence should remain permanently public, some should be accessible only to auditors, and some should expire. Good epistemic governance is not maximal data collection; it is disciplined collection with a clear reason for every retained signal.

Finally, the architecture should avoid making AI mandatory where it adds little value. A small theoretical workshop may need careful human discussion rather than an elaborate automated pipeline. A large grant call with thousands of submissions may benefit greatly from machine-assisted conflict checks and evidence extraction. The principle is subsidiarity: use the lightest mechanism that provides adequate accountability, and increase technical machinery only when scale or risk justifies it.

A pilot should also define an exit condition. If the new process adds substantial administrative cost without improving calibration, diversity of methods, reviewer satisfaction, or downstream research quality, it should be simplified or abandoned. Reform should not become a permanent bureaucracy merely because it was introduced under the banner of fairness or openness. The same standard applied to science should apply to the institutions that govern science: claims about better governance require evidence.

Decentralization also matters across institutions. A national funder, a university, a journal, and a professional society should not all need to purchase the same proprietary credibility score or depend on one opaque model provider. Shared standards for provenance and evidence can coexist with multiple implementations, local governance, and independent auditing. That arrangement is less convenient than one universal platform, but inconvenience is sometimes the price of avoiding a new form of institutional capture. The infrastructure of scientific assessment should make switching and comparison possible, so that a bad evaluator can itself be evaluated and replaced. A practical procurement rule can reinforce this architecture: publicly funded assessment systems should expose exportable decision records and stable interfaces rather than trapping evidence inside proprietary dashboards. This is not an argument that all software must be public. It is an argument that a public institution should remain capable of changing the software through which it exercises consequential epistemic authority without losing its own history.

There is a further political reason to resist centralization. Whoever controls a universal assessment layer acquires power not merely over publication but over hiring, funding, collaboration, and public legitimacy. Even a technically excellent score would become a target for lobbying, strategic optimization, and eventually capture. The safer design is federated: common protocols for exchanging evidence, multiple evaluators, local decision rights, and the ability to compare outputs. Interoperability should make pluralism cheaper, not create a single gate through which every research career must pass.

The most dangerous shortcut would be to replace journal prestige with one universal AI credibility score. That would preserve the original category error while concentrating it in technical infrastructure. Central services can provide identifiers, provenance, and interoperability, but consequential evaluations should remain reproducible by multiple systems and contestable by institutions with different mandates. The architecture should make it cheap to compare judgments, not cheap to impose one judgment everywhere.

Appendix C - Legal and Policy Design Principles

Procedure before prescription

A legal framework for scientific credibility should begin with restraint. Legislatures are poorly positioned to prescribe correct scientific methods across disciplines, and courts should not become routine appellate bodies for peer review. The legitimate role of law is to protect the conditions under which epistemic competition and institutional correction remain possible.

Publicly funded research should default toward inspectability. Publications should be accessible, and data, code, protocols, and provenance should be available when privacy, security, ethical obligations, and legitimate intellectual-property constraints permit. Where access must be restricted, institutions should preserve controlled routes for verification rather than treating restriction as equivalent to exemption from scrutiny.

Research-assessment procedures should be transparent enough to be understood by participants. Public funders and public universities should publish the criteria by which consequential scientific decisions are made, including the declared role of metrics and automated tools. This does not require disclosure of every confidential deliberation. It requires that the rules of the game exist before the result.

High-stakes automated evaluation should be subject to proportional requirements. Systems influencing grants, employment, promotion, or institutional funding should have documented purposes, validation evidence, known limitations, audit logs, human responsibility, contestability, and periodic bias testing. Models should not infer or use sensitive personal attributes unless there is a lawful, necessary, and explicitly justified reason.

Good-faith scientific criticism should receive legal and institutional protection. Whistleblower rules should provide confidential reporting channels and anti-retaliation mechanisms while preserving protections against knowingly false accusations. Retraction, correction, and post-publication criticism should be designed to make truthful updating easier rather than reputationally catastrophic.

Conflict rules should recognize intellectual and institutional conflicts as well as financial ones. The aim is not to exclude every competitor from review, which would make expert evaluation impossible. It is to disclose and manage conflicts proportionally, diversify decision pathways, and avoid repeated concentration of evaluative authority in tightly connected networks.

Procurement and infrastructure policy should favor interoperability. Public research systems should be wary of allowing a single vendor to control researcher identity, publication access, citation data, research analytics, and AI evaluation simultaneously. Portability and open standards are not only competition policies; they are safeguards against epistemic lock-in.

Finally, law should protect institutional experimentation. Funders should be able to test alternative allocation mechanisms, including masked stages, randomized components, and different review decompositions, under transparent protocols. Research assessment itself should be treated as an object of research.

The legal objective is modest but important: no institution should be required to believe a particular scientific conclusion, and no scientific institution exercising public power should be allowed to make its own mechanisms impossible to inspect.

Policy should therefore be evaluated with the same skepticism applied to scientific assessment. Does a disclosure rule expose a real conflict or merely add forms? Does an appeal process correct procedural error or create strategic delay? Does open access reduce barriers or shift costs onto authors without resources? Does a term limit broaden participation or remove scarce expertise too quickly? Good policy is not defined by the virtue of its intention but by the error profile it creates in practice.

Legal design should also preserve room for institutional experimentation. A procedural floor can require conflicts to be handled, public decisions to leave an auditable record, and good-faith integrity reporting to be protected, while allowing different disciplines to satisfy those obligations differently. Clinical research, pure mathematics, national-security research, and ethnography do not have identical disclosure or replication constraints. Regulation works best when it specifies the right being protected and the accountability outcome expected, then demands reasons for exceptions rather than pretending that one workflow is scientifically universal.

The same restraint should govern enforcement. Not every procedural deviation is misconduct, and not every bad scientific decision should become a legal dispute. Remedies can be graduated: correction of a record, recusal, rerunning a decision stage, independent audit, or, only for serious breaches, formal sanction. This preserves a distinction between ordinary uncertainty, institutional error, negligence, and deliberate wrongdoing. A legal framework that collapses those categories would encourage defensive bureaucracy and make genuine integrity cases harder to identify.

The policy boundary can be stated simply: law may require institutions exercising public power to explain and audit their procedures, but it should be extremely reluctant to declare which contested scientific conclusion must prevail. Requirements for data stewardship, conflict management, procurement transparency, access to reasons, due process, and protection from retaliation can strengthen the conditions of inquiry. Substantive scientific conclusions must remain corrigible by evidence rather than fixed by legal authority.

Economic rights and institutional mobility

One underused policy lever is portability. Researchers often depend on institutions not only for salary but for grants, data access, equipment, intellectual-property arrangements, and the administrative capacity to continue a project. Where feasible, funders can make a portion of support portable across host institutions, reducing the extent to which institutional affiliation becomes an irreversible career dependency. Portability will not suit every infrastructure-heavy domain, but as a principle it aligns bargaining power more closely with the people and teams who produced the scientific value.

Portability is only one part of institutional mobility. A researcher can also be locked in by data-use agreements, nonportable computing infrastructure, internal intellectual-property rules, or the loss of administrative support required to manage a large project. Funding agencies can reduce this dependency by standardizing transfer clauses, requiring timely decisions when investigators move, and distinguishing genuinely immovable infrastructure from arrangements that are merely administratively convenient for the host. The objective is not to turn every grant into personal property. Public and collaborative projects have obligations that outlive one investigator. The objective is to reduce situations in which institutional loyalty becomes a condition for continuing scientifically valuable work.

Mobility also affects competition among institutions. If people, grants, and research objects can move with proportionate friction, universities have stronger incentives to provide good environments rather than relying on prestige as a retention mechanism. Smaller institutions can compete for teams with demonstrated evidence rather than only for early-career talent that has not yet accumulated portable resources. This can modestly weaken cumulative advantage without pretending that all institutions can or should provide identical facilities. In capital-intensive domains, a major laboratory will still matter. The reform is to make the dependency explicit and specific: access to this instrument matters for this project, rather than the institution's general prestige silently mattering for every future judgment.

Legal and contractual design can support the same goal through transparent intellectual-property policies, clear ownership of research software and data, reasonable notice periods, and dispute procedures that do not leave a project frozen for years. These rules are mundane compared with debates about truth and merit, but they influence who can actually leave a prestigious center and continue working. A credibility system that invites intellectual independence while making economic independence impossible will reproduce the old hierarchy under new assessment language.

Finally, mobility should include the right to change direction. Long grants and permanent positions can protect ambitious research, but they can also entrench programs that no longer deserve their opportunity cost. Portability and long-horizon support therefore need periodic, evidence-based review at the level of the program rather than constant output pressure on every individual. The aim is a labor market in which scientists can take epistemic risks without becoming permanently dependent on one institution, while institutions retain the ability to stop supporting work whose rationale has genuinely disappeared. Mobility should also be measured rather than assumed. Funders can track how often grants transfer successfully, how long transfers take, which contractual clauses cause delay, and whether moves systematically disadvantage early-career investigators or researchers leaving elite centers. Without those data, portability can exist formally while remaining unusable in practice. The same principle recurs throughout the book: a right or reform matters only when its operational path can be inspected and used.

Legal and economic design also affects the ability to exit a captured environment. Portable grants, researcher-controlled data access where ethically possible, interoperable identities, reasonable intellectual-property terms, and labor protections can reduce the cost of moving between institutions. Mobility is not only a career benefit. It prevents a single employer, infrastructure provider, or disciplinary network from becoming the sole route through which a research program can survive.

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The scholarly sources below support the book's substantive empirical and conceptual claims. Appendix A contains direct Wikipedia links for accessible orientation to terminology; those pages are not treated as primary evidence. Their role is pedagogical, while the research literature below is the evidential spine of the argument.

This bibliography is an evidential spine rather than an exhaustive literature review. Where the book proposes mechanisms that have not been validated at scale, they are presented as institutional-design hypotheses that should be piloted, audited, and compared with existing practice.

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The Wikipedia links are included to lower the entry barrier for readers encountering unfamiliar terminology. They should be read as orientation pages that point outward to broader literature, not as substitutes for the research articles, policy documents, and primary sources cited above.